Attribution-NonCommercial-NoDerivatives 4.0 International
http://creativecommons.org/licenses/by-nc-nd/4.0/
https://doi.org/10.26219/heal.aueb.9726



Athens University of Economics & Business

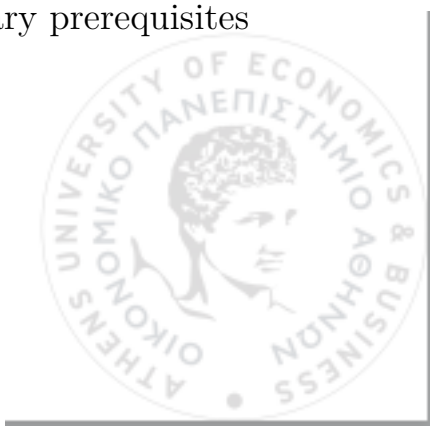
Machine Learning Framework for Expected Goals Inference: Bridging Statistical Modeling and Causal Understadning in Football Analytics

Michalis Troullinos

MSc - Business Economics with Analytics

Dissertation submitted in partial fulfillment of the necessary prerequisites for the acquisition of the MSc Degree

November 2025



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Abstract

Football analytics has evolved from descriptive statistics to sophisticated machine learning models, yet much of the existing research remains limited to isolated predictive or exploratory approaches. This study proposes an integrated analytical framework that combines supervised, unsupervised, and causal inference methodologies to advance the modeling of Expected Goals (xG) and deepen the understanding of shot quality and decision-making in football. Supervised models, including logistic regression, decision tree, random forests, gradient boosting, xgboost and neural network were employed to predict goal probabilities, while unsupervised methods such as Principal Component Analysis and K-Means clustering uncovered latent shot structures and tactical typologies. Isolation Forest identified rare or anomalous events, enhancing sensitivity to high-value but infrequent actions, and a Counterfactual Autoencoder introduced a causal layer that simulated "what-if" scenarios, quantifying how controlled changes in shot context, could affect the expected outcome. Experimental results demonstrate that ensemble models improve discriminative power, while unsupervised analyses reveal interpretable clusters that align with tactical behaviors on the pitch. The counterfactual component extends the framework beyond association, enabling causal insight into the determinants of chance creation. Collectively, these findings illustrate a progression from purely predictive modeling toward interpretable, context-aware, and prescriptive analytics. By bridging statistical accuracy with tactical relevance, this work contributes to the emerging paradigm of data-driven causal understanding in sports analytics, transforming raw performance data into actionable football intelligence.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Contents

1 Introduction 7
- **1.1 The Social Impact of a Global Phenomenon** 7
- **1.2 Data-Driven Insights into Scoring Opportunities** 7
- **1.3 The Role of xG in Performance Analysis** 8
- **1.4 Methodological Approaches to Team Offense** 9
- **1.5 Research Gap and Contributions** 9

2 Literature Review 11
- **2.1 Machine Learning for Sports Analytics** 11
- **2.2 Advances in Football Analytics** 12
- **2.3 Modeling Goal Probability with xG Models** 13

3 Exploratory Data Analysis 15
- **3.1 Dataset Overview** 15
- **3.2 Feature Description** 16
- **3.3 Descriptive Analysis** 19
- **3.4 Data Pre-Processing** 29

4 Methodology 33
- **4.1 Setup** 33
- **4.2 Supervised Approach** 33
  - **4.2.1 Logistic Regression** 34
  - **4.2.2 Decision Trees** 35
  - **4.2.3 Random Forests** 38
  - **4.2.4 Gradient Boosting** 40
  - **4.2.5 XGBoost** 42
  - **4.2.6 Neural Network** 43
  - **4.2.7 Evaluation metrics** 45
- **4.3 Unsupervised Approach** 48
  - **4.3.1 Principal Component Analysis and K-Means Clustering** 48
  - **4.3.2 Anomaly Detection with Isolation Forest** 50
- **4.4 Causal Reasoning Approach** 51

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

4.4.1 Counterfactual Autoencoder 51

5 Results and Findings **56**
5.1 Experiments **56**
5.2 Supervised Approach **56**
5.2.1 Logistic Regression 56
5.2.2 Decision Trees 59
5.2.3 Random Forests 62
5.2.4 Gradient Boosting 64
5.2.5 XGBoost 67
5.2.6 Neural Network 69
5.3 Unsupervised Approach **71**
5.3.1 Principal Component Analysis and K-Means Clustering . 71
5.3.2 Anomaly Detection with Isolation Forest 77
5.4 Causal Reasoning Approach **80**
5.4.1 Counterfactual Autoencoder 80
5.5 Comparative Insights into Shooting Dynamics **87**
5.5.1 Performance Assessment of Supervised Models 87
5.5.2 Performance Assessment of Unsupervised Models 92
5.5.3 Causal and Counterfactual Analysis 96

6 Conclusions and Recommendations **100**
6.1 Conclusions **100**
6.2 Practical Implications **101**
6.3 Future Directions **101**

A Additional Figures **103**

B Additional Tables **114**

Bibliography **124**



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

List of Figures

3.1 Outcome review of shot attempts 20
3.2 Temporal distribution of shot attempts 21
3.3 Spatial distribution of shot attempts 22
3.4 Relationship between shot quality and geometric factors 23
3.5 Shot attempts by body part of execution 23
3.6 Positional distribution of shot attempts 24
3.7 Shot attempts by play pattern 25
3.8 Situational context of shot execution 26
3.9 Correlation heatmap of shot attempt features 27
3.10 Expected goals shot map of Luis Suárez 28
3.11 Map of football player positions 31

4.1 Visualization of the 5-Fold cross-validation procedure 34
4.2 Schematic representation of a Decision Tree 36
4.3 Schematic representation of a Random Forest with three trees 38
4.4 Schematic representation of an Autoencoder architecture 52

5.1 ROC curve for Logistic Regression 58
5.2 Decision Tree structure 61
5.3 Precision–Recall curve of the Random Forest model 64
5.4 Feature Importance diagram of the Gradient Boosting model 65
5.5 PDP for the two most important features of the Gradient Boosting model 66
5.6 Feature Importance diagram of the XGBoost model 68
5.7 PDP for the two most important features of the XGBoost model 68
5.8 SHAP summary plot of the Neural Network output 71
5.9 K-Means clustering projected on 3D from multiple perspectives 74
5.10 Comparison of average xG and conversion rates across clusters 76
5.11 Feature importance (ΔAUC) for Isolation Forest 78
5.12 t-SNE visualisation of shot distribution (1% anomaly contamination) . . 79
5.13 Latent space visualization of shot embeddings 81
5.14 Visualization of the xG and goal directional vectors within the latent space 83
5.15 Counterfactual trajectory of Shot #48720 in latent space 85
5.16 Latent-Step Impact on xG values 86

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

5.17 Mean Feature Change (sample of 100 Counterfactuals) 87
5.18 ROC curves comparison among supervised learning models 89
5.19 Calibration curves of the supervised learning models 90
5.20 Precision–Recall curves for the supervised learning models 91
5.21 Accuracy–F1 relationship across supervised learning models 92
5.22 Cluster-wise distribution of anomalous (high-quality) shot instances . . . 95
5.23 Average counterfactual xG improvement per latent cluster 97
5.24 Mean feature Δchange between counterfactual and original shots 98

A.1 Pitch dimensions and coordinate system 103
A.2 Shooting angle for different shot positions on the pitch 104
A.3 Pairplot of numerical features (sample of 500 observations) 105
A.4 Mean ± SD of evaluation metrics (5-fold CV) for Logistic Regression . . 105
A.5 Confusion matrix for Logistic Regression 106
A.6 Gini index and Entropy curves 106
A.7 Cost–complexity pruning path of the Decision Tree 107
A.8 Learning curves of the Neural Network model 107
A.9 Feature importance based on SHAP values for the Neural Network 108
A.10 Scree plot of Cumulative Explained Variance for PCA components 109
A.11 Combined Elbow and Silhouette evaluation for optimal cluster selection . 109
A.12 K-Means clustering projected on 2D 110
A.13 Distribution of shot attempts across clusters 110
A.14 UMAP projection of Isolation Forest anomalies (1% contamination) . . . 111
A.15 Mean Squared Reconstruction Error across features for the CAE 111
A.16 Correlation matrix among latent dimensions 112
A.17 xG and Goal latent direction weights 112
A.18 Distribution of latent projections along the learned xG direction 113



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

List of Tables

3.1 Nico Williams' goal against England in UEFA Euro 2024 19
3.2 Comparison of Suárez and Džeko 29

4.1 Comparison of impurity measures used in Decision Trees. 38
4.2 Confusion matrix showing classification outcomes. 47
4.3 Dataset partitioning scheme 53

5.1 Average evaluation metrics for Logistic Regression 57
5.2 Key logistic regression results 59
5.3 Evaluation results of the Decision Tree model 59
5.4 Evaluation results of the Random Forest model 62
5.5 Evaluation results of the Gradient Boosting model 64
5.6 Evaluation results of the XGBoost model 67
5.7 Evaluation results of the Neural Network model 69
5.8 Most influential features for the first two principal components 72
5.9 Compact interpretation of the first two principal components 72
5.10 Summary of Isolation Forest anomaly detection results 77
5.11 Original shot #48720 84
5.12 Counterfactual reconstruction shot #48720 ($\alpha = 1.5$) 84
5.13 Comparison of Supervised Models performance 88
5.14 Comparative summary of dimensionality reduction approaches 94
5.15 Interpretation of clusters based on centroid profiles 94
5.16 Distribution of anomalous shots PCA–K-Means clusters 95

B.1 Overview of the modeling approaches employed in this study. 114
B.2 Key hyperparameters of the Random Forest algorithm 115
B.3 Key hyperparameters of the Gradient Boosting algorithm 115
B.4 Key hyperparameters of the XGBoost algorithm 116
B.5 Hyperparameter configuration of the Neural Network model 117
B.6 Evaluation metrics employed in the study 118
B.7 Key hyperparameters of PCA and K-Means clustering 118
B.8 Key hyperparameters of the Isolation Forest algorithm 119
B.9 Key hyperparameters of β–VAE 119

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

B.10 VIF values for continuous predictors 119
B.11 Full logistic regression results 120
B.12 Eigenvalues and explained variance ratio for each principal component . . 121
B.13 Full cluster centroids across all shooting features (standardised values). . 122
B.14 Isolation Forest anomaly detection full-table results 123



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Chapter 1

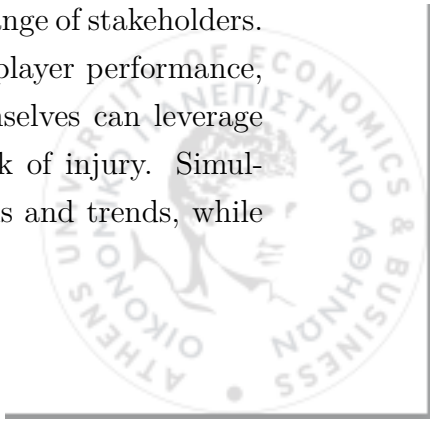
Introduction

1.1 The Social Impact of a Global Phenomenon

Football is indisputably the most popular sport worldwide, attracting fans who form an integral part of daily life in every corner of the globe. Its appeal extends far beyond the boundaries of athletics, connecting societies, cultures, and generations through a shared language that transcends borders. Major tournaments, such as the FIFA World Cup, are estimated to be followed by billions of viewers, highlighting the unique significance of the sport. Moreover, football functions as a social bond, fostering a sense of community, identity, and cultural heritage, whether through local clubs that enhance cohesion within communities or national competitions that inspire pride and unity at a national level. Consequently, football is not limited to the dimension of spectacle but has evolved into a phenomenon of substantial social and cultural importance in the contemporary era, exerting a profound impact on both local and international communities. In this context, performance metrics and statistical indicators now play a decisive role in understanding and evaluating performance at both the individual and team levels. The rapid advancement of technology and the increasing availability of data have fundamentally transformed the way the game is analyzed, ushering in a new era in which scientific approaches and data analytics are increasingly integrated with the practice of sport.

1.2 Data-Driven Insights into Scoring Opportunities

In recent years, data analysis has become an indispensable tool for understanding and interpreting football, offering new insights and opportunities to a wide range of stakeholders. Coaches and technical staff utilize performance metrics to evaluate player performance, design tactical strategies, and prepare for opponents. Players themselves can leverage individualized data to optimize their training and minimize the risk of injury. Simultaneously, analysts and researchers examine data to identify patterns and trends, while



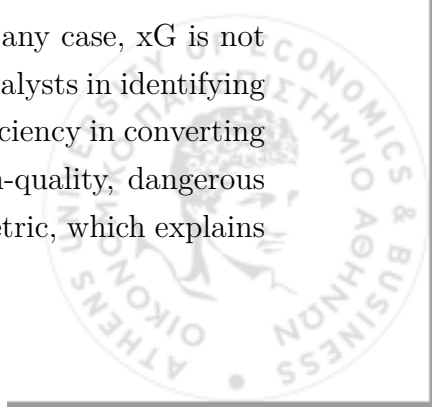
Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

fans gain a more comprehensive viewing experience through statistical tools and applications. The growing significance of these practices is also reflected in club investments, with specialized data analysis departments being established and cutting-edge technologies adopted, recognizing that scientific knowledge can provide a competitive advantage. This type of analysis is not limited to players' physical condition or tactical decisions but extends to more fundamental aspects of the game, such as the creation and exploitation of scoring opportunities. Within this context, as scoring remains the ultimate objective in football, research increasingly focuses on how goal probability can be evaluated and quantified, leading to more accurate assessments of team performance.

1.3 The Role of xG in Performance Analysis

Within this context, the Expected Goals (xG) metric has emerged as one of the most important tools for evaluating team performance. This indicator goes beyond simply recording whether a scoring opportunity resulted in a goal; it estimates the probability of success for each attempt based on factors such as shot location, shooting angle, defensive pressure, and other contextual elements of the play. The indicator is defined on a scale between 0 and 1, where a value closer to 0 denotes a very low likelihood of scoring, while a value closer to 1 reflects a significantly higher probability of a successful outcome. In this way, xG provides a more objective assessment of the quality of opportunities created or conceded by a team, irrespective of the final score. The significance of xG lies in its ability to distinguish whether a strong or weak performance is due to circumstantial factors, such as luck, or to substantive weaknesses and strengths. A higher xG value for a team or player indicates that the chances produced are more dangerous, as xG effectively measures the probability that a given attempt will result in a goal. Conceptually, it is as if, at the moment of the shot, time "freezes" and all relevant factors are simultaneously considered to yield a numerical probability that reflects the true value of the opportunity.

The interpretation of xG becomes particularly valuable when compared with a team's actual goals scored. If a team exhibits a high xG but low goal output, this typically signals issues with finishing or the players' ability to convert chances into goals. Conversely, a low xG combined with few goals indicates difficulty in creating high-quality opportunities, pointing to a need for improvements in offensive development and tactical execution. Finally, when the number of goals significantly exceeds the xG prediction, this may be attributed either to exceptional finishing performance or to circumstantial factors, such as luck, which tend to even out over the long term. In any case, xG is not merely an abstract statistical figure but a practical tool that guides analysts in identifying areas for improvement, whether related to opportunity creation or efficiency in converting chances. Over the long term, teams that consistently generate high-quality, dangerous opportunities are expected to achieve the goals "promised" by the metric, which explains



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

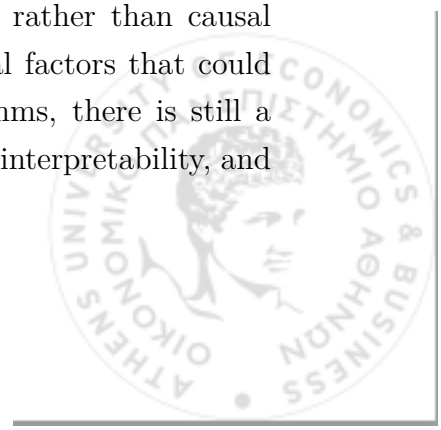
why xG has become a cornerstone of football analytics and is widely employed both for research purposes and for strategic decision-making within clubs

1.4 Methodological Approaches to Team Offense

Relevant contextual and situational factors are fed into mathematical or computational models, which are trained on historical match data to estimate the probability of success for each attempt. The methodological approach may vary from simple statistical models to more sophisticated machine learning techniques, yet the objective remains the same: to produce a metric that accurately reflects the quality of scoring opportunities. In the international literature, a variety of models have been proposed and employed to calculate Expected Goals, differing in terms of complexity and the type of data utilized. Classical examples include logistic regression models, which allow the estimation of the probability of success for a given attempt based on specific independent variables. With the advancement of technology, more complex machine learning techniques, such as random forests and gradient boosting machines, have been applied, enabling the modeling of nonlinear relationships and interactions between variables. In recent years, Neural Networks have also gained prominence, leveraging large volumes of data and providing the potential for more precise predictions, particularly when combined with player tracking data. The choice of the most appropriate model depends on both the availability and quality of the data, as well as the research objective, rendering this field highly flexible and rich in research opportunities.

1.5 Research Gap and Contributions

Recent literature in football analytics has demonstrated significant progress, particularly through the application of machine learning techniques for outcome prediction and xG estimation. In parallel, a range of studies has adopted more descriptive or statistical approaches to identify tactical patterns, assess player performance, and capture the structural dynamics of the game. Nevertheless, the majority of this research remains methodologically fragmented, typically focusing either on supervised predictive modeling or on unsupervised exploratory analysis. Only a few attempts have sought to integrate these perspectives or extend them toward causal interpretation. Moreover, existing xG models tend to rely on associative relationships between variables rather than causal ones, often overlooking the influence of contextual or counterfactual factors that could alter an outcome. Despite the increasing sophistication of algorithms, there is still a lack of comprehensive frameworks that combine predictive accuracy, interpretability, and tactical understanding in a unified analytical structure.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

This study aims to bridge these gaps by proposing an integrated analytical framework that combines supervised, unsupervised, and causal inference methodologies. Supervised learning models are employed to estimate goal-scoring probabilities, while unsupervised methods such as Principal Component Analysis (PCA) and K-Means clustering are used to uncover latent shot patterns and structural similarities. Anomaly detection via Isolation Forest enhances the capacity to identify rare or high-value events that conventional models typically underrepresent. Finally, a counterfactual autoencoder introduces a causal dimension by simulating "what-if" scenarios, examining how interpretable changes could affect the expected outcome. Through this synthesis, the study advances football analytics beyond purely statistical prediction toward a more interpretable, context-aware, and tactically meaningful understanding of chance creation and decision-making.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Chapter 2

Literature Review

A growing body of literature has emphasized the increasing role of data analytics in football, highlighting a notable rise in studies that integrate machine learning techniques. While football, which is widely regarded as the most popular sport globally, has received the majority of scholarly attention, there is also an emerging interest in extending such approaches to other sports. This expansion reflects a broader trend within the research community towards exploring innovative applications of analytical and computational methods across the sporting domain.

2.1 Machine Learning for Sports Analytics

With respect to research on team sports, Bunker, Susnjak (2022) employed machine learning techniques to predict match outcomes, proposing a classification of sports into two subcategories, invasion and striking. Their findings suggest that the former category, encompassing sports where two teams compete to achieve the highest score, is inherently more difficult to predict due to the greater uncertainty of outcomes. By contrast, striking sports, which are structured around innings, tend to yield more predictable results, that appears to be closely related to the underlying nature of the games themselves. This line of inquiry is further developed by Mgaya, Liu, Zhang (2021), who investigated the use of deep learning techniques, with a particular focus on football and basketball. These sports are widely acknowledged as prototypical cases of multi-agent interactions, which reinforces their suitability for the application of advanced computational methodologies.

In addition, Vec, Tomazic, Kos, Umek (2024) conducted a systematic review of the field, reporting that 64% of studies combining sports and artificial intelligence were published after 2020. This evidence underscores the rapid growth of the domain and confirms its increasing scholarly relevance. The review also highlights that Deep Learning methods are generally regarded as more effective than classical machine learning or traditional statistical approaches. Nevertheless, their successful implementation requires adequate data

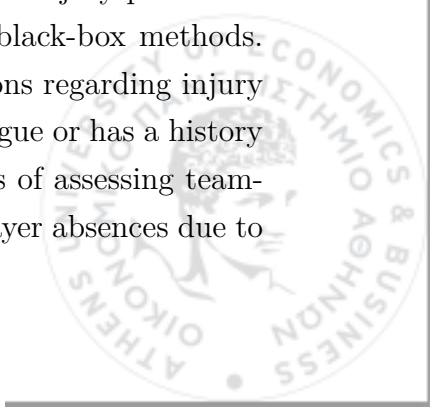
Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

pre-processing, particularly in terms of feature extraction, in order to mitigate noise. In this context, the careful selection of relevant features appears to be a more decisive factor than the size of the available dataset. Taken together, although the majority of current research remains focused on football, there is a growing recognition of the need to develop in-game applications with specialized models tailored to the specific characteristics of individual sports.

2.2 Advances in Football Analytics

Several studies have adopted straightforward approaches, without engaging in highly complex methodologies. A notable example is the work of Peña, Touchette (2012), who applied network theory to analyze team strategy. In this context, they identified game patterns, high-activity areas, and team tactical weaknesses, while also evaluating individual player performance. The methodology relied on representing passes as edges between nodes, corresponding to players, thus capturing the criticality of each player within the style of play. These findings suggest that player performance can be effectively assessed with clear practical applications. Moreover, Van Haaren (2021) estimated the probability of a shot resulting in a goal using generalized additive modelling (GAM). This approach highlights the potential for making predictive analyses more accessible to practitioners who are not specialists in analytics or machine learning. The models are constructed without relying on features; instead, the playing field is partitioned into 16 zones, represented by coordinates transformed into fuzzy assignment vectors. Additionally, the model considers the body part used to execute the shot toward the target, thereby avoiding highly complex mathematical computations. Overall, it produces models that are both interpretable and methodologically manageable, providing actionable insights while maintaining relative simplicity.

Conversely, several studies have concentrated on the application of machine learning techniques. For instance, Markopoulou, Papageorgiou, Tjortjis (2024) employed a range of models, including linear regression, ridge regression, random forest, xgboost, gradient boosting, and multilayer perceptron algorithms (MLP), so as to estimate the probability of a goal. These findings underscore the potential of these methods in football data analytics and indicate that the analytical scope in this domain can be further expanded to broader critical aspects of the sport, with opportunities to incorporate advanced representations. Rossi, Pappalardo, Cintia (2022) proposed a guideline for injury prediction using machine learning techniques, employing both white-box and black-box methods. It is intended to diminish errors that could lead to misleading decisions regarding injury prevention, particularly in cases where a player exhibits excessive fatigue or has a history of medical conditions. Additionally, the framework provides a means of assessing team-related costs by reducing potential financial losses associated with player absences due to



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

injury. In a broader examination of team-related costs, Rodriguez, Hassan, Coad (2019) identified the key factors determining the market value of a football player. Their analysis considered not only the spectrum of on-field actions using feature clustering, but also external player characteristics, such as age and nationality. The study initially employed bayesian model averaging (BMA) and subsequently adopted instrumental variable BMA (IVBMA) to address issues of endogeneity, bridging the methodological gap between sports analytics and economic theory. Finally, an innovative line of research has shifted towards the prediction of subsequent events within the dynamic flow of a football match. Mendes-Neves, Meireles, Mendes-Moreira (2024) introduced a novel methodology that utilizes large event models, inspired by the architecture of large language models (LLMs), to forecast which action is likely to follow a preceding one, representing one of the first attempts to combine the advancements of LLM-inspired architectures with the unique analytical challenges of football.

2.3 Modeling Goal Probability with xG Models

Since goalscoring represents the most decisive aspect of football, considerable interest has been directed toward evaluating the likelihood and contextual risk associated with a scoring attempt. The first systematic effort in this area was conducted by Pollard, Ensum, Taylor (2004), who manually recorded 1,096 kicked shots during the FIFA World Cups from 1986 to 2002 and applied logistic regression to model goal probability. The dependent variable was a binary indicator, taking the value 1 if the attempt resulted in a goal and 0 otherwise. Independent variables included the distance from the goal, the angle of execution, and a binary variable, space, equal to 1 if the shooter was positioned more than one meter away from the nearest defender at the moment of execution, and 0 otherwise, indicating that he was under immediate defensive pressure. The results demonstrated that for every additional yard of shooting distance, the probability of scoring decreased by 15%. Similarly, for each one-degree deviation from the central axis of the goal, the scoring probability decreased by 2%. Conversely, when a player executed the shot without defensive pressure, the probability of scoring increased by 2.22%. All three factors were found to be statistically significant, underscoring their critical role in determining goal probability.

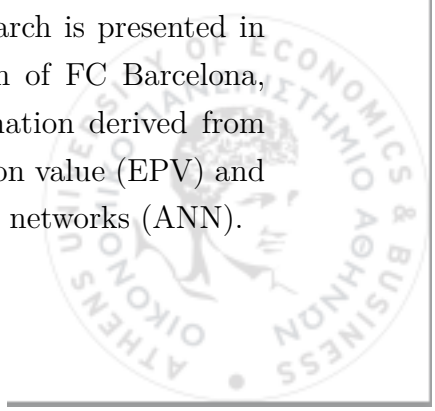
Moving to the contemporary era, considering that the statistical services of football competitions record loads of metrics for each match, efforts have been made to optimize the accuracy of existing indicators, with the aim of improving the XGoal model. Using machine learning techniques, Mead, O'Hare, McMenemy (2023) initially incorporated additional features as independent variables, reaching a total of 20. Beyond the core variables, examples of the newly introduced ones include the shooter's team Elo Rating, his market value, his cumulative PlayeRank at the time of the match, the current league

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

ranking of the team, and even stadium segmentation during the game. The methodology employed neural networks, specifically MLP, as well as adaBoost, xgboost, and random forest. Furthermore, techniques such as logarithmic loss, hyperparameter tuning, and cross-validation were applied, ultimately providing a more practical tool for teams and analysts to enhance tactical analysis and, consequently, inform their economic decisions. XGoal metric is shown to be preferred over traditional statistics for predicting the future performance of a team and its likelihood of success.

Cavus, Biecek (2022) not only combined machine learning but also explainability techniques to enhance the interpretability and practical utility of the xGoal model. As such models are often considered black-box, Explainable AI (XAI) was employed to provide transparency, interpretability of predictions is improved through shapley additive explanations (SHAP), partial dependence plots (PDP) and individual conditional expectation (ICE), illustrating how predictions change when a specific feature is modified. For example, when the distance from the goal increases or the body part used for execution changes. Regarding modeling techniques, random forest and xgboost outperform logistic regression, adaboost, boosting machine, catboost, and decision tree. With random forest, prediction accuracy is increased and logarithmic loss is minimized, while the risk of overfitting is reduced. With xgboost, the model is better able to handle high-dimensional data, calibration is improved as measured by the brier score, and the model becomes faster, more flexible and interpretable. Although measurements indicate that decreasing the execution angle increases the xGoal, the PDP show the opposite trend, due to the bias of successful slightly angled executions and to interactions with other factors such as the ball speed prior to execution. Nevertheless, the general principle remains that wider execution angles reduce the phase danger, as reflected by the observation that for execution angles greater than 45 degrees, xGoal decreases, whereas for angles below 45 degrees, the execution approximates a shot from the small box height, where the trajectory toward the goal is nearly straight.

Davis, Robberechts (2024) sought to address this form of bias, emphasizing that it distorts the estimation of shot danger, leading either to underestimation or overestimation of finishing outcomes. To diminish this issue, they proposed an AI fairness approach for calibrated xG models across player subgroups, highlighting the primacy of quality over quantity. Lucey, Bialkowski, Monfort, Carr, Matthews (2015) focused on the quality of opportunities by incorporating strategic features and spatiotemporal data to achieve more accurate predictions. This was accomplished through the analysis of play patterns, hot-spots, and weaknesses in team efficiency. A similar line of research is presented in the dissertation of Pardo (2020), conducted under the supervision of FC Barcelona, which developed a new model integrating qualitative player information derived from tracking data. The implementation was applied to expected possession value (EPV) and performance metrics, utilizing logistic regression and artificial neural networks (ANN).



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Chapter 3

Exploratory Data Analysis

3.1 Dataset Overview

The dataset employed in the present study originates from the StatsBomb platform, which is widely regarded as one of the most reliable and recognized providers of advanced football statistics. StatsBomb has established itself as a leading source in the field of sports analytics, offering highly detailed and systematically collected event data from professional football matches. These data are publicly available and provided with the explicit purpose of facilitating research, innovation, and the advancement of analytical methods within the academic and professional community. The value of the dataset lies in both its breadth and its granularity, as it provides event-level annotations that capture not only the outcome of specific on-field actions but also a wide array of contextual attributes, including spatial and temporal coordinates, player and team identifiers, and play patterns. It comprises a total of 50,677 observations and 17 selected variables for each shot, with each observation corresponding to a single scoring attempt from open play. That denotes actions within the continuous flow of the match, excluding all restart or set-piece situations that led directly to goals, such as penalties, direct or indirect free kicks and direct corners, whose tactical context and scoring probabilities differ fundamentally from open-play events. For each shot, both spatial information is recorded and categorical attributes are provided. Hence, this collection contains every recorded shot and spans a wide range of high-level competitions, including:

1. Domestic leagues of the Top 5 European leagues for the 2015/16 season: Premier League, La Liga, Serie A, Ligue 1, and Bundesliga.

2. UEFA Champions League finals from the 2003/04 to the 2018/19 seasons.

3. International tournaments: FIFA World Cup 2018 and 2022, UEFA Euro 2020 and 2024, Copa América 2023, and the African Cup of Nations 2023.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

This dataset is considered particularly suitable for the development and evaluation of Expected Goals (xG) models and machine learning algorithms, as it combines a substantial volume of observations that ensure statistical reliability with a wide spectrum of explanatory variables. These include spatial, temporal, and categorical dimensions that are essential for quantifying the probability of scoring. Furthermore, the dataset covers a diverse range of competitions and match contexts, allowing for broader generalization and meaningful comparisons across different football environments. Finally, the high quality and granularity of the information provided, offer the necessary detail to train robust and reliable predictive models.

3.2 Feature Description

The final dataset employed in this study consists of 25 variables describing each shot attempt. These variables encompass spatial, temporal, and categorical dimensions, offering a comprehensive representation of the contextual and technical characteristics associated with every shooting event.

Match ID – Numerical (Integer). A unique code identifying each match. It links every shot to its corresponding game and enables grouping and analysis at the match level.

Player Name – Categorical (String). Records the full name of the player attempting the shot.

Team Name – Categorical (String). Records the name of the team of the player attempting the shot. It enables team-level aggregation and facilitates comparative analyses across different clubs.

Opponent Name – Categorical (String). Records the name of the opposing team against, which the shot was attempted.

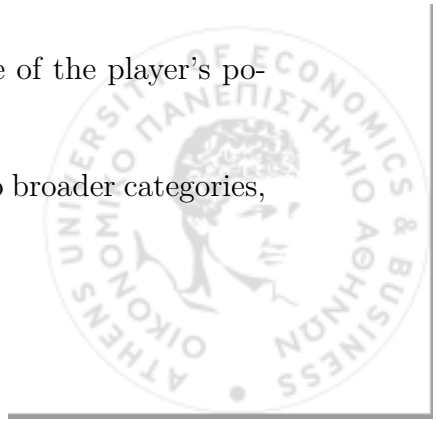
Competition Name – Categorical (String). Specifies the competition in which the match took place.

Season Name – Categorical (String). Indicates the football season to which the match belongs.

Position ID – Numerical (Integer). A numeric code representing the player's position on the pitch, as defined by the StatsBomb data specification.

Position Name – Categorical (String). Provides the detailed name of the player's position in the starting lineup

Position Group – Categorical (String). Groups player positions into broader categories, such as Attack, Midfield, or Defence.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Position Code – Numerical (Integer). A simplified numerical code provided to flag and classify player positions into broader categories; primarily a supporting variable.

Minute – Numerical (Integer). Records the match minute in which the shot attempt occurred.

X – Numerical (Float). Denotes the X coordinate of the pitch location from which the shot was taken, according to StatsBomb.

Y – Numerical (Float). Denotes the Y coordinate of the pitch location from which the shot was taken, according to StatsBomb.

Distance – Numerical (Float). Represents the calculated distance from the shot location to the center of the goal, measured in pitch units.

Angle – Numerical (Float). Represents the calculated shooting angle relative to the goalmouth, expressed in radians.

Outcome Name – Categorical (String). Specifies the outcome of the scoring opportunity.

Is Goal – Binary (Integer). Indicates whether the shot resulted in a goal or not.

Body Part Name – Categorical (String). Records the part of the body used to take the shot.

Body Part Code – Numerical (Integer). A numeric code corresponding to the body part recorded by StatsBomb; primarily a supporting variable.

Play Pattern Name – Categorical (String). Describes how the attacking phase leading to the shot originated.

Play Pattern Code – Numerical (Integer). A numeric code corresponding to the play pattern category; provided as an auxiliary variable.

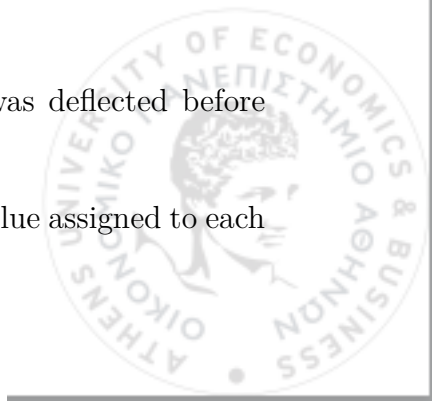
Under Pressure – Binary (Integer). Flags whether the shooter was under defensive pressure at the time of the attempt or not.

Shot First Time – Binary (Integer). Indicates whether the shot was taken directly without controlling the ball or not.

Shot One-on-One – Binary (Integer). Identifies whether the shot situation was a one-on-one with the goalkeeper or not.

Shot Deflected – Binary (Integer). Indicates whether the shot was deflected before reaching the goal or not.

StatsBomb xG – Numerical (Float). Provides the expected-goals value assigned to each



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

shot, expressed as a probability between 0 and 1 of it resulting in a goal. This metric is generated by StatsBomb using statistical and machine learning models trained on extensive historical data of shot outcomes. The calculation incorporates multiple factors, in order to approximate the likelihood of scoring in similar situations.

The dataset employed in this study includes all core variables commonly used in xG modeling. ensuring methodological comparability with prior research. Variables related to match importance, current team ranking, or home/away status were intentionally omitted due to the heterogeneity of the dataset, which spans multiple competitions of different types, including international tournaments, domestic leagues, and final-stage matches, where such contextual factors are either inconsistently recorded or not directly comparable.

A notable advantage of this dataset lies in its enriched contextual structure. It incorporates Position Group and Code, which allow for player role aggregation and position, based performance analysis, features rarely available in earlier studies. Similarly, the inclusion of Play Pattern Name and Code provides insight into the tactical origin of each shot, aligning with the advanced event representations proposed in more recent works. Moreover, the binary indicators introduce situational depth, capturing critical aspects of shot context that are often absent from standard xG datasets. Collectively, these features extend beyond traditional statistical modeling, offering a richer and more interpretable foundation for both predictive and causal analyses of shot quality in football.

To facilitate visualization and enhance interpretability, Table 3.1 presents a subset of the variables corresponding to the equalizing goal of Nico Williams in the UEFA Euro 2024 final. This goal corresponded to Spain's opening strike in the final, a match that concluded with a 2–1 victory and ultimately secured the team's triumph in the tournament. For clarity, several auxiliary variables that were included in the dataset and previously described have been omitted from this example, as they primarily serve illustrative rather than analytical purposes.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

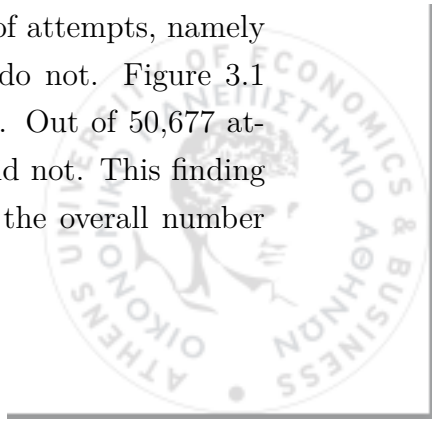
Feature	Type	Value
Player Name	Categorical	Nicholas Williams
Team Name	Categorical	Spain
Opponent Name	Categorical	England
Competition Name	Categorical	UEFA Euro
Season Name	Categorical	2024
Position Name	Categorical	Left Wing
Position Group	Categorical	Attack
Minute	Numerical	46
Distance	Numerical	15.68
Angle	Numerical	0.365
Outcome Name	Categorical	Goal
Body Part Name	Categorical	Left Foot
Play Pattern Name	Categorical	From Throw In
Under Pressure	Binary	0
Shot First Time	Binary	1
Shot One-on-One	Binary	0
Shot Deflected	Binary	0
StatsBomb xG	Numerical	0.112

Table 3.1: Nico Williams' goal against England in UEFA Euro 2024

3.3 Descriptive Analysis

In this section, a descriptive analysis of the data is conducted with the aim of understanding the main characteristics of the sample and the factors that influence the likelihood of a shot attempt being successful. Through graphical and statistical representations, a comprehensive overview of the distribution of attempts is provided, both in terms of temporal and spatial patterns, as well as the conditions of execution and the specific attributes of each shot.

The analysis begins with an examination of the overall outcome of attempts, namely the proportion of shots resulting in goals compared to those that do not. Figure 3.1 illustrates the distribution of outcomes across all recorded attempts. Out of 50,677 attempts, 4,904 (9.7%) resulted in goals, while the remaining 90.3% did not. This finding confirms that scoring a goal is a relatively rare event compared to the overall number



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

of attempts, which is consistent with evidence reported in the international literature. Presenting this ratio constitutes the first step towards understanding the importance of the factors examined in the following sections, as they help explain why some attempts have a higher probability of success than others.

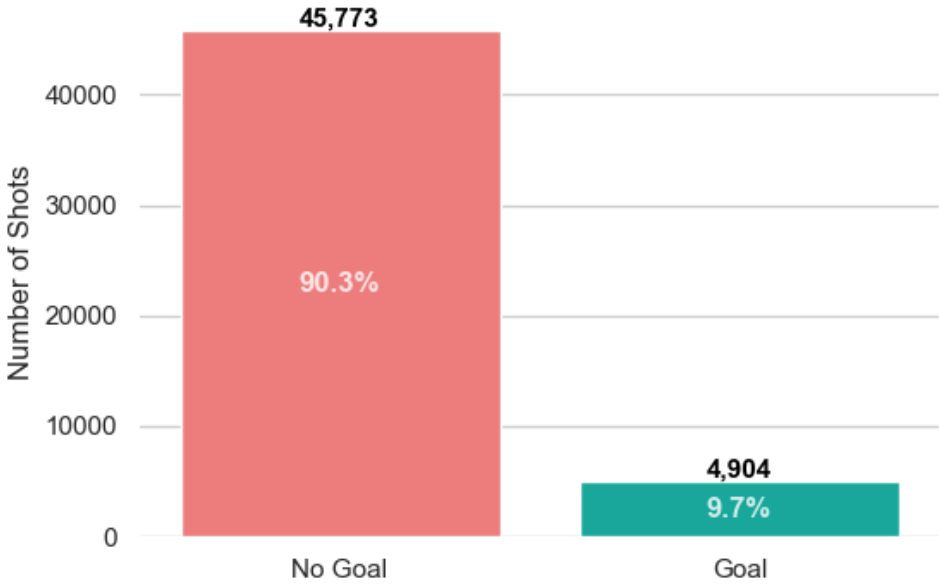


Figure 3.1: Outcome review of shot attempts

The temporal distribution of shot attempts is illustrated in Figure 3.2. It can be observed that shots are relatively evenly spread throughout the course of the match, with a noticeable increase during stoppage time of each half, particularly just before the 45th and 90th minute. This pattern is expected, as teams tend to intensify their offensive efforts in the final moments of each period in order to take advantage of the remaining time. Moreover, it becomes evident that teams are more offensively active in the second half compared to the first. This phenomenon may be attributed to the fact that, following the first half, teams possess a more comprehensive understanding of the dynamics of the game and are consequently more inclined to assume greater tactical risk. In contrast, the early stages of the match are often characterized by a more conservative approach, as teams seek to assess and adapt to the unfolding conditions of play. In addition, the increased attacking intensity may also be linked to the introduction of fresh players through substitutions, who bring renewed energy and contribute to offensive play. This observation highlights the importance of temporal factors in shaping the rhythm of chance creation. Regarding extra time, a smaller yet significant number of attempts can be observed, reflecting both the reduced duration and the physical fatigue of the players. Finally, the peaks observed at the 45th and 90th minute should be interpreted with caution, as they are largely an artifact of stoppage time being recorded within those minute bins.

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

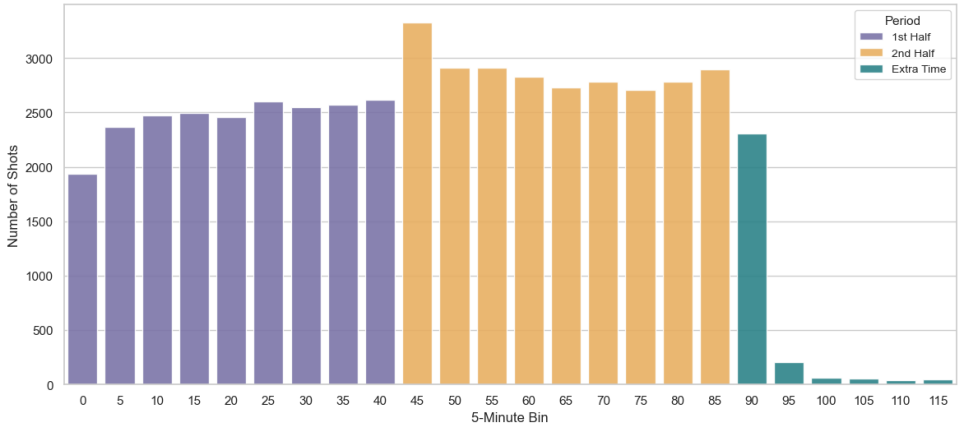


Figure 3.2: Temporal distribution of shot attempts

As expected, the vast majority of shots originate from within or around the penalty area, with a pronounced concentration in the central region directly in front of the goal, as illustrated by the spatial distribution of shot attempts in Figure 3.3. The highest intensity of activity is observed around and slightly ahead of the penalty spot, corresponding to the so-called "golden zone," where xG values typically peak due to optimal distance and shooting angle. A secondary concentration appears just outside the penalty arc, indicating medium-range attempts or second-ball situations that arise when teams fail to penetrate the defensive block. As distance from goal increases, shooting locations become more central, reflecting players' natural preference to maintain alignment with the goal frame in order to maximize scoring probability. From a tactical perspective, this spatial pattern also suggests that when teams are unable to progress through structured build-up play, they often switch to more direct strategies, exploiting crosses and aerial deliveries into the contact zone in front of goal. Such direct deliveries frequently reproduce dangerous phases of play: for instance, deflections after blocked shots, rebounds following goalkeeper interventions, or scrambles after contested aerial balls, which still take place within the most valuable shooting areas identified by xG analysis. Overall, the density map demonstrates a clear spatial bias of attacking play toward central, high-probability areas, highlighting the strategic importance of the penalty zone as the dominant source of goal-scoring opportunities.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

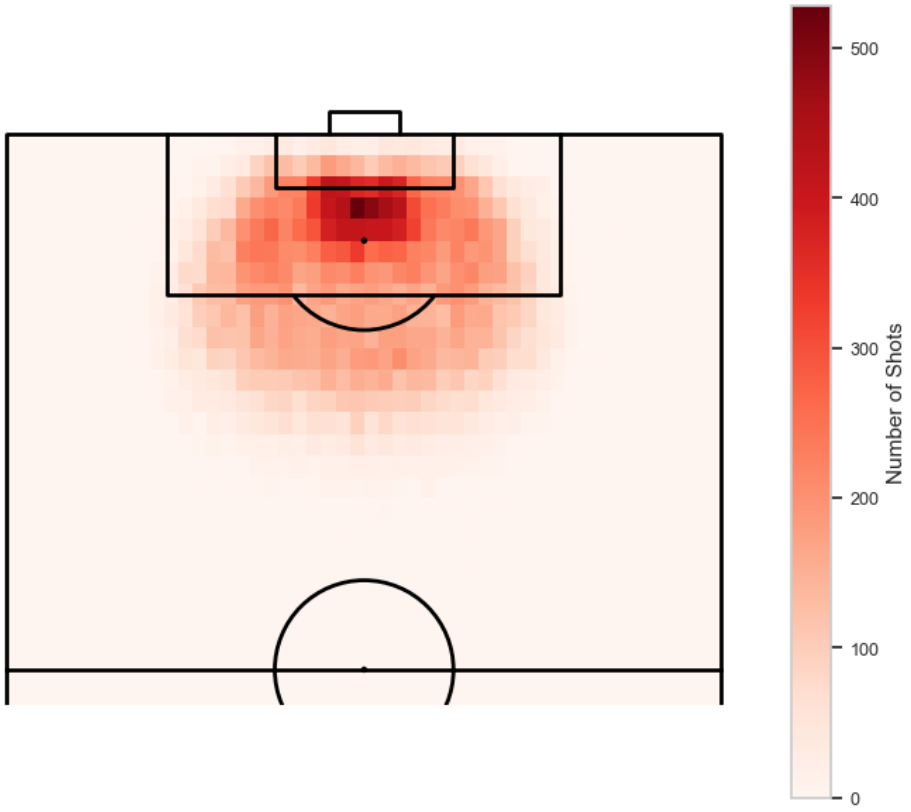


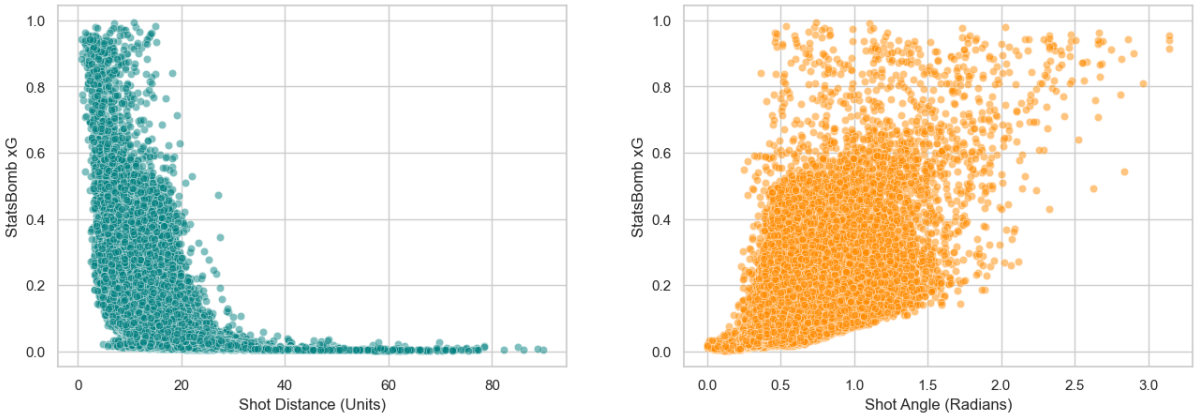
Figure 3.3: Spatial distribution of shot attempts

Building on this, The relationship between shot quality and geometric characteristics is illustrated in the following figures. Figure 3.4a presents the association between shooting distance and the probability of success, as expressed through the xG metric, thereby reflecting the danger posed by a given attempt. A clear decreasing trend is observed: as the distance from the goal increases, the likelihood of converting the attempt into a goal declines. This distribution confirms that long-range efforts rarely yield high scoring probabilities.

Similarly, Figure 3.4b demonstrates the link between shooting angle and xG. Attempts originating from central positions, offering a wider field of view towards the goal, are associated with higher probabilities of success. In contrast, shots taken from narrow angles exhibit lower xG values, highlighting their reduced effectiveness due to limited target visibility and increased execution difficulty. This analysis underscores the fundamental importance of spatial positioning in determining shot success, as both distance and angle, play a decisive role in shaping the probability of goal conversion.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/



(a) Shot quality as a function of distance (b) Shot quality as a function of angle

Figure 3.4: Relationship between shot quality and geometric factors

The distribution of shot attempts by body part is presented in Figure 3.5. The majority of attempts are executed with the right foot, accounting for more than 50% of the total, a finding that reflects not only the numerical predominance of right-footed players in professional football but also the tactical adaptation of teams to their dominant foot. Left-footed attempts represent 31.0% of the total, indicating that although left-footed players are less common, they nevertheless provide an important complementary dimension to offensive play. Headers account for 17.7% of all attempts, underscoring the relevance of aerial duels and crosses in chance creation, particularly in set-piece situations. Finally, shots executed with other parts of the body are extremely rare, typically arising from circumstantial events rather than deliberate or structured choices of execution.

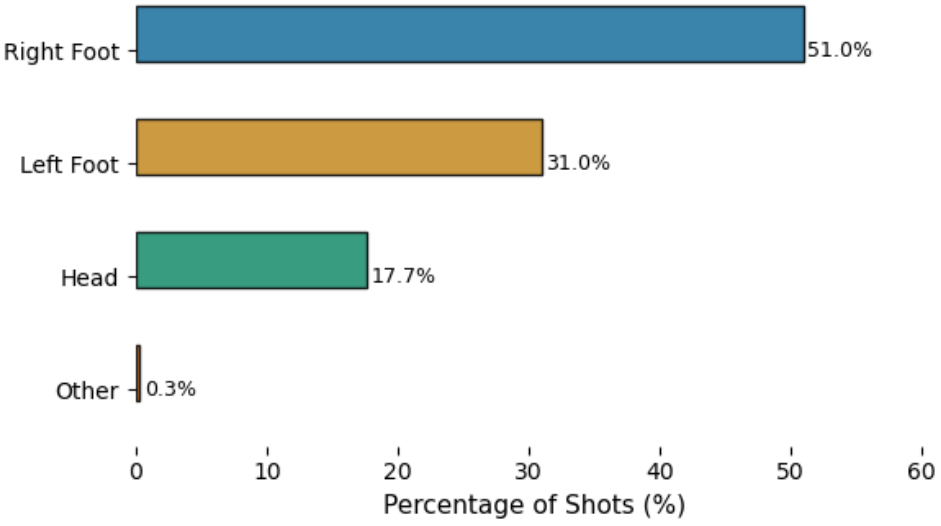


Figure 3.5: Shot attempts by body part of execution

It is also important to examine how shot attempts are distributed across player roles. The distribution of shot attempts by player position is shown in Figure 3.6. As expected,

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

the majority of attempts are generated by center forwards, who represent the primary offensive outlet for most teams. A substantial contribution is also observed from wingers, who exploit wide areas to create opportunities, as well as from central midfielders, who frequently participate in both the buildup and the execution of attacking plays.

In contrast, defenders account for a smaller share of attempts, with certain positional exceptions. Specifically, left and right backs, as well as right and left center backs in a back-four system, record more attempts compared to pure center backs. This can be attributed to the fact that most teams operate with four defenders rather than three, thus increasing the relative frequency of actions from those positions. Similarly, the limited contribution of the secondary striker position is explained by the decreasing prevalence of formations employing two traditional forwards, resulting in fewer observations for this role in the dataset. Overall, it reflects the tactical organization of modern teams, highlighting that positions with direct access to the opponent's goal constitute the main source of shot attempts, while other roles contribute conditionally depending on formation and match strategy.

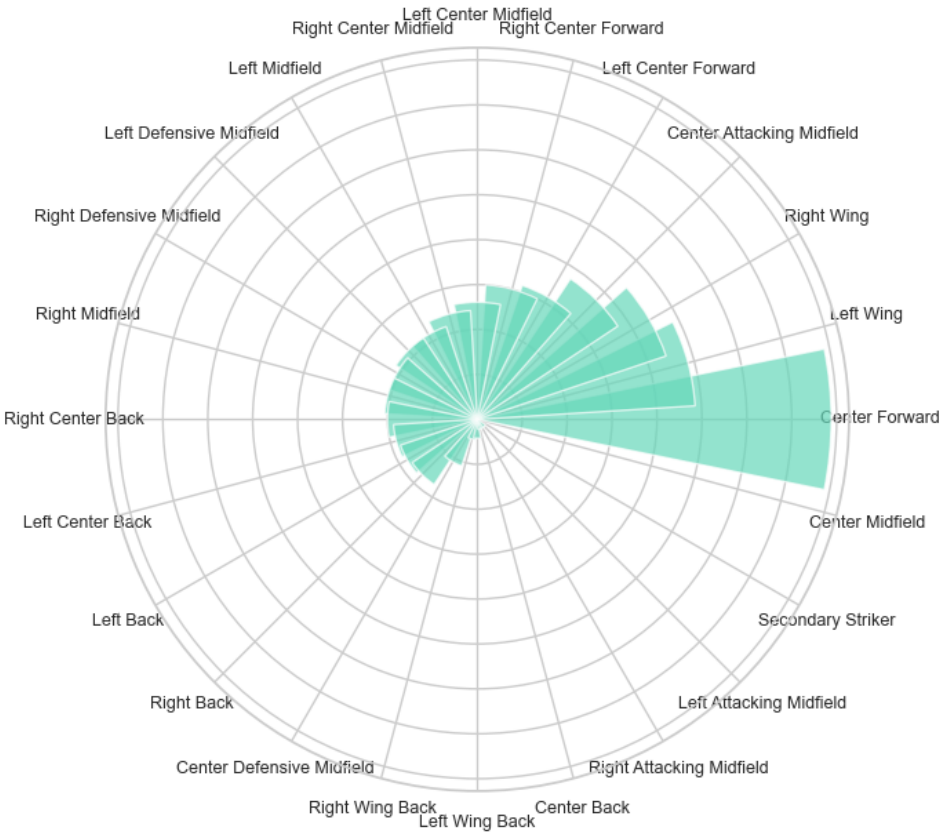
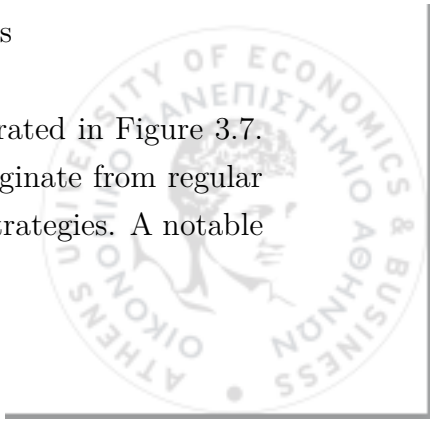


Figure 3.6: Positional distribution of shot attempts

The contextual framework in which shot attempts occur is illustrated in Figure 3.7. The analysis reveals that the overwhelming majority of attempts originate from regular open play, which is expected given its central role in team offensive strategies. A notable



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

proportion also stems from set pieces, which create dangerous situations and allow for structured attacking opportunities.

Alternatively, other contexts such as shots following counter attacks, goal kicks, or kick-offs are far less frequent, reflecting the rarity of these situations within the dynamics of match play. The particularly low share of counter-attack attempts indicates that the conditions required for such actions are relatively uncommon in the dataset, as they depend on rapid transitions and unstructured defensive setups. Nevertheless, when these situations do arise, they may produce high-quality opportunities due to the temporary defensive disorganization of the opponent.

This distribution also reflects broader tactical tendencies in modern football. The predominance of open play highlights the reliance of most teams on sustained possession and collective buildup patterns as the principal means of generating chances. At the same time, set pieces, while less frequent, carry significant tactical weight because they represent controlled environments that can be systematically exploited, often proving decisive in tightly contested matches. From a training perspective, this explains why teams invest considerable preparation time in rehearsed routines for both open play movements and set-piece executions, whereas contexts such as counters or kick-offs remain opportunistic rather than systematic sources of attempts.

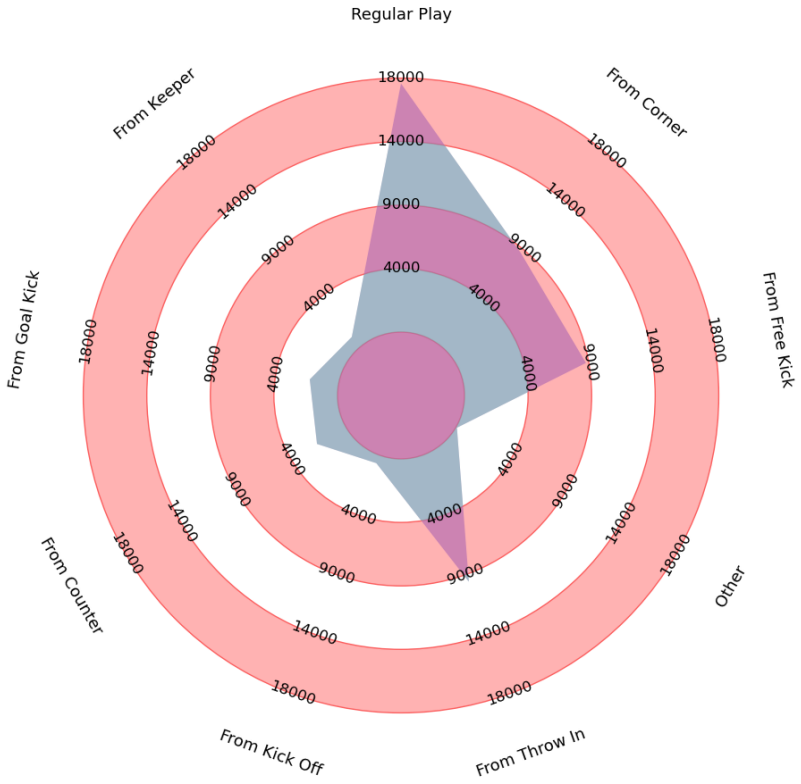


Figure 3.7: Shot attempts by play pattern



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

As illustrated in Figure 3.8, 32.9% of the attempts were executed first time, without prior control of the ball, while 28.0% were taken under defensive pressure. One-on-one situations with the goalkeeper account for 5.1%, whereas deflected efforts represent only 1.4%. These figures highlight the varying situational contexts under which shots are attempted, reflecting differences in execution conditions and potential difficulty.

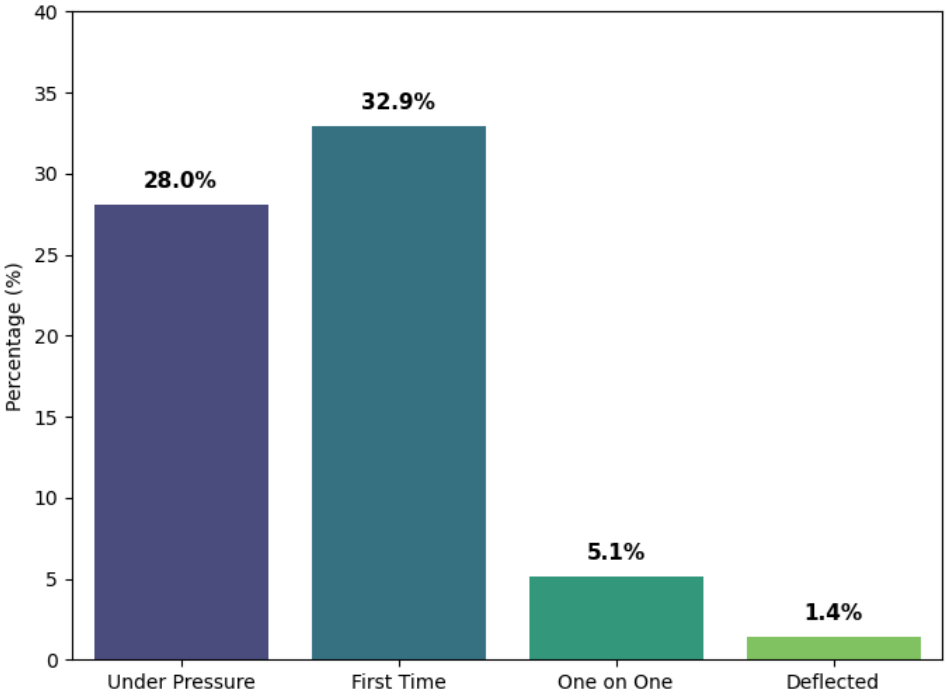


Figure 3.8: Situational context of shot execution

Therefore, Figure 3.9 presents the correlation heatmap of the variables in the sample. As expected, the geometric parameters exhibit strong interdependencies: distance is negatively correlated with angle ($r = -0.76$), as more lateral shooting positions tend to be associated with greater distance from the goal. Equally important is the positive correlation between angle and xG ($r = 0.68$), which confirms the decisive contribution of shooting angle to the likelihood of scoring. At the same time, distance shows a strong negative correlation with xG ($r = -0.55$), indicating that long-range efforts correspond to lower-quality chances. The remaining variables display low or moderate correlations, suggesting that the information they provide does not substantially overlap, which justifies their inclusion in the analytical model. Overall, this visualization highlights the key relationships among the features that describe shot attempts, with spatial parameters in particular emerging as fundamental determinants of scoring probability.

Accordingly, Figure A.3 illustrates the pairplot of the numerical variables, based on a random sample of 500 observations from the dataset. This visualization enables the simultaneous examination of both the distributions of individual variables and their pairwise relationships. On the diagonal, kernel density estimates, KDE plots, depict the shape of

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

each distribution, facilitating the detection of skewness or potential outliers. Off-diagonal scatterplots reveal distinct correlation patterns among the spatial parameters, while also indicating that the remaining variables do not exhibit strong interdependencies. Thus, the use of a representative subsample allows for an effective visual exploration of the key relationships without compromising the general validity of the conclusions.

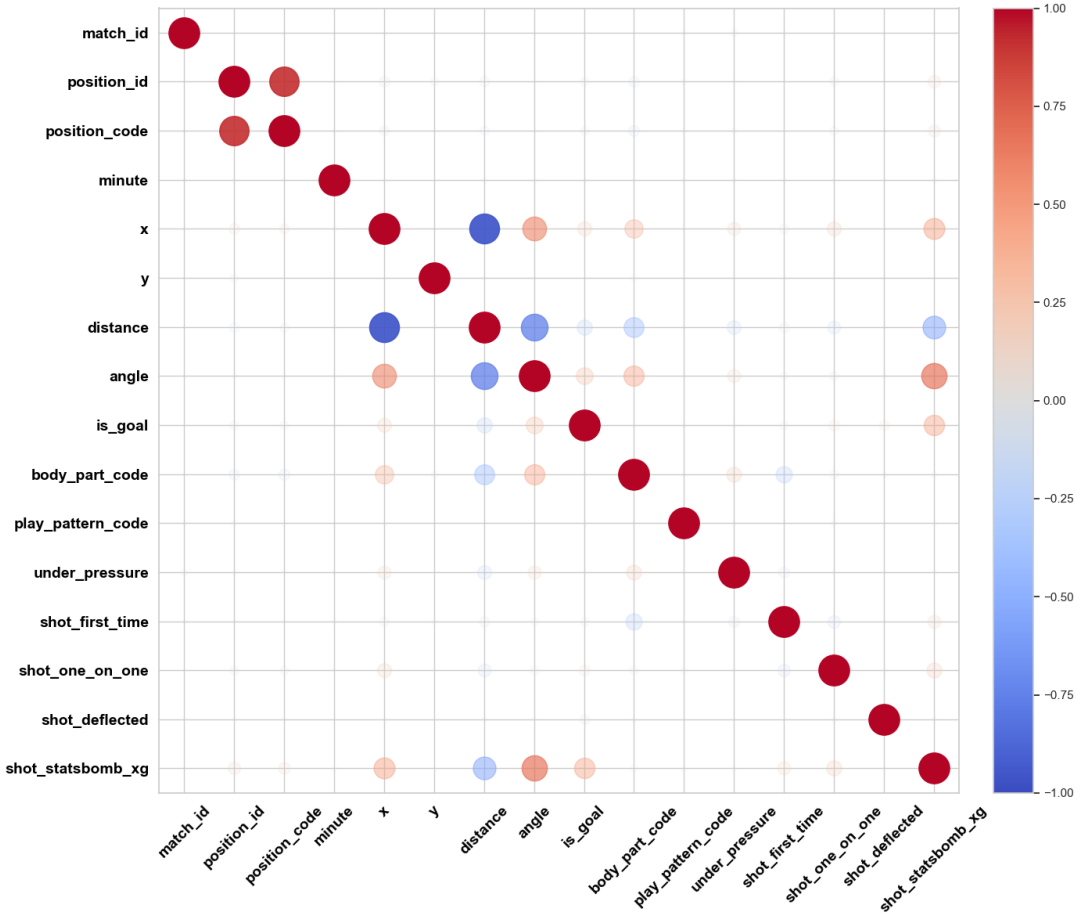


Figure 3.9: Correlation heatmap of shot attempt features

Beyond the aggregate analysis, particular interest lies in the application of the xG metric to individual player performance, as it can highlight differences that raw statistics alone may obscure. A representative case study is the 2015/16 season of Luis Suárez in La Liga, one of the most productive campaigns of his career. That year, Suárez won the Golden Boot (Pichichi Award) and emerged as the top scorer across Europe's top five leagues, setting a personal record of 40 league goals in a single season. His outstanding efficiency, both in chance creation and finishing, played a decisive role in the title-winning campaign of FC Barcelona, establishing him as a central pillar of the team's attacking play. Figure 3.10 illustrates the distribution and quality of Suárez's shot attempts in La Liga during the season under study. The shot map reveals a strong concentration of efforts within the penalty area and from central positions, reflecting both his playing style and his ability to occupy high-value scoring zones. Out of 133 total open-play attempts,

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Suárez scored 37 goals, significantly outperforming the model's cumulative expectation of 23.66 xG. This substantial overperformance emphasizes his exceptional finishing ability and consistent capacity to exceed statistical predictions. The 3 additional goals that complete his overall tally derived from penalty kicks and are therefore excluded from the present analysis.

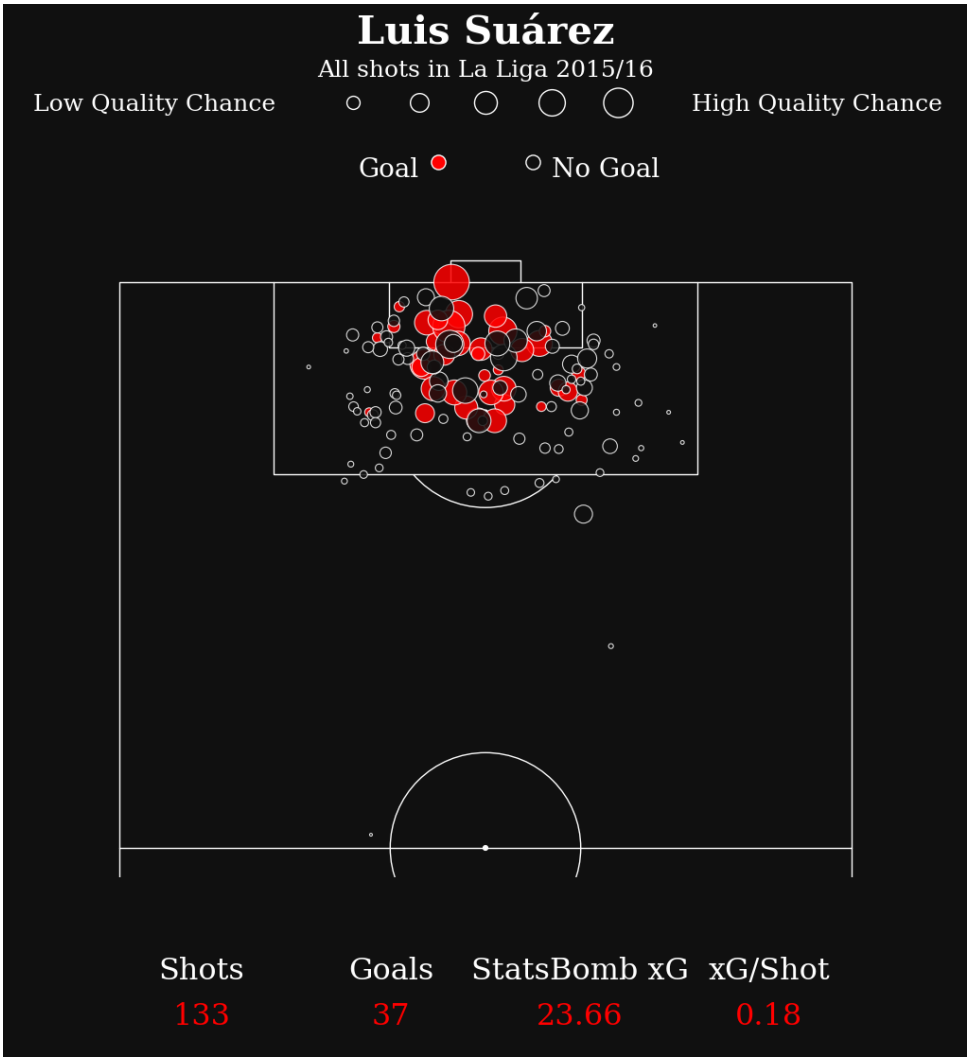


Figure 3.10: Expected goals shot map of Luis Suárez

In contrast to this remarkable efficiency, Edin Džeko's 2015/16 Serie A season with AS Roma illustrates a case of significant underperformance relative to the xG metric. Specifically, across 86 total open-play attempts, the Bosnian striker scored only 6 goals, whereas his cumulative expected goals amounted to 11.90. The resulting difference of –5.90 goals highlights a low finishing efficiency given the volume and quality of chances he generated. Nevertheless, this finding does not undermine the utility of the metric; rather, it reinforces its reliability over the long term. Although Džeko converted almost half as many goals as xG suggested during that campaign, he rebounded in the following season and emerged as Serie A's top scorer. This trajectory reflects a key property of xG:

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

deviations observed in individual periods tend to converge over time, providing a more accurate representation of true finishing ability and revealing weaknesses that traditional statistics may overlook.

Overall, the two case studies highlight the dual role of the xG metric in performance evaluation. As shown in Table 3.2, Suárez's season exemplifies how outperforming expected goals underscores extraordinary efficiency and consistency, whereas Džeko's case demonstrates that a substantial negative deviation reveals limitations in chance conversion despite high opportunity volume. Notably, the xG per shot metric does not vary as sharply as the conversion rate, indicating that two seasons with comparable chance quality may be perceived very differently, one as unremarkable and the other as outstanding.

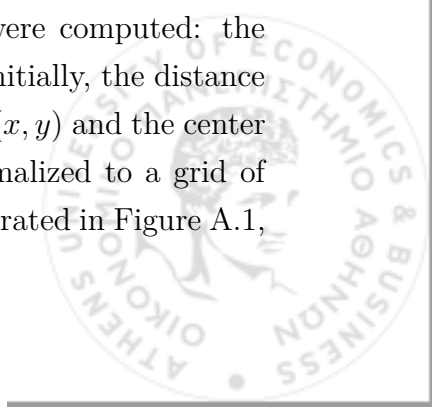
Table 3.2: Comparison of Suárez and Džeko

Key Metric	Luis Suárez	Edin Džeko
Competition	La Liga	Serie A
Season	2015/16	2015/16
Total Shots	133	86
Total Goals	37	6
Total xG	23.66	11.90
Goals–xG	+13.34	-5.90
Conversion Rate (%)	27.8	7.0
xG per Shot	0.178	0.138

This juxtaposition illustrates that xG extends beyond the mere reporting of quantitative aggregates, offering instead a deeper tool for individual performance analysis. At the club level, xG values are systematically leveraged by technical departments to assess the threat posed by a player's on-field presence, serving as a critical factor in contract renewals and transfer decisions.

3.4 Data Pre-Processing

In order to prepare the final dataset employed in the methodology, a series of transformations was applied to the raw event data and the previously constructed dataset, with the objective of deriving new, interpretable features suitable for subsequent analysis. Among these, for each shot attempt two fundamental geometric features were computed: the distance from the shooting location to the goal and the shot angle. Initially, the distance was defined as the Euclidean distance between the shooting location (x, y) and the center of the goal. In the StatsBomb coordinate system, the pitch is normalized to a grid of 120×80 units, with the center of the goal located at $(120, 40)$, as illustrated in Figure A.1,



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

which depicts the pitch along with its coordinate system. The calculation was performed using the following formula:

$$\text{distance} = \sqrt{(x_{\text{goal}} - x)^2 + (y_{\text{goal}} - y)^2}$$

The shot angle represents the visual angle formed by the shooter with respect to the two vertical goalposts from the shooting location. For its calculation, the goalposts were defined by the coordinates $(120, y_1)$ and $(120, y_2)$, where:

$$y_1 = 40 - \frac{8.61}{2}, \quad y_2 = 40 + \frac{8.61}{2}$$

The value 8.61 corresponds to the actual width of the goal (7.32 meters), which in the normalized StatsBomb coordinate system is represented as 8.61 pitch units, thereby preserving the correct geometric proportions. The angle was then computed as the absolute difference between the two arctangent values formed by the lines connecting the shooting location to each goalpost:

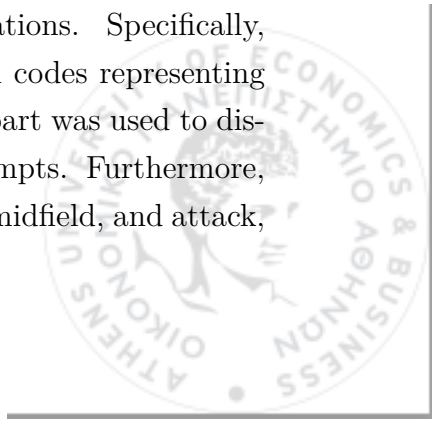
$$\text{angle} = \left| \tan^{-1}\left(\frac{y_2 - y}{x_{\text{goal}} - x}\right) - \tan^{-1}\left(\frac{y_1 - y}{x_{\text{goal}} - x}\right) \right|$$

The calculation of the angle was performed in *radians*, as the `arctan2` function returns values in this unit. For interpretability, radians can be converted to degrees using the relation:

$$1 \text{ rad} \approx 57.3^\circ$$

As illustrated in Figure A.2, shots taken closer to the center of the pitch are associated with wider shooting angles (e.g., $72.4^\circ \approx 1.26$ rad), whereas shots from more acute positions result in narrower angles (e.g., $20.9^\circ \approx 0.36$ rad).

Moving beyond features directly related to the spatial position of the player on the pitch, additional transformations were applied to encode contextual and event-specific information into a suitable format for modeling. Variables representing True/False conditions were converted into numerical binary form (0/1). Furthermore, the target variable was defined as the binary column, which takes the value 1 when the outcome of the shot is a goal and 0 otherwise. In this way, the predictive task was formally framed as a binary classification problem. In addition to the binary variables, several categorical features were also encoded to facilitate subsequent analysis and transformations. Specifically, the variable representing the play pattern was mapped to numerical codes representing different types of build-up, while the variable representing the body part was used to distinguish between right-foot, left-foot, header, or other types of attempts. Furthermore, player positions were grouped into three broader categories, defence, midfield, and attack,



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

as illustrated in Figure 3.11, which represents the players either in their starting line-up positions on the pitch or in the roles they assumed following substitutions. Notably, when tactical adjustments occur during a match, the dataset systematically updates the positional information to reflect the player's new role within the same game.

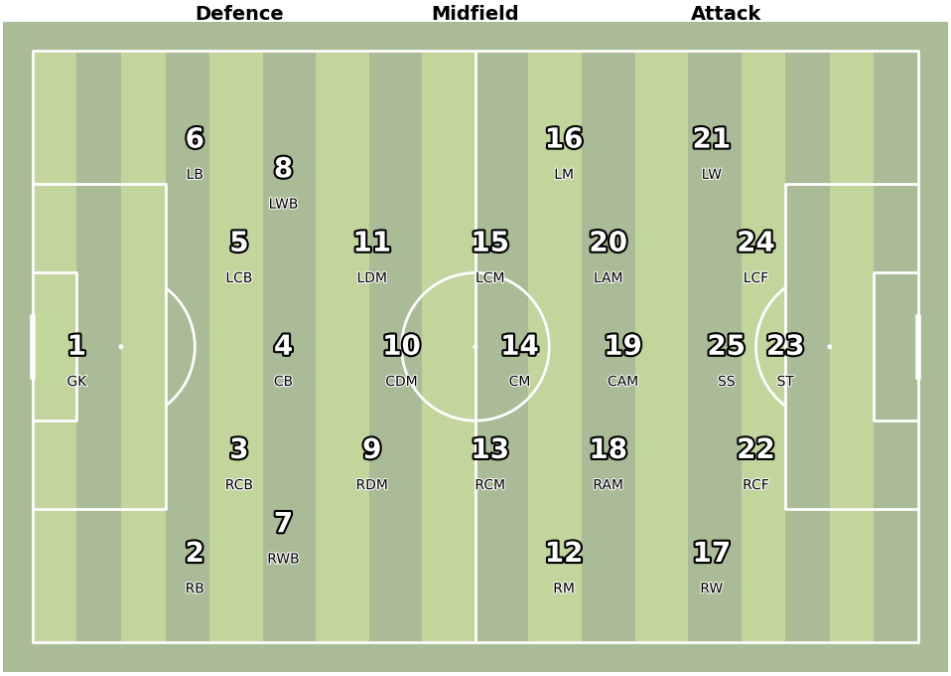
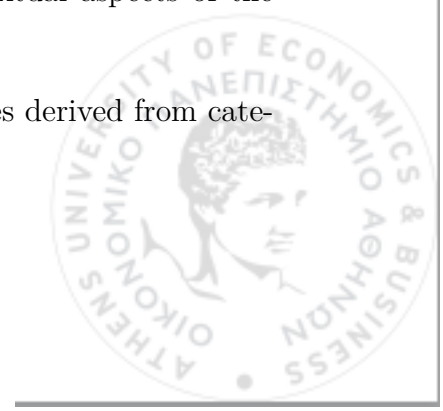


Figure 3.11: Map of football player positions

To consolidate the preprocessing steps and prepare the data for modeling, the final stage of the pipeline explicitly defined the feature set to be used for training. The objective at this stage was to retain only those variables with genuine predictive value for the problem under investigation, while ensuring consistency in their representation. Accordingly, the categorical variables that had previously been numerically encoded were further transformed using the *one-hot encoding* technique. This approach assigns a separate binary feature to each category, thereby preventing the artificial introduction of spurious ordinal relationships, eliminating any misleading hierarchy among categories, and yielding a dataset that is both more transparent and interpretable.

- **Numeric features:** 3 continuous variables describing the geometry and timing of the shot, *distance*, *angle*, and *minute*.
- **Binary features:** 4 dichotomous indicators capturing contextual aspects of the attempt, *under pressure*, *first time*, *one on one*, and *deflected*.
- **Categorical features (one-hot encoded):** dummy variables derived from categorical attributes:
  - 4 dummies representing the shooting *body part*,



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

- 9 dummies representing the *play pattern*,
- 4 dummies representing the player *position group*.

Overall, this procedure yielded a total of 24 explanatory features. In addition to these, the dataset also contained the binary target variable indicating whether or not a given shot resulted in a goal. Furthermore, an additional variable was included, namely the expected goals value provided by StatsBomb for each attempt. This variable will not be used as an explanatory feature in the training of the models but solely as a benchmark for comparison purposes. This comprehensive structure contributed to the construction of a homogenized and well-balanced dataset, suitable for the application of machine learning methods. By reducing complexity and ensuring the consistent representation of both numerical and categorical variables, the procedure enhanced not only the interpretability of the data but also the reliability and generalization capability of the predictive models to be developed.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Chapter 4

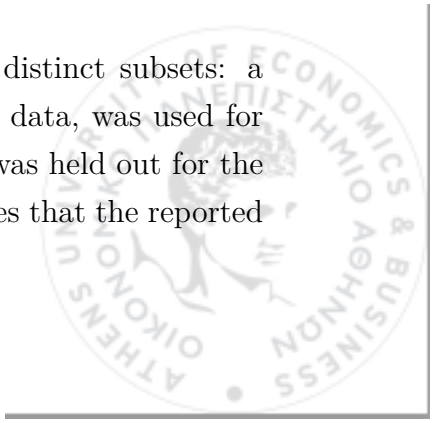
Methodology

4.1 Setup

For the supervised models, the training data were used to learn the relationship between shot characteristics and goal conversion. Each model predicted the binary goal outcome from multidimensional shot features, with k-fold cross-validation applied for hyperparameter tuning and overfitting prevention. Evaluation metrics were computed on the held-out test set to assess generalization and calibration performance. The unsupervised models, by contrast, were trained without any access to the outcome-related variables, the goal indicator and the StatsBomb xG values, ensuring that the latent structure and cluster formation depended solely on intrinsic shot characteristics. This design prevents information leakage and preserves the descriptive nature of unsupervised analysis. Finally, in the causal and counterfactual analysis stage, the learned latent representations were employed to generate controlled perturbations in feature space, simulating minimal, interpretable interventions on shot characteristics. These counterfactual instances were then re-evaluated using the trained xG models to estimate how small changes could affect the predicted probability of scoring. Together, this multi-stage framework ensures methodological rigor and provides complementary insights across predictive, descriptive, and causal domains. The different modeling paradigms employed in this study are summarized concisely in Table B.1, highlighting their respective use of outcome variables and analytical objectives.

4.2 Supervised Approach

For the experimental procedure, the dataset was divided into two distinct subsets: a training set and a test set. The training set, comprising 80% of the data, was used for model fitting and representation learning, while the remaining 20% was held out for the independent evaluation of model performance. This separation ensures that the reported



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

results reflect the generalization ability of the models to unseen data rather than their capacity to memorize training examples. Within the training data, a k-fold cross-validation scheme ($k = 5$) was employed, following the approach of Kohavi (1995). In each iteration, four folds were used for training and one for validation, rotating until every fold had served as a validation set once. Performance metrics were then averaged across folds to obtain a robust estimate and to guide hyperparameter optimization. This approach mitigates overfitting, reduces variance associated with a single split, and provides a stable basis for final evaluation on the held-out test set. An illustration of the cross-validation process is shown in Figure 4.1, where the alternating roles of the folds in training and validation are clearly depicted.



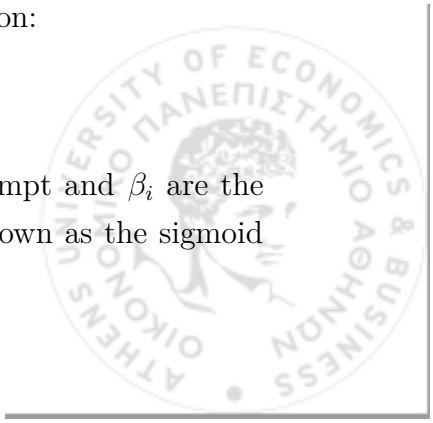
Figure 4.1: Visualization of the 5-Fold cross-validation procedure

4.2.1 Logistic Regression

In the experimental procedure, logistic regression was employed with the objective of estimating the probability that each shot attempt would result in a goal, following the formulation described in Hosmer, Lemeshow, and Sturdivant (2013). The model was repeatedly trained on five stratified training subsets and evaluated on the corresponding validation subsets, yielding five independent estimates of performance. This procedure enabled the computation of stable average values of the evaluation metrics, thus providing a more reliable assessment of overall model performance. Logistic regression models the probability of the positive outcome (goal) through the logistic function:

$$P(y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)}}$$

where $X = (x_1, x_2, \ldots, x_p)$ represents the features of each shot attempt and β_i are the coefficients estimated during training. The logistic function, also known as the sigmoid



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

function, maps any real-valued input into the range [0,1], thereby allowing the output to be interpreted as a probability. For each iteration of the procedure, the evaluation metrics presented in the corresponding section were computed. The final reported values correspond to the averages across the five folds.

To check for potential multicollinearity among the continuous predictors used in the logistic regression model, we computed the Variance Inflation Factor (VIF) O'Brien (2007).

4.2.2 Decision Trees

Decision Trees are among the most widely used supervised learning methods for both classification and regression problems. Their fundamental logic lies in the recursive partitioning of the dataset into smaller subsets, aiming to create nodes that are as "pure" as possible with respect to the distribution of the dependent variable.

The structure of a Decision Tree comprises three main types of nodes: the root node, which represents the entire dataset and serves as the starting point of the model, the internal nodes, where splitting rules are applied based on selected features in order to minimize the chosen impurity measure and the leaf nodes, or terminal nodes, which are not further subdivided and correspond to the final class prediction. The tree is constructed recursively, starting from the root and continuing with successive partitions until predefined stopping criteria are met, such as the maximum allowable depth or the minimum number of samples required in a node. The resulting model exhibits a hierarchical structure, in which internal nodes represent decision rules and the leaves provide the final classification outcome, as illustrated schematically in Figure 4.2.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

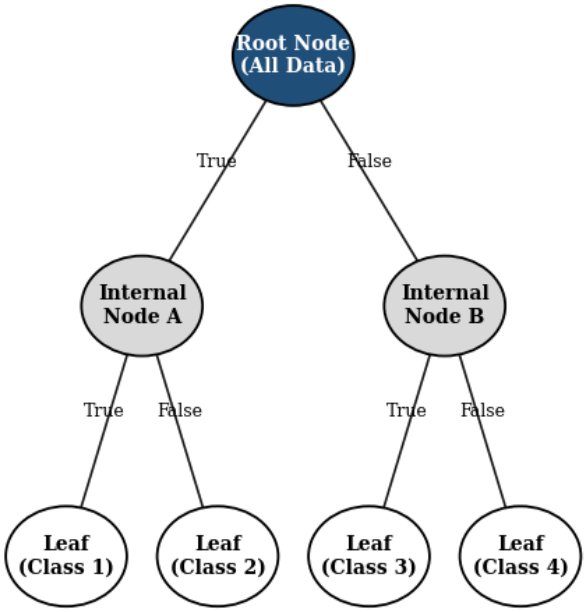
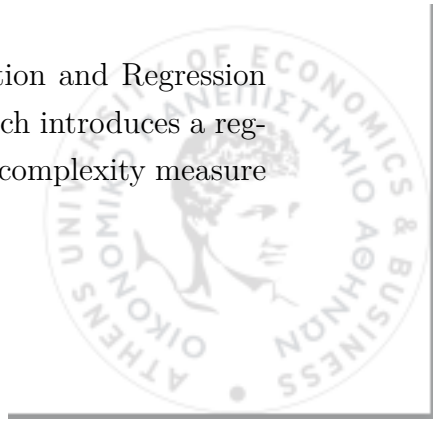


Figure 4.2: Schematic representation of a Decision Tree

The selection of the optimal feature and split point is based on an impurity measure, which quantifies the heterogeneity of the classes within a node. Commonly used impurity measures are the Gini Index and Entropy, which guide the tree-building process by favoring partitions that increase node purity.

Decision Trees are attractive due to their simplicity, interpretability, and their ability to handle both categorical and continuous variables without strict parametric assumptions. Nonetheless, they also present important drawbacks: they are prone to overfitting, often unstable with respect to small perturbations in the data, and may underperform compared to more advanced ensemble models. To mitigate this issue, pruning techniques are employed, which aim to reduce the size of the tree by removing branches that contribute little to predictive performance. Two main approaches that exist, are pre-pruning, also known as early stopping, where the growth of the tree is halted once a stopping condition is met, and post-pruning, where a fully grown tree is subsequently simplified by removing nodes that do not improve generalization according to a validation criterion. Pruning thus leads to simpler, more interpretable, and more stable models, while simultaneously reducing the risk of overfitting.

To further formalize the pruning process, the CART (Classification and Regression Trees) algorithm employs the so-called *cost–complexity criterion*, which introduces a regularization parameter α that penalizes the size of the tree. The cost–complexity measure



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

of a subtree T is defined as

$$R_\alpha(T) = R(T) + \alpha \cdot |T|,$$

where $R(T)$ denotes the empirical error of the tree on the training set and $|T|$ represents the number of terminal nodes (leaves). As the value of α increases, additional branches are pruned, resulting in smaller and more interpretable trees that achieve a better trade-off between predictive accuracy and generalization performance.

In the CART algorithm, the most frequently employed impurity measures are the Gini index and Entropy, as discussed in James, G, Witten, D, Hastie, T, & Tibshirani, R (2021). The Gini index is defined as

$$Gini(t) = 1 - \sum_{i=1}^{C} p_i^2$$

where C denotes the total number of classes and p_i is the proportion of samples belonging to class i within node t. The value of the Gini index ranges from 0, corresponding to a pure node where all observations belong to a single class, to a maximum of 0.5 in the case of binary classification when the two classes are equally represented (50–50 distribution).

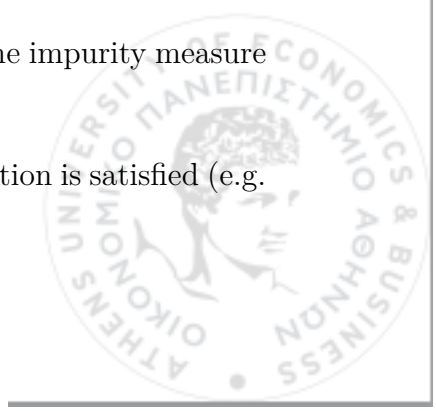
An alternative impurity measure, derived from Information Theory, is the entropy. Like the Gini index, entropy quantifies the degree of impurity of a node by assessing the uncertainty associated with the class distribution. Its value equals 0 when the node is pure and increases as the distribution of the samples among classes becomes more balanced. In binary classification problems, the entropy reaches its maximum value of 1 when the two classes are equally probable. Although entropy and the Gini index often lead to similar split selections, minor differences in the resulting tree structure may occur. The entropy is defined as:

$$Entropy(t) = -\sum_{i=1}^{C} p_i \cdot \log_2(p_i),$$

where C is the number of classes and p_i denotes the proportion of samples belonging to class i in node t.

In this thesis, the Decision Tree is implemented using the Gini index, as it is the default and computationally more efficient criterion of the CART algorithm. Thus, the training of a Decision Tree proceeds as follows:

1. Define the dataset with independent variables (X) and the target variable (y).
2. At each node, select the feature and split point that minimize the impurity measure (Gini index).
3. Recursively continue the splitting process until a stopping condition is satisfied (e.g. maximum depth, minimum number of samples per leaf).



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

4. The final model is a tree whose internal nodes contain decision rules and whose leaves correspond to the predicted classes, “Goal” and “No Goal”.

Consequently, Table 4.1 summarizes the key properties of the two most common impurity measures.

Criterion	**Formula**	**Range (binary case)**	**Maximum at $p = 0.5$**
Gini	$1 - \sum p_i^2$	$[0, 0.5]$	0.5
Entropy	$-\sum p_i \log_2(p_i)$	$[0, 1]$	1

Table 4.1: Comparison of impurity measures used in Decision Trees.

4.2.3 Random Forests

Random Forests represent one of the most powerful and widely adopted supervised learning algorithms, belonging to the family of ensemble methods. The central idea underlying this model is the combination of multiple simple learners, specifically Decision Trees, in order to achieve improved generalization and to mitigate the risk of overfitting, as illustrated schematically in Figure 4.3, where an indicative ensemble of three trees is depicted.

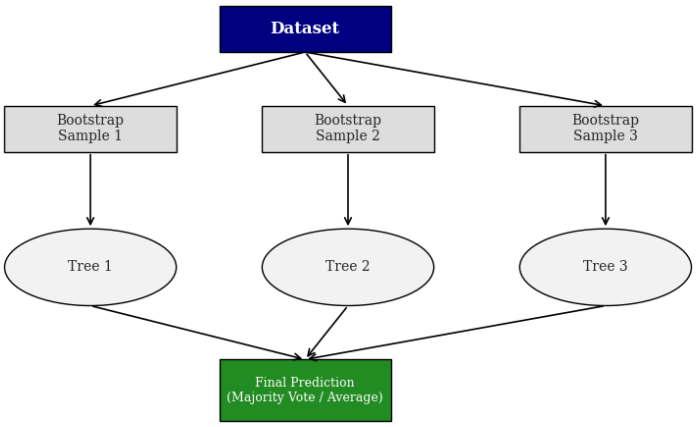


Figure 4.3: Schematic representation of a Random Forest with three trees

Unlike a single Decision Tree, which often exhibits high variance and sensitivity to small perturbations in the data, the Random Forest constructs an ensemble of trees, each trained on a random bootstrap sample of the original dataset and considering a random subset of features at each split. The final prediction is obtained through majority voting in classification problems or through averaging in regression problems, resulting in more stable and reliable outcomes.

The training procedure of a Random Forest can be summarized as follows:

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

1. Generate random bootstrap samples from the original dataset using sampling with replacement.

2. Train a Decision Tree on each bootstrap sample. At each split, only a randomly selected subset of features is considered, which increases the diversity among trees.

3. Repeat the process for a predefined number of trees ($n_estimators$).

4. Aggregate the predictions of all trees, either through majority vote in classification tasks or through averaging in regression tasks.

In this study, the Random Forest model was implemented with a number of carefully selected hyperparameters. The number of trees was set to $n_estimators = 300$, a choice that balances predictive stability with computational efficiency. A maximum depth of 15 levels was imposed on each tree to prevent overfitting while maintaining sufficient flexibility. The Gini index was adopted as the splitting criterion due to its computational efficiency and its wide applicability in classification tasks. In addition, class_weight="balanced" was enabled to address the inherent imbalance between goal and no-goal outcomes in the dataset. Finally, $n_jobs = -1$ allowed the model to exploit all available processing cores during training.

Typical values for these hyperparameters, as reported in the literature Breiman (2001), Hastie, Tibshirani, Friedman (2009), James, Witten, Hastie, Tibshirani (2021), and in widely used software implementations such as Pedregosa et al. (2011), are summarized in Table B.2.

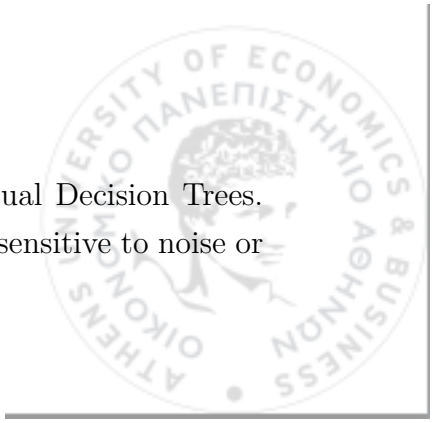
The use of randomization in both the sampling of observations and the selection of features constitutes the key principle that differentiates Random Forests from single Decision Trees, as it reduces the correlation among trees and enhances the predictive accuracy of the ensemble. Mathematically, the final prediction in a binary classification setting can be expressed as:

$$\hat{y} = \arg\max_{c \in \{0,1\}} \sum_{m=1}^{M} \mathbb{1}\{h_m(x) = c\},$$

where M denotes the total number of trees, $h_m(x)$ is the prediction of the m-th tree for observation x, and $\mathbb{1}$ is the indicator function. Thus, the final decision corresponds to the class that receives the majority of votes across all trees. In regression problems, the prediction is given by the average of the predictions of all trees:

$$\hat{y} = \frac{1}{M} \sum_{m=1}^{M} h_m(x).$$

Random Forests present several advantages compared to individual Decision Trees. They provide significantly improved predictive accuracy and are less sensitive to noise or



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

small variations in the training data. They can effectively handle large-scale datasets and high-dimensional feature spaces, and they offer measures of feature importance, which allow for the interpretation of variable relevance in the model. Nevertheless, Random Forests also present certain limitations. Their computational cost is higher compared to single trees, their interpretability is restricted due to the complexity of the ensemble, and they require careful hyperparameter tuning in order to achieve optimal performance.

4.2.4 Gradient Boosting

Gradient Boosting belongs to the family of ensemble methods and constitutes one of the most powerful approaches in supervised learning. The core idea of the model is the construction of an ensemble of simple base learners, typically shallow decision trees, trained sequentially so that each new learner attempts to correct the errors of the previous ones. This iterative procedure leads to a strong final model with significantly improved generalization ability. In contrast to Random Forests, where trees are trained independently and their outputs are combined through majority voting or averaging, Gradient Boosting follows a sequential training scheme. Each new tree is added to the ensemble in order to reduce the residual error of the previous stage. This procedure is called boosting, while the term gradient refers to the fact that the updates are guided by the direction of steepest descent of the loss function, that is, the negative gradient.

The training process of Gradient Boosting can be summarized in the following steps:

1. Initialize the model with a simple prediction (e.g., the mean in regression or the log-odds in binary classification).
2. Compute the residuals, that is, the difference between the true values and the predictions.
3. Fit a weak learner (typically a decision tree) to the residuals.
4. Update the overall model by adding the new tree, scaled by the learning rate parameter.
5. Repeat *steps 2-4* for a predetermined number of iterations M.

Mathematically, the procedure can be expressed as follows:

$$F_0(x) = \arg\min_{\gamma} \sum_{i=1}^{N} L(y_i, \gamma),$$

$$r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{m-1}(x)},$$



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

$$h_m(x) \approx r_{im}, \quad m = 1, \ldots, M,$$

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x),$$

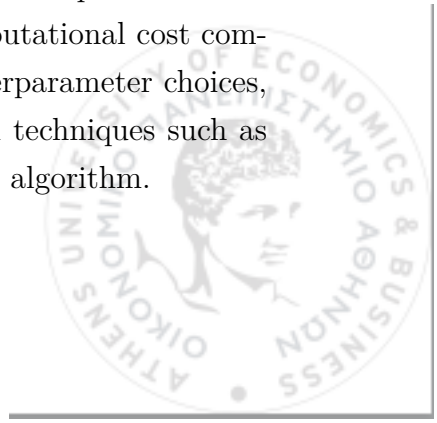
where L is the loss function (e.g., log loss in binary classification), r_{im} are the pseudo-residuals, $h_m(x)$ is the weak learner fitted to the residuals, and ν is the learning rate.

Gradient Boosting involves several critical hyperparameters that strongly influence its performance, as highlighted in Friedman, J. H. (2001) and Friedman, J. H. (2002). The most important are summarized in Table B.3.

In this study, the selection of Gradient Boosting hyperparameters was not based solely on the literature, as there is no universally optimal configuration applicable across all datasets. For this reason, a Grid Search with 5-fold cross-validation was employed, aiming to maximize the AUC score, which is particularly suitable for imbalanced binary classification tasks. This procedure was implemented using the `scikit-learn` library Pedregosa et al. (2011). For each combination of parameters, the training set was partitioned into five folds, with four used for training and one for validation in rotation, in order to obtain a more reliable performance estimate. The parameter combination that achieved the highest mean AUC across folds was selected as optimal, corresponding to $n_estimators = 200$, $learning_rate = 0.1$, $max_depth = 2$ and $subsample = 1.0$. These values were subsequently used for training the final Gradient Boosting model on the complete training set.

The final model with the optimal configuration was retrained on the entire training set and subsequently evaluated on the independent test set. The evaluation employed metrics such as Log Loss, for probabilistic calibration,, Brier Score, for probability reliability, , AUC, for discriminative ability, as well as Accuracy, Precision, Recall, and F1-score, thereby providing a comprehensive assessment of performance. In addition, feature importance scores were extracted to investigate the relative influence of predictors. To further enhance interpretability, Partial Dependence Plots (PDPs) were generated for the most important features, illustrating their marginal effect on the predicted goal probability while averaging over the distribution of other variables. This approach enabled the model to be interpreted not merely as a black-box predictor, but also as a tool offering insights into how specific factors shape the expected goals estimation.

Gradient Boosting provides high predictive accuracy and can model complex nonlinear relationships between variables. However, it exhibits increased computational cost compared to simpler models and is more sensitive to inappropriate hyperparameter choices, which may lead to overfitting. Careful tuning through optimization techniques such as Grid Search is therefore essential to fully exploit the potential of the algorithm.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

4.2.5 XGBoost

XGBoost (Extreme Gradient Boosting) is one of the most widely adopted and efficient implementations of tree-based boosting algorithms, with extensive applications in machine learning and large-scale data analysis. It was developed with the primary objectives of improving predictive accuracy, reducing training time, and incorporating additional regularization mechanisms to mitigate overfitting.

The fundamental principle of XGBoost remains consistent with classical Gradient Boosting: trees are added sequentially so that each new tree corrects the errors of its predecessors by fitting to the negative gradient of the loss function. However, XGBoost introduces significant improvements in terms of optimization, parallelization, and regularization, which make it one of the most effective and computationally efficient boosting frameworks available.

The training process of XGBoost can be summarized in the following steps:

1. Define the objective function as the sum of a differentiable loss function and a regularization term controlling model complexity.
2. At iteration t, compute the first- and second-order gradients of the loss function with respect to the current predictions.
3. Construct a new regression tree $f_t(x)$ by optimizing a regularized objective function based on the computed gradients.
4. Update the ensemble prediction by adding the newly fitted tree, scaled by the learning rate parameter.
5. Repeat *steps 2–4* for a predetermined number of iterations M or until convergence.

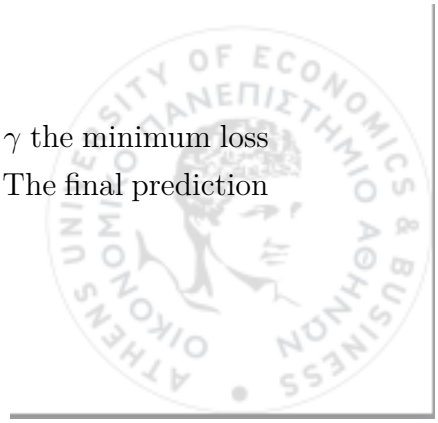
Mathematically, the procedure can be expressed as follows:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{N} l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t),$$

where l is the differentiable loss function, $\hat{y}_i^{(t-1)}$ is the prediction of the ensemble up to iteration $t-1$, f_t is the tree added at iteration t, and $\Omega(f_t)$ is the regularization term defined as:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2,$$

with T denoting the number of leaves in the tree, w_j the leaf weights, γ the minimum loss reduction required for a split, and λ the L2 regularization parameter. The final prediction after M iterations is therefore given by:



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

$$\hat{y}_i^{(M)} = \sum_{t=1}^{M} f_t(x_i).$$

XGBoost includes a wide range of hyperparameters that govern both model training and regularization. The most important are summarized in Table B.4, following the description of Chen & Guestrin (2016).

In this study, hyperparameter optimization for XGBoost was carried out using Grid Search with 5-fold cross-validation, targeting the maximization of the AUC metric, which is particularly appropriate for imbalanced binary classification problems. Compared to classical Gradient Boosting, the optimization process involved a broader set of parameters, including not only *n_estimators*, *learning_rate*, *max_depth*, and *subsample*, but also *colsample_bytree*, which controls the proportion of features considered at each tree. The best-performing configuration was subsequently retrained on the complete training set and evaluated on the independent test set.

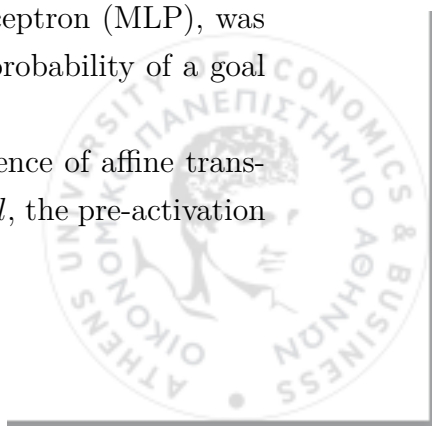
Model performance was assessed using a comprehensive set of evaluation metrics, namely Log Loss for probabilistic calibration, Brier Score for probability reliability, AUC for discriminative ability, and Accuracy, Precision, Recall, and F1-score for classification quality. In addition, feature importance scores were extracted to quantify the relative influence of predictors, while Partial Dependence Plots (PDPs) were generated for the most influential features. These interpretability tools allowed for a deeper understanding of how key variables affect the probability of scoring a goal, complementing the predictive power of the model.

XGBoost offers high predictive accuracy, strong regularization capabilities to prevent overfitting, and substantial training speed improvements due to optimized algorithms and parallelization. Nevertheless, it requires careful hyperparameter tuning to fully leverage its potential and is more complex to use and interpret compared to simpler models such as single decision trees.

4.2.6 Neural Network

Artificial Neural Networks (ANNs) constitute a class of supervised learning algorithms inspired by the structure and operation of biological neural systems. They consist of layers of interconnected computational units (neurons) that transform input data through a series of weighted linear combinations and nonlinear activation functions. In this study, a feedforward neural network, also referred to as a Multi-Layer Perceptron (MLP), was constructed using the `TensorFlow/Keras` framework to model the probability of a goal event given contextual and positional features.

The forward propagation in an MLP can be described as a sequence of affine transformations followed by nonlinear activations. For a neuron j in layer l, the pre-activation



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

value is defined as:

$$z_j^{(l)} = \sum_{i=1}^{n_{l-1}} w_{ij}^{(l)} a_i^{(l-1)} + b_j^{(l)},$$

where $a_i^{(l-1)}$ are the activations from the previous layer, $w_{ij}^{(l)}$ and $b_j^{(l)}$ denote the learnable weights and biases, and n_{l-1} is the number of neurons in the preceding layer. The activation of neuron j is then computed using a nonlinear function $\sigma(\cdot)$:

$$a_j^{(l)} = \sigma(z_j^{(l)}).$$

The network as a whole represents a parametric function $f_\theta : \mathbb{R}^d \to [0, 1]$, parameterized by $\theta = \{W, b\}$, which maps input features x_i to an estimated probability $\hat{y}_i$ of the target class.

For binary classification, the Binary Cross-Entropy (BCE) loss function was employed to measure the divergence between predicted probabilities $\hat{y}_i$ and true outcomes y_i:

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^{N} \Big[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \Big].$$

Model parameters were optimized through gradient-based minimization using the Adaptive Moment Estimation (Adam) optimizer Kingma, Ba (2014), which adaptively adjusts learning rates per parameter using first- and second-order moment estimates of the gradients. The learning rate η was set to 0.001, a widely adopted value in the literature for ensuring stable convergence across a variety of network architectures Goodfellow et. al. (2016).

The implemented MLP architecture comprised two fully connected (*Dense*) hidden layers with 64 and 32 neurons, respectively, using the Rectified Linear Unit (ReLU) activation function:

$$\text{ReLU}(z) = \max(0, z),$$

which provides nonlinearity and mitigates the vanishing gradient problem Nair, Hinton (2010). The output layer consisted of a single neuron with a sigmoid activation function:

$$\sigma(z) = \frac{1}{1 + e^{-z}},$$

suitable for mapping outputs to probabilities in the range $[0, 1]$ for binary classification.

To reduce overfitting and enhance generalization, a dropout layer was applied after the first hidden layer with a rate $p = 0.3$. Dropout randomly deactivates a fraction p of neurons during training, effectively approximating an ensemble of subnetworks and reducing co-adaptation among units Srivastava, Hinton, Krizhevsky, Sutskever, Salakhutdinov (2014). Additionally, an *Early Stopping* criterion was integrated to monitor the validation loss and halt training when no improvement was observed for a predefined number of epochs (patience = 10). Upon termination, the model restored the weights from the epoch yield-

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

ing the minimum validation loss, thereby ensuring convergence near the optimal point while avoiding unnecessary training iterations.

The dataset was partitioned into training and test sets with an 80–20 ratio, preserving class balance through stratified sampling. Within the training data, 20% was further reserved as a validation subset to monitor learning progress. The model was trained for a maximum of 100 epochs using a batch size of 64, a setting that balances gradient stability and computational efficiency. Throughout training, both the training and validation losses were recorded to generate learning curves, offering a diagnostic overview of the model's convergence behavior and overfitting tendencies.

To support model interpretability, SHAP (SHapley Additive exPlanations) values Lundberg, Lee (2017) were computed post-training. SHAP provides a theoretically grounded method based on cooperative game theory for quantifying each feature's contribution to the model output. For an instance x, the prediction $f(x)$ can be decomposed additively as:

$$f(x) = \phi_0 + \sum_{j=1}^{M} \phi_j,$$

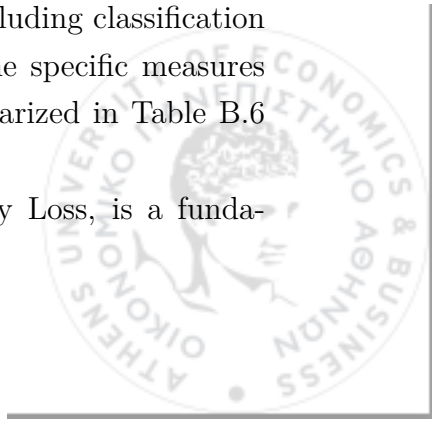
where ϕ_j denotes the contribution (Shapley value) of feature j and ϕ_0 represents the average model output across the dataset. Features with higher absolute ϕ_j values exert greater influence on the predicted probability. Both summary and bar plots were employed to visualize the distribution and magnitude of SHAP values across input variables, enabling the identification of dominant predictors and the directionality of their effects. Table B.5 summarizes the key hyperparameters adopted for the neural network. These were selected based on best practices from the deep learning literature and empirical validation for stable optimization and effective regularization.

In summary, the designed neural network constitutes a feedforward supervised learning model that captures complex nonlinear relationships among input variables through multiple hidden layers and nonlinear activations. Regularization techniques such as dropout and early stopping were incorporated to ensure robust generalization, while SHAP-based explainability was used to attribute the model's probabilistic predictions to individual input features in a theoretically transparent manner.

4.2.7 Evaluation metrics

The performance of all models was evaluated using a comprehensive set of evaluation metrics that capture complementary aspects of predictive ability, including classification accuracy, discrimination capacity, and probabilistic calibration. The specific measures adopted in this study, together with their interpretation, are summarized in Table B.6 Powers (2011), Brier (1950), Bishop (2006).

The Logarithmic Loss (Log Loss), also known as Cross-Entropy Loss, is a funda-



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

mental metric for evaluating probabilistic binary classification models. It measures the divergence between the predicted probabilities and the actual outcomes. A lower value of the index indicates that the model produces well-calibrated predictions and is therefore more reliable. Conversely, a higher Log Loss value suggests that the model assigns probabilities that deviate significantly from reality, particularly when it predicts an outcome with high confidence that ultimately proves to be incorrect.

$$\text{Log Loss} = -\frac{1}{N}\sum_{i=1}^{N}\Big[y_i \cdot \log(p_i) + (1-y_i)\cdot \log(1-p_i)\Big]$$

The Brier Score computes the mean squared deviation between the predicted probability and the actual outcome, with the important property of penalizing larger errors more severely due to the squaring of deviations. This metric captures both the accuracy and the calibration of probabilistic predictions. A lower value of the index implies that the predicted probabilities are closer to the true outcomes. In contrast, higher values indicate that the model systematically assigns probabilities that diverge from reality, thereby reducing its reliability.

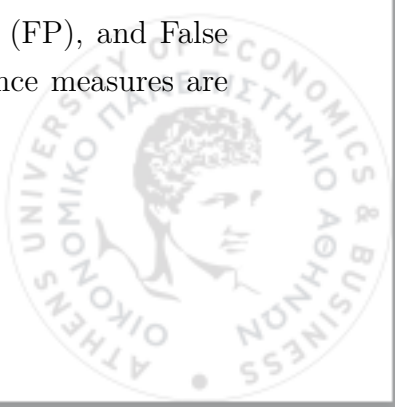
$$\text{Brier Score} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2$$

The Area Under the Curve (AUC) refers to the area under the Receiver Operating Characteristic (ROC) curve, which depicts the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR) across all possible decision thresholds. The AUC quantifies the overall discriminative ability of the model to correctly distinguish between the two classes (goal vs. no goal). A higher value of the index indicates stronger discriminative performance. Specifically, AUC = 0.5 implies that the model performs no better than random guessing, while AUC = 1.0 represents perfect classification. Formally, the value of the AUC corresponds to the probability that a randomly chosen positive instance (X^+) will receive a higher predicted score than a randomly chosen negative instance (X^-), where $f(X)$ denotes the predicted probability produced by the model:

$$\text{AUC} = P\Big(f(X^+) > f(X^-)\Big)$$

All of the following evaluation metrics are defined in terms of the confusion matrix, which provides a systematic summary of classification performance by comparing predicted labels with actual outcomes. The confusion matrix records four fundamental quantities: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Based on these quantities, several standard performance measures are derived, as described below. Specifically:

- True Positives (TP): correctly predicted goals,



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

- True Negatives (TN): correctly predicted no goals,
- False Positives (FP): no goals incorrectly predicted as goals,
- False Negatives (FN): goals incorrectly predicted as no goals.

The presentation of the confusion matrix, either in tabular or graphical form, can significantly enhance the understanding of these evaluation metrics and often provides a comprehensive overview of model performance, as illustrated in Table 4.2.

	Predicted Positive	**Predicted Negative**
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Table 4.2: Confusion matrix showing classification outcomes.

Accuracy represents the proportion of correct predictions, including both goals and no goals, relative to the total number of predictions made by the model. While it provides a general indication of performance, it can be misleading in cases of class imbalance, as a model may achieve high accuracy by predominantly predicting the majority class.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

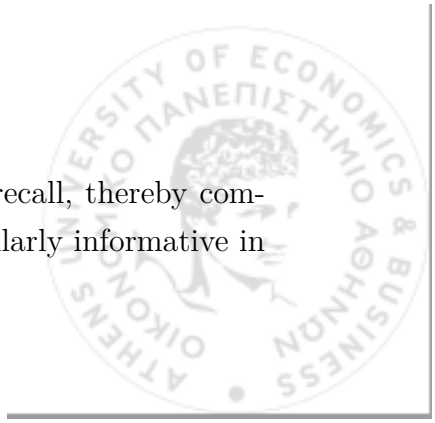
Precision denotes the proportion of predicted positive instances that correspond to actual positive outcomes. It reflects the reliability of the model when assigning positive labels. In the context of goal prediction, precision indicates the fraction of shots predicted as goals that indeed resulted in goals. Higher values of precision correspond to fewer false positive predictions.

$$Precision = \frac{TP}{TP + FP}$$

Recall denotes the proportion of actual positive instances that are correctly identified by the model. It reflects the ability of the model to capture all positive outcomes. In the context of goal prediction, recall indicates the fraction of shots that truly resulted in goals and were correctly classified as such. Higher values of recall correspond to fewer false negative predictions.

$$Recall = \frac{TP}{TP + FN}$$

The F1-score is defined as the harmonic mean of precision and recall, thereby combining both aspects of performance into a single metric. It is particularly informative in



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

the presence of class imbalance, as it penalizes models that achieve high performance on one dimension while performing poorly on the other. Higher F1-score values indicate a more balanced performance across precision and recall.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

4.3 Unsupervised Approach

In contrast to supervised learning, where the training process relies on labeled outcomes, the unsupervised approach focuses on the exploration of data without the requirement of such labels. Its objective is to uncover hidden patterns and latent structures that emerge from the intrinsic properties of the dataset, thereby enabling the identification of clusters of observations with shared characteristics. At the same time, it facilitates the discovery of association rules that reveal meaningful relationships among variables, as well as the derivation of reduced and more interpretable representations of the original feature space. In addition to clustering and dimensionality reduction, the framework also incorporates anomaly detection techniques, specifically, the Isolation Forest algorithm, in order to identify statistically atypical events that may correspond to exceptional shot situations. This component complements the clustering analysis by highlighting high-impact outliers that conventional grouping methods may overlook. Through this integrated approach, unsupervised learning provides a comprehensive means of data exploration that enhances the understanding of complex shot dynamics and offers complementary insights beyond the predictive scope of supervised methods.

4.3.1 Principal Component Analysis and K-Means Clustering

Unsupervised learning was employed in order to uncover latent structure in the dataset without relying on the outcome variable `is_goal` and xG value given by StatsBomb. The analytical pipeline consisted of two consecutive steps: Dimensionality reduction via Principal Component Analysis (PCA) and clustering of the reduced feature space using the K-Means algorithm. This methodology aims to identify groups of shots with similar statistical profiles and to assess whether such clusters correspond to meaningful differences in expected goal values. Given a centered data matrix $X \in \mathbb{R}^{n \times p}$, PCA projects the original features into a lower-dimensional orthogonal subspace that maximizes the retained variance, thereby facilitating the subsequent clustering stage.

Formally, PCA solves the following optimization problem:

$$w_k = \arg\max_{\|w\|=1} \mathrm{Var}(Xw),$$

which is equivalent to the eigenvalue decomposition



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

$$X^\top X w = \lambda w,$$

where w denotes the eigenvector (principal component) and λ the corresponding eigenvalue. Each eigenvalue directly measures the variance explained by its associated component, expressed as:

$$\lambda_k = \mathrm{Var}(X w_k).$$

The proportion of variance explained by the k-th component, known as the explained variance ratio, is then defined as:

$$\mathrm{EVR}_k = \frac{\lambda_k}{\sum_{j=1}^{p} \lambda_j}.$$

To determine the appropriate dimensionality, multiple diagnostics were considered, including the magnitude of eigenvalues, the cumulative explained variance ratio, and the elbow criterion derived from the scree plot. These criteria provided a principled basis for selecting a reduced subspace of principal components for subsequent analysis, while ensuring that the majority of the variance present in the original data was preserved. The loadings of each component were further inspected in order to identify the variables contributing most strongly to the principal directions.

The reduced representation was then clustered using K-Means. The K-Means algorithm partitions the dataset into k non-overlapping clusters $\{C_1, \ldots, C_k\}$ by minimizing the within-cluster sum of squares (WCSS):

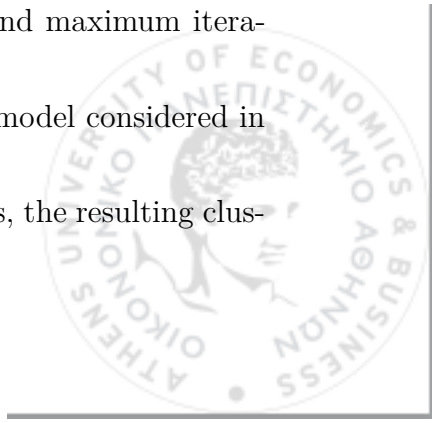
$$\min_{C_1,\ldots,C_k} \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2,$$

where μ_j denotes the centroid of cluster C_j. The optimization proceeds iteratively, alternating between assigning each data point to the nearest centroid and recomputing the centroids until convergence.

The choice of the number of clusters k was informed by both the Elbow method, which inspects the marginal reduction in WCSS Thorndike (1953), and internal validation metrics such as the Silhouette coefficient, which measures cluster cohesion and separation Kaufman, Rousseeuw (1990). For centroid initialization, the k-means++ strategy was adopted to improve convergence Arthur, Vassilvitskii (2007), while other hyperparameters such as the number of random restarts (n_init) and maximum iterations (max_iter) were set following standard practice.

Table B.7 summarizes the key hyperparameters of the K-Means model considered in this study.

Although unsupervised learning does not rely on labeled outcomes, the resulting clus-



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

ters were subsequently mapped back to the original feature space by applying the inverse PCA transformation to the cluster centroids. This facilitated interpretation of the typical shot characteristics within each cluster. Furthermore, post hoc analysis examined the distribution of expected goal values (xG) and conversion rates (goals/shots) across clusters. The purpose of this procedure was to evaluate whether the clustering process had effectively learned to partition shots into groups that correspond to different scoring probabilities. In other words, the analysis aimed to assess whether latent structures discovered in the feature space coincide with systematic differences in expected goals and conversion rates, thereby providing evidence that unsupervised learning can reveal meaningful patterns aligned with the overarching xG modeling framework.

4.3.2 Anomaly Detection with Isolation Forest

In order to identify rare or unusual shot patterns that may strongly influence scoring probability, anomaly detection was applied using the Isolation Forest algorithm Liu et. al. (2008). Isolation Forest is an ensemble-based method that detects anomalies by constructing a collection of random decision trees, where anomalies are expected to be isolated closer to the root nodes due to their scarcity and attribute irregularity. Unlike distance-based methods such as k-nearest neighbors, Isolation Forest does not rely on pairwise distances, which makes it particularly well suited for high-dimensional datasets where distance metrics often become less informative. In addition, the recursive partitioning induced by random trees allows the method to capture non-linear structures in the data without requiring an explicit dimensionality reduction step.

Formally, the anomaly score of an observation x is based on the average path length $E[h(x)]$ required to isolate it across all trees in the ensemble:

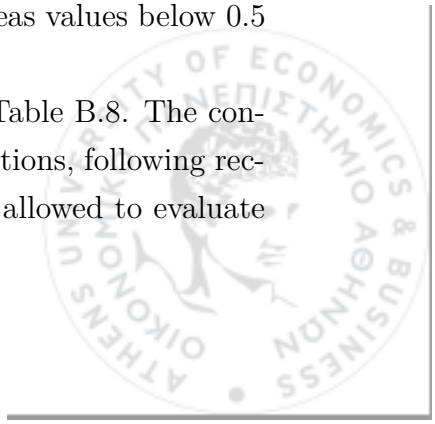
$$s(x, n) = 2^{-\frac{E[h(x)]}{c(n)}},$$

where n is the number of samples, $E[h(x)]$ the average path length, and $c(n)$ a normalization factor representing the average path length in a binary search tree with n samples. The normalization factor is given by:

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n},$$

where $H(i)$ denotes the i-th harmonic number, $H(i) \approx \ln(i) + \gamma$ with γ being the Euler–Mascheroni constant. Scores closer to 1 indicate anomalies, whereas values below 0.5 correspond to normal points.

The key hyperparameters of Isolation Forest are summarized in Table B.8. The contamination parameter was varied to explore different anomaly proportions, following recommendations in the literature for imbalanced data scenarios. This allowed to evaluate



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

the robustness of anomaly detection at different thresholds.

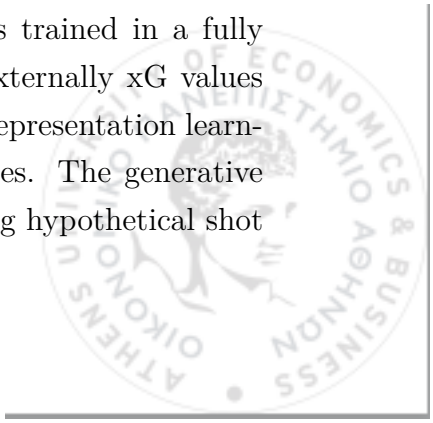
Detected anomalies were mapped back to the original feature space and compared against normal observations in terms of distance, angle, expected goals, and goal conversion rate. This procedure aimed to assess whether the anomalies highlighted by the model correspond to qualitatively distinct and rare shot patterns, such as close-range attempts from unusual angles, headers under pressure, or counter-attack situations. The overarching purpose of this analysis was to highlight precisely those patterns that generate high-quality and dangerous scoring opportunities, and to evaluate whether these rare events not only exhibit elevated xG values, but also maintain equally high conversion rates. In this way, anomaly detection was leveraged as a tool to uncover rare but decisive shot types that may play a disproportionate role in football match outcomes.

4.4 Causal Reasoning Approach

Beyond descriptive and predictive modeling, the study integrates a causal inference perspective through the implementation of a Counterfactual Autoencoder. This type of analysis is designed to learn a compact and disentangled latent representation of shot events, capturing the underlying relationships between contextual features and positional attributes. Trained in an unsupervised manner, the model operates without direct access to the goal outcome during training, focusing instead on reconstructing shot characteristics through a reduced latent space. Once trained, the latent representation enables the generation of counterfactual scenarios, hypothetical and minimally altered versions of a given shot, that simulate how specific feature changes could influence the expected scoring probability. By reintegrating these reconstructed samples into the supervised xG models, the framework moves from correlation-based analysis toward causal interpretation, identifying the controllable factors that most effectively improve shot success. This causal layer bridges the gap between data-driven modeling and tactical prescription, offering actionable insights into the mechanisms that drive goal creation.

4.4.1 Counterfactual Autoencoder

The purpose of the Counterfactual Autoencoder (CAE) is to learn a compact, interpretable, and disentangled latent representation of shot events, such that variations along specific latent dimensions correspond to meaningful, controllable changes in shot characteristics. Unlike conventional supervised xG models, the CAE is trained in a fully unsupervised fashion: it has no access to the goal outcome or to externally xG values provided, during training. These targets are only reintroduced after representation learning to probe the latent space and to generate counterfactual samples. The generative mechanism of the CAE facilitates causal-style reasoning by producing hypothetical shot



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

instances that correspond to the smallest feasible perturbations in feature space leading to an increased estimated probability of goal conversion. Such counterfactual reasoning frameworks have been extensively applied in representation learning and causal inference Louizos, Shalit, Mooij, Sontag, Zemel, Welling (2017), Kingma, Welling (2014), Kusner, Loftus, Russell, Silva (2017). The underlying structure of an autoencoder is illustrated in Figure 4.4. The network consists of two symmetrical components, an encoder and a decoder, connected through a compact latent space. The encoder compresses the input shot features into a low-dimensional latent representation, while the decoder reconstructs the original input from this encoded form. By enforcing this bottleneck, the model learns to capture only the most salient and generalizable aspects of shot characteristics. In the context of the Counterfactual Autoencoder, this latent space becomes the foundation for causal exploration. Small, interpretable perturbations along latent dimensions can be decoded into synthetic shot variations reflecting alternative conditions, illustrating how minimal feature changes might alter the expected goal probability.

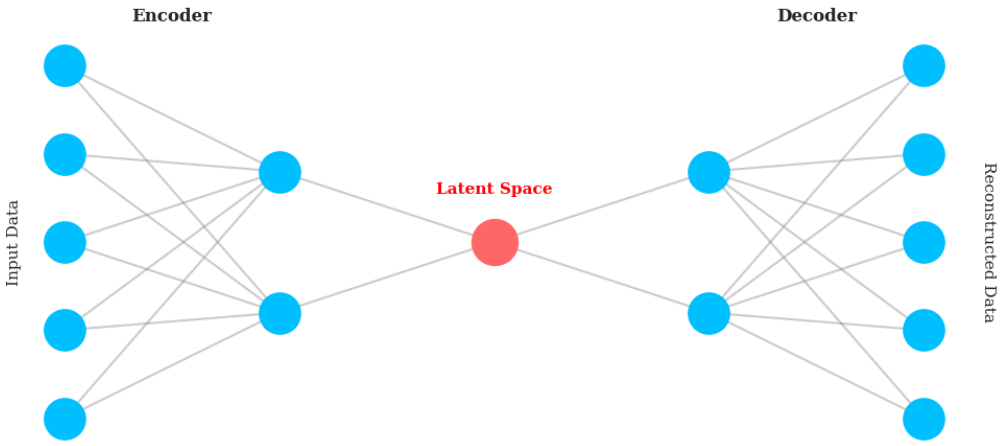


Figure 4.4: Schematic representation of an Autoencoder architecture

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i, s_i)\}_{i=1}^{n}$ denote the collection of n shot events, where:

- $\mathbf{x}_i \in \mathbb{R}^p$ is the feature vector representing the shot's spatial, kinematic, and contextual properties,
- $y_i \in \{0, 1\}$ is the binary outcome variable indicating whether the shot resulted in a goal (`is_goal`),
- $s_i \in [0, 1]$ is the externally estimated expected-goals value (`shot_statsbomb_xg`).

During unsupervised training, both y_i and s_i are excluded. Let $X \in \mathbb{R}^{n \times p}$ be the

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

feature matrix defined as:

$$X = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix}.$$

The CAE is then trained solely on X, learning a latent mapping $f_\theta : \mathbb{R}^p \to \mathbb{R}^{d_z}$ and its inverse $g_\phi : \mathbb{R}^{d_z} \to \mathbb{R}^p$ that jointly minimize reconstruction error, while implicitly capturing low-dimensional structure in the data. The remaining feature set comprises exclusively descriptive numerical variables, while categorical attributes are assumed to have been one-hot encoded. Moreover, a verification step confirms that all retained features are numeric, otherwise the pipeline halts execution, preserving methodological rigor and reproducibility.

Formally, the model operates on the purified feature space

$$\{\mathbf{x}_i\}_{i=1}^n \subset \mathbb{R}^p, \quad p = \text{number of numeric variables after exclusion.}$$

To ensure unbiased model evaluation, the dataset is partitioned into disjoint index subsets:

$$\mathcal{I}_{\text{train}},\ \mathcal{I}_{\text{val}},\ \mathcal{I}_{\text{test}},$$

such that

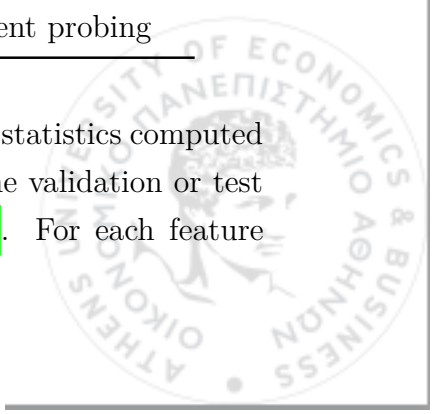
$$\mathcal{I}_{\text{train}} \cup \mathcal{I}_{\text{val}} \cup \mathcal{I}_{\text{test}} = \{1, \ldots, n\}, \qquad \mathcal{I}_a \cap \mathcal{I}_b = \emptyset,\ a \neq b.$$

The dataset was divided into training, validation, and test subsets, following common practice in unsupervised deep learning Goodfellow, Bengio, Courville (2016), ensuring sufficient data for model training, tuning, and evaluation. Let $X_{\text{train}}, X_{\text{val}}, X_{\text{test}}$ denote the respective feature matrices, while the held-out variables y and s are partitioned analogously but remain unused during training. The relative share of the total sample size, as shown in Table 4.3.

Table 4.3: Dataset partitioning scheme

Split	**Proportion (%)**	**Purpose**
Training	70	Fit scaler and train autoencoder
Validation	15	Early stopping and hyperparameter tuning
Test	15	Held-out evaluation and latent probing

All features are standardized to zero mean and unit variance using statistics computed exclusively on the training set, ensuring that no information from the validation or test sets influences model learning LeCun, Bottou, Orr, Müller (2012). For each feature



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

$j \in \{1, \ldots, p\}$, the empirical mean and standard deviation are defined as:

$$\mu_j = \frac{1}{|\mathcal{I}_{\text{train}}|} \sum_{i \in \mathcal{I}_{\text{train}}} x_{ij}, \qquad \sigma_j = \sqrt{\frac{1}{|\mathcal{I}_{\text{train}}| - 1} \sum_{i \in \mathcal{I}_{\text{train}}} (x_{ij} - \mu_j)^2}.$$

Each feature value is then normalized as:

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j + \varepsilon}, \quad \varepsilon > 0$$

where ε is a small stabilizing constant (e.g., 10^{-8}) to prevent division by zero. The resulting standardized matrices are denoted by

$$\tilde{X}_{\text{train}}, \quad \tilde{X}_{\text{val}}, \quad \tilde{X}_{\text{test}}.$$

Let $\mathbf{x} \in \mathbb{R}^p$ denote a standardized shot feature vector. The encoder f_θ maps $\mathbf{x}$ to the parameters of a Gaussian latent distribution:

$$(\boldsymbol{\mu}_z, \log \boldsymbol{\sigma}_z^2) = f_\theta(\mathbf{x}),$$

from which latent codes are sampled via the reparameterization trick:

$$\mathbf{z} = \boldsymbol{\mu}_z + \boldsymbol{\sigma}_z \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, I).$$

The decoder g_ϕ reconstructs the input as $\hat{\mathbf{x}} = g_\phi(\mathbf{z})$. Both encoder and decoder are fully connected networks with two hidden layers (128 and 64 units) and ReLU activations. The model is optimized with Adam (10^{-3} learning rate, batch size 256), minimizing the β–VAE loss:

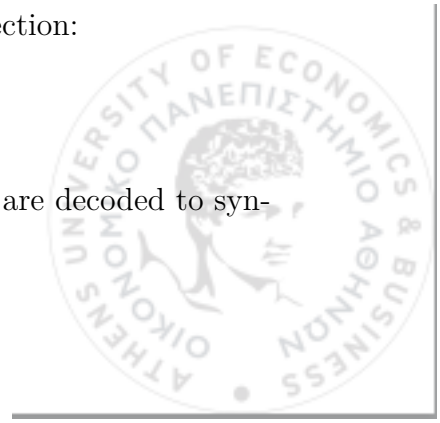
$$\mathcal{L} = \mathbb{E}\left[\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2\right] + \beta \, D_{\text{KL}}(q_\theta(\mathbf{z}|\mathbf{x}) \,\|\, \mathcal{N}(0, I)) \,.$$

The β–VAE is an unsupervised generative model extending the standard Variational Autoencoder (VAE) by introducing a regularization coefficient β that encourages disentangled latent representations. Early stopping based on validation reconstruction loss prevents overfitting.

After training, the encoder's mean output $\boldsymbol{\mu}_z$ provides the latent representation of each shot. Once the unsupervised model is converged, the outcome variables (xG and goal) are reintroduced to estimate semantic directions within the latent space. Counterfactual examples are produced by perturbing latent codes along the xG direction:

$$\mathbf{z}' = \mathbf{z} + \alpha \, \mathbf{w}_{xG},$$

with α controlling the traversal magnitude. The perturbed codes $\mathbf{z}'$ are decoded to syn-



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

thetic shot configurations $\hat{\mathbf{x}}'$, representing minimal, plausible changes in shot features that correspond to altered scoring potential. A post-processing constraint restores one-hot encoded categorical groups to ensure semantic validity of generated samples. Table B.9 summarizes the principal hyperparameters adopted for training the β–VAE and for implementing the counterfactual generation procedure.

The latent dimensionality $d_z = 8$ was chosen as a balanced configuration that captures sufficient variability in shot features while maintaining interpretability and stability in downstream regressions. Lower latent dimensions ($d_z < 6$) led to underrepresentation of meaningful shot variability, resulting in higher reconstruction loss, whereas larger latent dimensions ($d_z > 10$) produced redundant factors with low mutual information, in line with the Information Bottleneck principle Higgins et al. (2017) and and Alemi et al. (2017). A moderate regularization value $\beta = 4$ was adopted to encourage meaningful factor separation without compromising reconstruction quality, following similar heuristics discussed in Burgess et al.(2018) and Locatello et al. (2019).

In summary, the Counterfactual Autoencoder establishes an unsupervised generative framework for learning compact and interpretable latent representations of shot events. By disentangling representation learning from outcome supervision, the model captures intrinsic regularities of shooting behavior while enabling post-hoc causal interpretation. Linear projections fitted in the latent space define semantically meaningful directions along which variations correspond to systematic changes in expected goal probability and scoring likelihood. Counterfactual samples generated through controlled traversals along these directions allow the examination of causal mechanisms underlying shot success. Specifically, how minimal and interpretable modifications in spatial or contextual features alter the offensive danger of a given play. This methodology thus provides an interpretable bridge between generative modeling and causal performance analysis in football analytics.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Chapter 5

Results and Findings

5.1 Experiments

This section presents the experimental evaluation of the proposed framework, encompassing supervised, unsupervised, and causal inference components. A series of experiments was conducted to assess model performance in predicting goal probabilities, uncovering latent shot structures, and detecting rare events. The supervised models were evaluated through standard classification metrics, while unsupervised techniques provided structural and contextual insights through representation learning. Finally, the counterfactual analysis introduced a causal perspective, enabling the examination of hypothetical changes in match conditions and their effect on expected outcomes. Together, these experiments offer a comprehensive understanding of model behavior, interpretability, and tactical relevance.

5.2 Supervised Approach

5.2.1 Logistic Regression

Logistic regression was applied in order to estimate the probability that each shot attempt would result in a goal. Model training and evaluation were performed through five-fold stratified cross-validation, ensuring that class proportions were preserved across folds. The average evaluation metrics obtained are summarized in Table 5.1.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Table 5.1: Average evaluation metrics for Logistic Regression

Metric	**Logistic Regression**
Log Loss	0.260
Brier Score	0.074
AUC	0.802
Accuracy	0.907
Precision	0.601
Recall	0.118
F1 Score	0.197

As shown in Figure A.4, the barplot summarizes the average performance of the logistic regression model across the five cross-validation folds, with error bars indicating one standard deviation. Where the error bars are narrow, the results are highly consistent across folds, which reflects the stability of the model. This is evident for metrics such as Log Loss, Brier Score, Accuracy, and AUC. Precision shows somewhat higher variability, likely due to the limited number of positive cases (goals), which makes this metric more sensitive to differences in fold composition. Recall and F1 Score remain consistently low but with limited variance, indicating that the model is stably underperforming in identifying goals. Overall, the plot confirms that while the model is stable and reliable in terms of calibration and overall accuracy, its ability to capture positive outcomes remains a significant weakness.

The results indicate that the model achieves good calibration of predicted probabilities, as evidenced by the low Log Loss and Brier Score, and it reaches a very high overall accuracy of 90.7%. In contrast, Recall is limited to 0.118 and the F1 Score to 0.197, which shows that the model struggles to identify positive cases (goals). The AUC value of 0.802 demonstrates reasonably good discriminative power, although this is not fully reflected in the balance between Precision and Recall. The ROC curve presented in Figure 5.1 further illustrates the model's ability to discriminate between positive and negative cases across different classification thresholds. The curve shows the trade-off between the True Positive Rate (Recall) and the False Positive Rate across varying thresholds, with an AUC of 0.802, indicating reasonably good discriminative ability despite the limitations observed in Recall and F1 Score. The False Positive Rate (FPR) represents the proportion of non-goal attempts that the model incorrectly predicts as goals.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

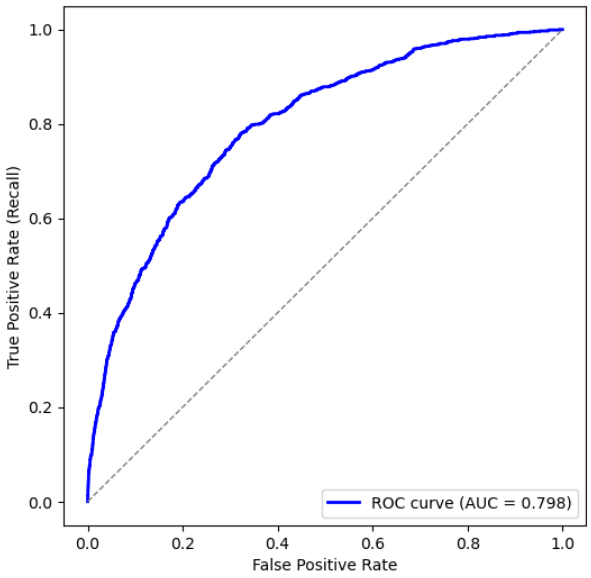


Figure 5.1: ROC curve for Logistic Regression

Moreover, Fiagure A.5 presents the confusion matrix derived from the evaluation. The model correctly classifies the majority of non-goal attempts, but it fails to capture a large proportion of actual goals, which explains the low Recall score. So as to facilitate interpretation, Table 5.2 presents the key regression results, focusing on the predictors that exhibit statistically significant and substantively meaningful effects on scoring probability. Variables with limited statistical significance or unstable estimates, including categorical indicators referring to body part name and play pattern name, have been omitted from this table to maintain clarity and interpretability. However, the complete regression output including all estimated coefficients is provided in Table B.11, ensuring full transparency of the modeling process. The estimated coefficients provide further insight into the determinants of goal probability.

The `distance` has a negative and statistically significant effect (coef $= -0.107$, $p < 0.001$), consistent with the intuition that the likelihood of scoring decreases as the distance from goal increases. Conversely, the `angle` exhibits a strong positive effect (1.628, $p < 0.001$), showing that more favorable shooting angles increase the chance of scoring. Being `under_pressure` is negatively associated with goal probability (-0.239, $p < 0.001$), in line with the expectation that defensive pressure reduces shooting success. The variable `shot_one_on_one` has a significant positive effect (0.758, $p < 0.001$), reflecting the higher likelihood of scoring in one-on-one situations with the goalkeeper. Finally, `shot_deflected` shows a very strong positive association (2.125, $p < 0.001$), suggesting that deflections can substantially increase the probability of a goal. Many categorical predictors did not yield statistically significant coefficients. This is likely due to the limited number of observations within certain categories rather than multicollinearity. As shown in Table B.10, all continuous variables had Variance Inflation Factor (VIF) values

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

below the commonly accepted threshold of 5, suggesting that multicollinearity is not a concern in this dataset. It is also worth noting that the optimization produced a convergence warning. This may indicate the need for additional iterations or the use of an alternative solver in order to obtain more stable coefficient estimates. The Pseudo R^2 of 0.184 is relatively low in absolute terms; however, it is considered acceptable for data characterized by a high degree of randomness, such as scoring attempts. In any case, the other evaluation metrics provide a more substantive assessment of the model's predictive performance.

Table 5.2: Key logistic regression results

	Coefficient	Std. Error
distance	-0.107***	(0.005)
angle	1.628***	(0.081)
under_pressure	-0.239***	(0.038)
shot_one_on_one	0.758***	(0.057)
shot_deflected	2.125***	(0.089)
minute	0.001	(0.001)
shot_first_time	-0.035	(0.043)
Observations	50,677	
Pseudo R^2	0.184	

Note: *p<0.1; **p<0.05; ***p<0.01

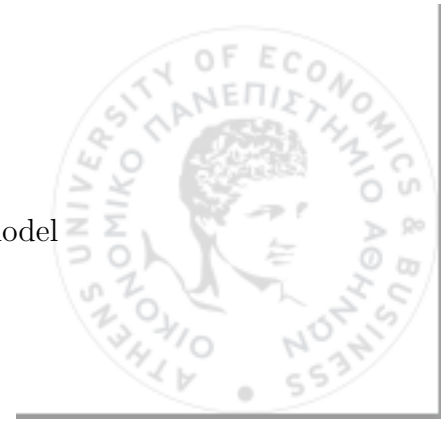
In sum, logistic regression provides interpretable and statistically sound insights into the factors affecting the probability of scoring. However, its limitations in detecting positive cases highlight the potential need for more flexible machine learning approaches.

5.2.2 Decision Trees

The performance of the Decision Tree model was evaluated using five-fold cross-validation. The training procedure employed the Gini index as the splitting criterion, with the maximum depth set to five. The evaluation metrics are summarized in Table 5.3.

Metric	**Decision Tree**
Log Loss	0.272
Brier Score	0.076
AUC	0.782
Accuracy	0.906
Precision	0.576
Recall	0.098
F1 Score	0.168

Table 5.3: Evaluation results of the Decision Tree model



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

The results indicate that the Decision Tree achieves a high overall accuracy of 90.6%, although this figure is somewhat misleading as it largely reflects the predominance of the majority class, "No Goal." The model shows very low recall, reaching only 9.8%, which demonstrates its inability to adequately identify the minority class, "Goal." As a result, the F1 score is limited to 0.168, highlighting the poor balance between precision and recall. The AUC value of 0.782 suggests that, despite these limitations in classification performance, the model retains a reasonably good discriminative ability between the two classes when assessed through predicted probabilities. In addition, the Log Loss of 0.272 and the Brier Score of 0.076 indicate that the probability estimates are relatively consistent, though not optimal.

Overall, while the Decision Tree provides interpretable and transparent decision rules, its main weakness lies in detecting rare "Goal" cases. This limitation stems from the class imbalance, as the dominance of "No Goal" outcomes and the Gini criterion favor splits that prioritize the majority class. The restriction on tree depth further limits the model's ability to capture complex patterns, leading to overfitting towards the majority and highlighting the need for more advanced methods or class balancing techniques to improve recall without severely reducing accuracy.

In order to aid interpretation, Figure 5.2 displays the learned Decision Tree (max depth = 5). The root split occurs on shot distance, indicating that distance is the dominant determinant of outcome: very short attempts are treated differently from medium and long-range shots. On the right branch, distance > 18, leads to nodes with very low Gini impurity and predictions of No Goal, reflecting the rarity of successful long-range attempts. Within shorter and medium ranges, the tree further splits on the body part used for the shot, distinguishing headers from non-headers, and for longer shots on whether the attempt was deflected, suggesting that both the shooting body part and deflections modulate the baseline effect of distance. Notably, most terminal nodes still predict No Goal, which is consistent with the global class imbalance observed in the data.

The depth was restricted to five levels as a form of pre-pruning, in order to control model complexity, improve generalization, and preserve readability. This early-stopping choice prevents the tree from growing idiosyncratic branches that overfit rare patterns, while keeping the learned decision rules transparent.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

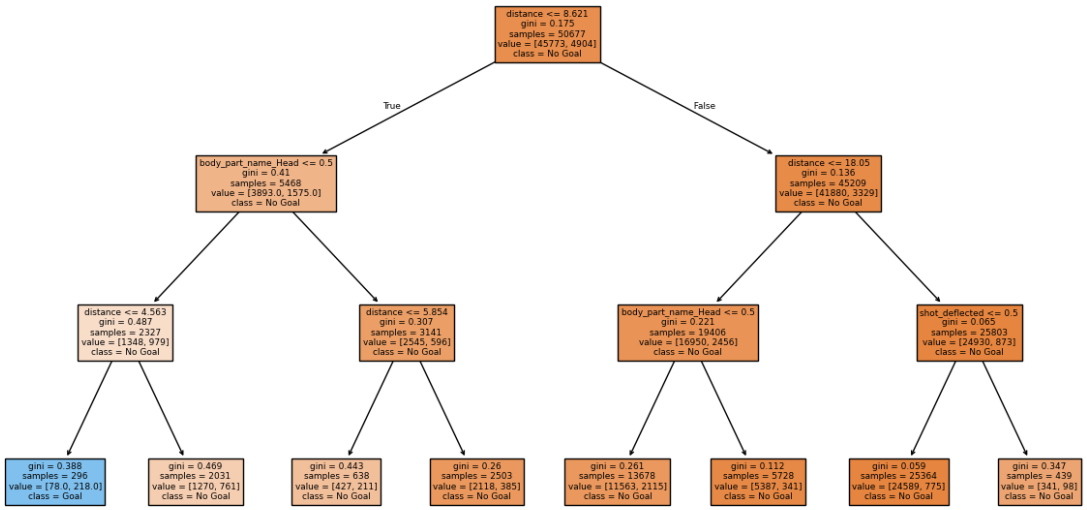


Figure 5.2: Decision Tree structure

The tree illustrates how variables such as shooting distance, the use of the head, and whether the shot was deflected influence the likelihood of scoring. Long-range attempts are almost invariably classified as No Goal, confirming the strong negative effect of distance. For shorter attempts, the shooting body part becomes relevant: headers tend to have a lower probability of conversion compared to shots taken with the foot. In certain cases, deflections slightly alter the prediction, but the overall tendency still points to a missed attempt. In summary, the Decision Tree highlights distance as the dominant determinant, while other attributes, such as headers or deflections, act as secondary modifiers. Given the rarity of goals in the dataset, the majority of leaf predictions remain biased toward the 'No Goal' class.

The comparison between the two most commonly used impurity criteria, namely the Gini index and Entropy, is illustrated in Figure A.6, showing the imputy curves as a function of the probability of belonging to the "Goal" class. Both measures reach their maximum value when the probability of belonging to each of the two classes is equal. Consequently, their scales differ: the Gini index ranges up to 0.5, while Entropy reaches a maximum of 1. Despite these differences, in practice they usually lead to very similar tree structures, with the Gini index being more frequently adopted due to its computational efficiency.

Furthermore, Figure A.7 presents the cost–complexity pruning path of the Decision Tree. As the value of the regularization parameter ccp_α increases, the complexity of the tree is progressively reduced through the removal of branches. This pruning process leads to simpler and more interpretable models, at the expense of a gradual increase in total impurity. In the present case, the curve shows that for very small values of ccp_α (close to zero), the impurity remains low but the tree retains a high level of complexity, which

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

risks overfitting. As ccp_α increases, larger jumps in impurity occur when entire subtrees are removed, resulting in a clear simplification of the model. The relatively smooth progression of the curve up to approximately $ccp_\alpha \approx 0.002$ indicates that several pruning steps can be applied with only limited cost in impurity. Beyond this point, however, pruning leads to sharper increases in impurity, suggesting that the model begins to lose predictive power. From a practical perspective, this diagram highlights how the tuning of ccp_α governs the trade-off between model fit and generalization. Selecting an intermediate value allows the construction of a smaller and more stable tree, which better generalizes to unseen data compared to the fully grown tree.

In summary, Decision Trees deliver reasonably good results but remain less competitive compared to more advanced ensemble methods. Nevertheless, their strength lies in the balance between interpretability and simplicity, particularly when pruning is applied to enhance generalization and reduce model instability. This interpretability makes them especially valuable in football analytics, serving as a bridge between statistical modeling and applied tactical analysis. Their tree-structured representation allows analysts and coaches without specialized knowledge in machine learning to understand how specific factors shape shot outcomes, thereby facilitating the explanation of match situations and highlighting recurring patterns that give rise to high-quality scoring opportunities. In applied settings, such clarity and communicability can be as important as raw predictive accuracy.

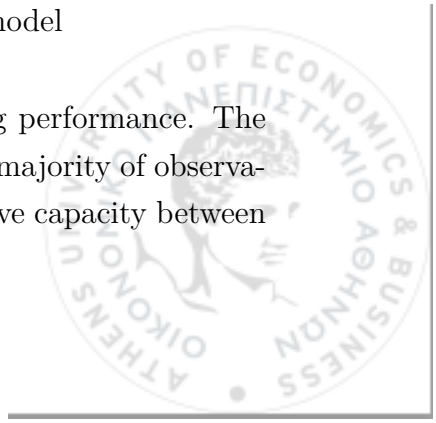
5.2.3 Random Forests

The performance of the Random Forest model was evaluated using five-fold cross-validation. The results are summarized in Table 5.4.

Metric	**Random Forest**
Log Loss	0.375
Brier Score	0.116
AUC	0.861
Accuracy	0.830
Precision	0.276
Recall	0.439
F1 Score	0.339

Table 5.4: Evaluation results of the Random Forest model

The results show that the Random Forest achieves overall strong performance. The accuracy of 83% indicates that the model correctly classifies the vast majority of observations, while the AUC value of 0.861 demonstrates a high discriminative capacity between



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

the two classes. The Log Loss of 0.375 and the Brier Score of 0.116 confirm that the probability estimates produced by the model are well calibrated and can be interpreted with confidence. Of particular interest is the assessment of precision and recall. The model attains a precision of 27.6% and a recall of 43.9%. This indicates that the Random Forest is able to identify almost half of the positive cases that correspond to goals, although a substantial proportion of its positive predictions do not materialize. The F1 score, equal to 0.339, reflects the overall balance between accuracy and detection capability, capturing the combined effect of moderate precision and enhanced recall.

The Precision–Recall curve, in Figure 5.3, provides a more detailed view of the model's behavior. Its shape illustrates the expected trade-off between precision and recall: as the ability to identify positive cases increases, the accuracy of those predictions decreases. The Average Precision (AP), with a value of 0.292, summarizes quantitatively the performance of the Random Forest and confirms its ability to detect positive cases satisfactorily, even though some of these predictions correspond to false positives.

The continuous curve represents the performance of the Random Forest across different thresholds, showing how precision decreases as recall increases. The shaded area beneath the curve serves as a supportive visual element that emphasizes the region covered by the model's performance and facilitates the interpretation of the overall results. The dashed line represents the no-skill classifier, which predicts solely according to the proportion of the positive class in the dataset. In practice, this line corresponds to the probability of scoring if predictions were made randomly in proportion to the frequency of positive cases. The fact that the Random Forest curve lies significantly above this baseline confirms the superiority of the model compared to random guessing. The Average Precision expresses the mean value of precision across the full range of recall and serves as the analogue of the AUC for Precision–Recall analysis. Higher values of this measure indicate a more favorable balance between precision and recall, which makes the metric particularly useful for the evaluation of models under conditions of severe class imbalance. In the present context, a higher Average Precision reflects the ability of the model to capture a larger proportion of goal attempts while limiting false detections, which highlights its practical value for expected goals estimation.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

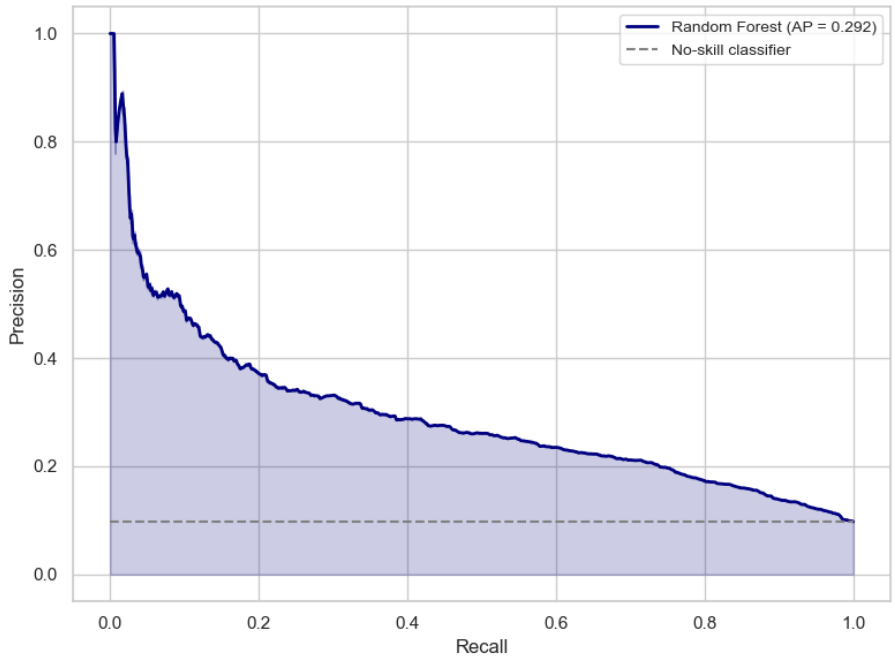


Figure 5.3: Precision–Recall curve of the Random Forest model

Overall, the Random Forest exhibits the characteristics of a stable and effective ensemble method. It demonstrates strong discriminative ability, provides reliable probability estimates, and shows an enhanced capacity to identify rare events. Despite the relatively modest precision, its ability to capture a larger share of positive cases makes the model particularly valuable for estimating scoring probabilities.

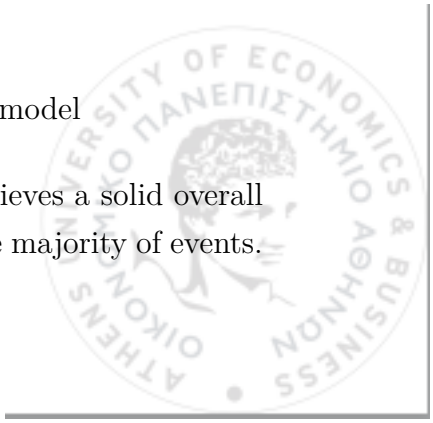
5.2.4 Gradient Boosting

The performance of the Gradient Boosting model was evaluated using five-fold cross-validation. The results are reported in Table 5.5.

Metric	Gradient Boosting
Log Loss	0.284
Brier Score	0.080
AUC	0.772
Accuracy	0.869
Precision	0.344
Recall	0.385
F1 Score	0.363

Table 5.5: Evaluation results of the Gradient Boosting model

The results indicate that the tuned Gradient Boosting model achieves a solid overall accuracy of 86.9%, demonstrating its capacity to correctly classify the majority of events.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

The AUC value of 0.772 reflects a moderate ability to discriminate between goal and non-goal outcomes, suggesting that while the model captures meaningful patterns, its ranking performance remains improvable. The relatively low Log Loss and Brier Score values confirm that the predicted probabilities are acceptably calibrated and consistent with the observed outcomes. Regarding the balance between precision and recall, the model attains a precision of 34.4%, meaning that approximately one in three predicted goals corresponds to an actual goal, and a recall of 38.5%, indicating that it successfully identifies a moderate portion of true positive cases. The resulting F1 score of 0.363 illustrates a more balanced performance compared to the untuned version, mainly due to the optimized decision threshold that improved sensitivity to rare goal events without severely compromising precision.

So as to enhance the understanding of Gradient Boosting, the Feature Importance diagram is presented in Figure 5.4. This result illustrates which features are used by the model to predict the probability of scoring and the relative weight of each one. In other words, the analysis quantifies the contribution of each predictor to the decision-making process of the model.

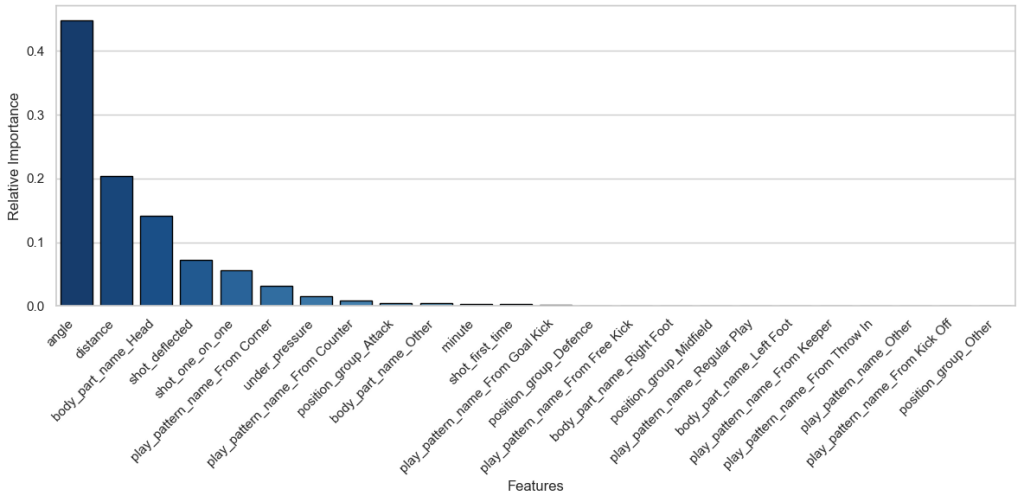


Figure 5.4: Feature Importance diagram of the Gradient Boosting model

The analysis demonstrates that the shot angle emerges as the most influential factor, with a substantially higher impact compared to all other variables. It is followed, with significant contribution, by the shooting distance and the information on whether the attempt was made with a header. Additional determinants of scoring probability include whether the shot was taken under defensive pressure, whether the ball was deflected on its trajectory towards the goal, and whether the attempt originated from a set piece situation. From a tactical perspective, the model captures patterns of play that are consistent with football knowledge and practical experience. It is noteworthy that goals arising from counterattacks or from attempts executed with body parts other than the head or foot, although relatively infrequent in the dataset, are still recognized as factors

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

influencing the probability of success. This indicates that Gradient Boosting does not only rely on the most frequent patterns but also identifies rarer tactical circumstances that can prove decisive.

Further interpretation is provided by the Partial Dependence Plots (Figure 5.5), which illustrate the marginal effect of the two most important features on the probability of scoring a goal. The curve corresponding to the variable *angle* indicates that, as the shooting angle relative to the goal increases, the probability of a successful attempt also rises. In contrast, the curve for the variable *distance* exhibits a clearly decreasing trend, suggesting that greater shooting distance reduces the likelihood of scoring. These findings are consistent with football theory and practical experience, confirming that the model captures essential game patterns. Moreover, the black vertical lines at the base of the plots represent the actual observations of the features in the training sample. In the absence of such indications in certain regions of the axis, the model would resort to predictions outside the training domain, thereby reducing the reliability of the estimates. In this case, however, the presence of lines across almost the entire range of values demonstrates that the model's predictions are grounded in real data and are therefore considered largely reliable.

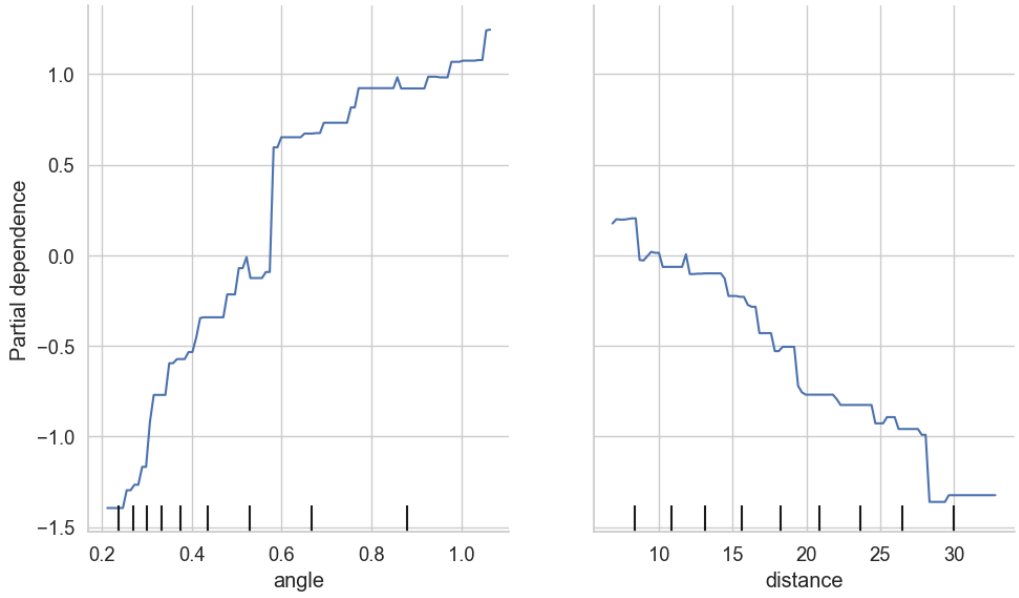


Figure 5.5: PDP for the two most important features of the Gradient Boosting model

Overall, the tuned Gradient Boosting model emerges as a robust predictive tool, offering well-calibrated probability estimates and high overall accuracy in identifying goal events. Compared to its untuned version, the model demonstrates a more balanced trade-off between precision and recall, achieving improved sensitivity to positive cases without a substantial loss in overall accuracy. Furthermore, its emphasis on identifying the most influential variables and the capacity to visualize their partial effects render Gradient Boosting not only accurate but also interpretable. This dual capability highlights its dis-

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

tinct value for xG analysis, where both predictive performance and model explainability are essential.

5.2.5 XGBoost

The evaluation of the XGBoost model was conducted using five-fold cross-validation, and the results are summarized in table 5.6.

Metric	**XGBoost**
Log Loss	0.284
Brier Score	0.080
AUC	0.778
Accuracy	0.863
Precision	0.330
Recall	0.404
F1 Score	0.363

Table 5.6: Evaluation results of the XGBoost model

The results indicate that the tuned XGBoost model achieves an overall accuracy of 86.3%, demonstrating its ability to correctly classify the majority of cases. The AUC value of 0.778 reflects a moderate discriminative capacity between the two classes, suggesting that while the model effectively distinguishes between goal and non-goal events, there is still room for improvement. The relatively low Log Loss and Brier Score confirm that the probability estimates produced by the model are satisfactorily calibrated and can be considered reliable for interpretation. In terms of classification balance, the model attains a precision of 33.0% and a recall of 40.4%, indicating a more equitable relationship between accuracy and completeness compared to the untuned version. The resulting F1 score of 0.363 reflects this improved trade-off, demonstrating that the tuned XGBoost model achieves a more balanced performance in identifying positive (goal) outcomes, particularly after optimizing the decision threshold.

For a deeper understanding of the model's behavior, feature importance metrics as well as Partial Dependence Plots were examined. In the feature importance diagram, shown in figure 5.6, it is observed that attempts executed with a header emerge as the most decisive factor, followed by the shooting angle at the moment of execution towards the goal. In addition, the shooting distance is found to have a significant contribution, along with contextual factors such as whether the shot was one-touch, whether the ball was deflected, and whether the attempt originated from a set piece situation such as a corner kick. Finally, spatio-temporal characteristics are also taken into account, since not only the player's role in the lineup but also the minute of the match in which the

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

shot occurred are found to contribute to the model. These results suggest that the model does not rely solely on fundamental geometric variables but also incorporates the broader tactical and contextual conditions of the attempt.

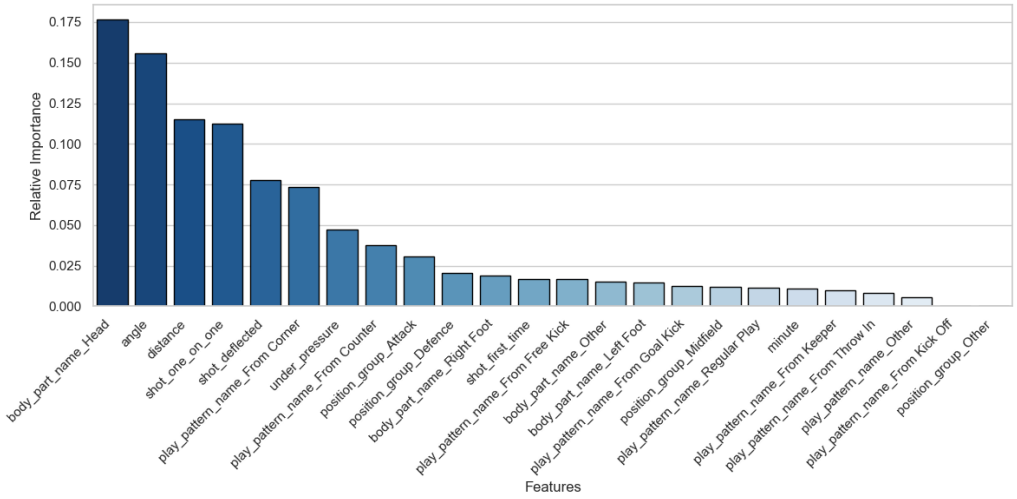


Figure 5.6: Feature Importance diagram of the XGBoost model

The analysis with Partial Dependence Plots(Figure 5.7) provides additional interpretability. For the variable referring to attempts executed with the head, a negative effect on scoring probability is observed compared to attempts taken with the foot, confirming that headers generally have a lower likelihood of success. Furthermore, with respect to the shooting distance from the goal, a strongly decreasing relationship is identified: as the execution distance increases, the probability of scoring decreases accordingly. This finding reflects well-established tactical patterns and aligns with the practical understanding of the game.

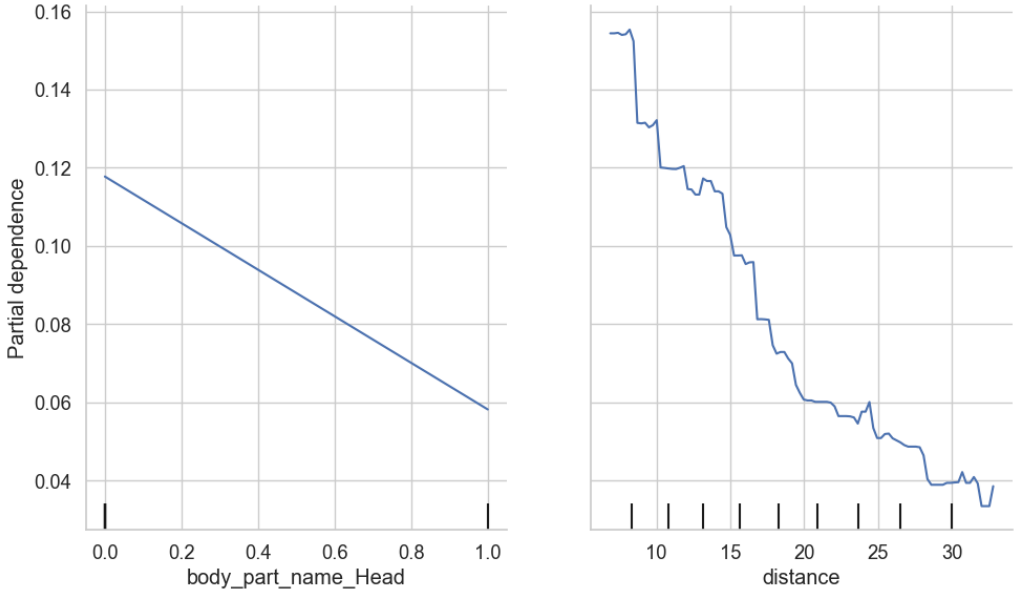


Figure 5.7: PDP for the two most important features of the XGBoost model

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Overall, the tuned XGBoost model proves to be an effective framework for the estimation of xG, offering well-calibrated probability estimates and strong predictive capability. Although its performance still reflects the inherent difficulty of detecting all goal events, the model maintains a reasonable balance between precision and recall, achieving a stable and interpretable outcome. Moreover, by highlighting the most influential factors contributing to goal likelihood and enabling the visualization of their partial effects, XGBoost provides both predictive accuracy and analytical transparency. These characteristics establish it as a valuable tool for football data analysis across both tactical and match-level contexts.

5.2.6 Neural Network

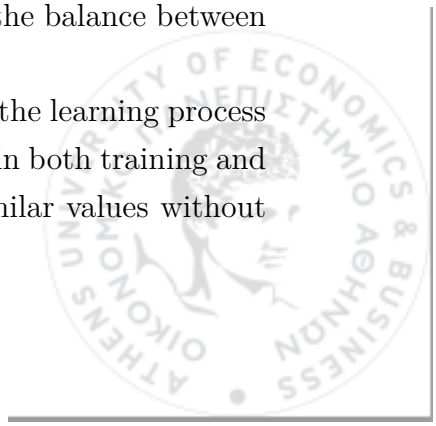
The performance of the Neural Network model was evaluated on the test set using standard probabilistic and classification metrics. The results are presented in Table 5.7.

Metric	**Neural Network**
Log Loss	0.264
Brier Score	0.075
AUC	0.792
Accuracy	0.906
Precision	0.595
Recall	0.080
F1 Score	0.140

Table 5.7: Evaluation results of the Neural Network model

The model exhibits an overall accuracy of approximately 90.6%, confirming its ability to correctly classify the majority of cases. The AUC value of 0.792 indicates a strong discriminative capacity, suggesting that the neural network effectively distinguishes between successful and unsuccessful shots. Similarly, the relatively low Log Loss and Brier Score demonstrate that the predicted probabilities are well calibrated, with limited deviation between predicted and true outcome probabilities. As with most classification models in imbalanced sports datasets, precision, 59.5%, surpasses recall, 8.0%, reflecting the challenge of correctly identifying all goal events while maintaining probabilistic reliability. The resulting F1 Score summarizes this trade-off, capturing the balance between precision and sensitivity.

The evolution of the training and validation loss functions during the learning process is illustrated in Figure A.8. The learning curves show a rapid decline in both training and validation loss during the initial epochs, stabilizing thereafter at similar values without



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

significant divergence. This indicates that the model achieved effective convergence without evidence of overfitting. The smooth parallel evolution of the two curves suggests that the implemented regularization mechanisms, such as dropout and early stopping, successfully prevented excessive memorization of the training data while maintaining consistent generalization on unseen samples. Moreover, the steady plateau of both curves at relatively low loss values suggests that the model is not underfitting, as it has adequately captured the underlying patterns in the data.

To enhance interpretability, SHAP (SHapley Additive exPlanations) analysis was applied to quantify the contribution of individual features to the model output. Figure A.9 presents the mean absolute SHAP values, which represent the average magnitude of each variable's impact on the predicted probability of scoring a goal. The results reveal that the *distance* from the goal is by far the most influential predictor, followed by the *angle* of the shot. These two geometric factors dominate the decision process, reflecting their fundamental role in determining scoring probability. Among the categorical and contextual variables, the body part used for the shot, such as right foot, left foot, or head, the type of play pattern, including actions from corners, counterattacks, or free kicks, as well as situational aspects like deflected shots and one-on-one situations, also show measurable contributions to the model's output.

The detailed SHAP summary plot in Figure 5.8 provides deeper insight into how each feature influences the model's predictions. The horizontal dispersion of SHAP values captures both the strength and direction of each variable's effect. For the *distance* feature, higher distances correspond predominantly to negative SHAP values (blue points on the left), meaning that as distance increases, the predicted probability of scoring decreases. Conversely, smaller distances (red points) are associated with positive SHAP values, indicating higher scoring likelihood. A similar but less pronounced pattern is observed for the *angle*, where smaller angles, more acute shot trajectories) are linked to lower goal probabilities. Furthermore, contextual variables such as *From Corner* and *One-on-One* show localized but significant positive contributions, implying that such scenarios increase the expected success rate when compared to standard open-play attempts.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

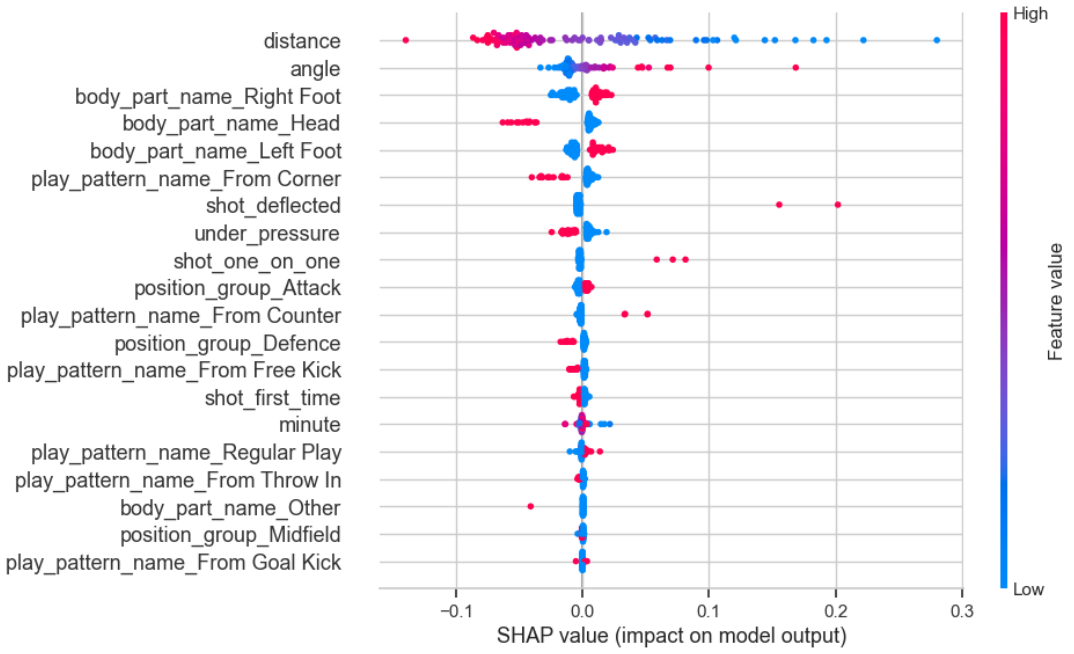


Figure 5.8: SHAP summary plot of the Neural Network output

Overall, the neural network demonstrates a robust capacity to learn meaningful non-linear patterns from shot-related data. Its learned feature structure aligns closely with domain knowledge in football analytics, where spatial geometry dominates the scoring probability, while contextual aspects exert secondary but interpretable influences. The integration of SHAP explainability provides additional transparency into the network's internal decision logic, ensuring that its predictive behavior remains interpretable and consistent with tactical intuition.

5.3 Unsupervised Approach

5.3.1 Principal Component Analysis and K-Means Clustering

To reduce the dimensionality of the dataset while retaining the majority of the variance, we applied PCA. Figure A.10 presents the cumulative explained variance, where the first 15 principal components capture 83.11% of the total variance Abdi, H., Williams, L. J. (2010).

In addition to the cumulative variance criterion, the eigenvalues and the explained variance ratio of each component were also examined in order to determine the optimal number of retained dimensions. Table B.12 summarizes the eigenvalues and corresponding variance explained by the first 24 principal components.

Table 5.8 highlights the most influential features for the first two principal components. From a football perspective, PC1 (12.33%) captures the dimension of shot execution. It is positively associated with headers and wider shooting angles, while it is negatively weighted by distance and right-footed attempts. This indicates that PC1 con-

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

trasts aerial or wide-angle efforts, which are typically harder to convert, with close-range or more controlled footed shots, which tend to be higher-quality chances. In other words, higher values of PC1 represent technically demanding finishing attempts, whereas lower values correspond to more favourable shooting opportunities. PC2 (7.79%), in contrast, represents the context of chance creation. It is strongly influenced by the position of the player, positive for attackers, negative for midfielders and defenders, and by the type of play pattern, such as corners and regular open play. This suggests that PC2 differentiates between shots arising from specialist attacking roles and structured set-pieces versus those taken by deeper-lying players in open play situations.

Feature	**PC1 Loadings**
Body part: Head	0.472
Angle	0.446
Distance	-0.442
Right Foot	-0.290
From Corner	0.287

Top features for PC1

Feature	**PC2 Loadings**
Attack Position	0.675
Midfield Position	-0.445
Defensive Position	-0.324
From Corner	-0.275
Regular Play	0.186

Top features for PC2

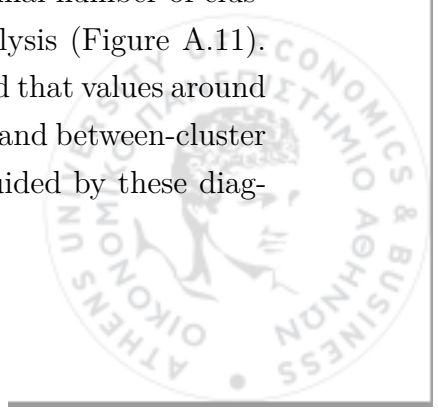
Table 5.8: Most influential features for the first two principal components

Together, as summarized in Table 5.9, PC1 and PC2 disentangle two fundamental dimensions of shot-taking: the technical difficulty of execution and the tactical origin of the attempt. These latent factors not only reduce data complexity but also align with intuitive football interpretations of chance quality.

Principal Component	**Interpretation**
PC1 (Shot Execution)	High: Headers, wide angles, technically demanding shots Low: Close-range, controlled footed shots, favourable angles
PC2 (Context of Creation)	High: Shots by attackers, structured play Low: Shots by midfield/defence, open play situations

Table 5.9: Compact interpretation of the first two principal components

After PCA projection, K-Means clustering was applied. The optimal number of clusters was assessed using both the Elbow method and Silhouette analysis (Figure A.11). While neither method provided a uniquely sharp cutoff, both suggested that values around $k = 5$ strike a reasonable balance between within-cluster compactness and between-cluster separation. Therefore, the choice of $k = 5$ was made empirically, guided by these diagnostics as well as interpretability considerations.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Clusters were visualized through both two-dimensional and three-dimensional projections of the PCA space. In the two-dimensional view based on PC1 and PC2 (Figure A.12), the separation between clusters appears less distinct, as several groups overlap one another. This is expected, since the clustering was performed in a 15-dimensional space, whereas the projection compresses the information into only two components.

As a result, part of the variance that differentiates clusters is inevitably lost in this representation. When moving to the three-dimensional visualization (Figure 5.9), the boundaries between groups become slightly clearer, although some overlap remains. This again reflects the complexity of the data, where clusters are better distinguished in the full multi-dimensional space rather than in low-dimensional projections. Nevertheless, these visualizations provide intuitive insight into the relative spread and interaction of the clusters.

Figure A.13 illustrates the distribution of shots across the five clusters. The sample sizes are highly heterogeneous, with certain clusters containing a large proportion of observations while others are represented only by a small minority. Such imbalances likely reflect underlying structural patterns in the data, where some types of attempts occur frequently whereas others are considerably rarer. This uneven distribution suggests that certain clusters capture dominant shooting behaviours, while others represent more specialised or less common situations.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

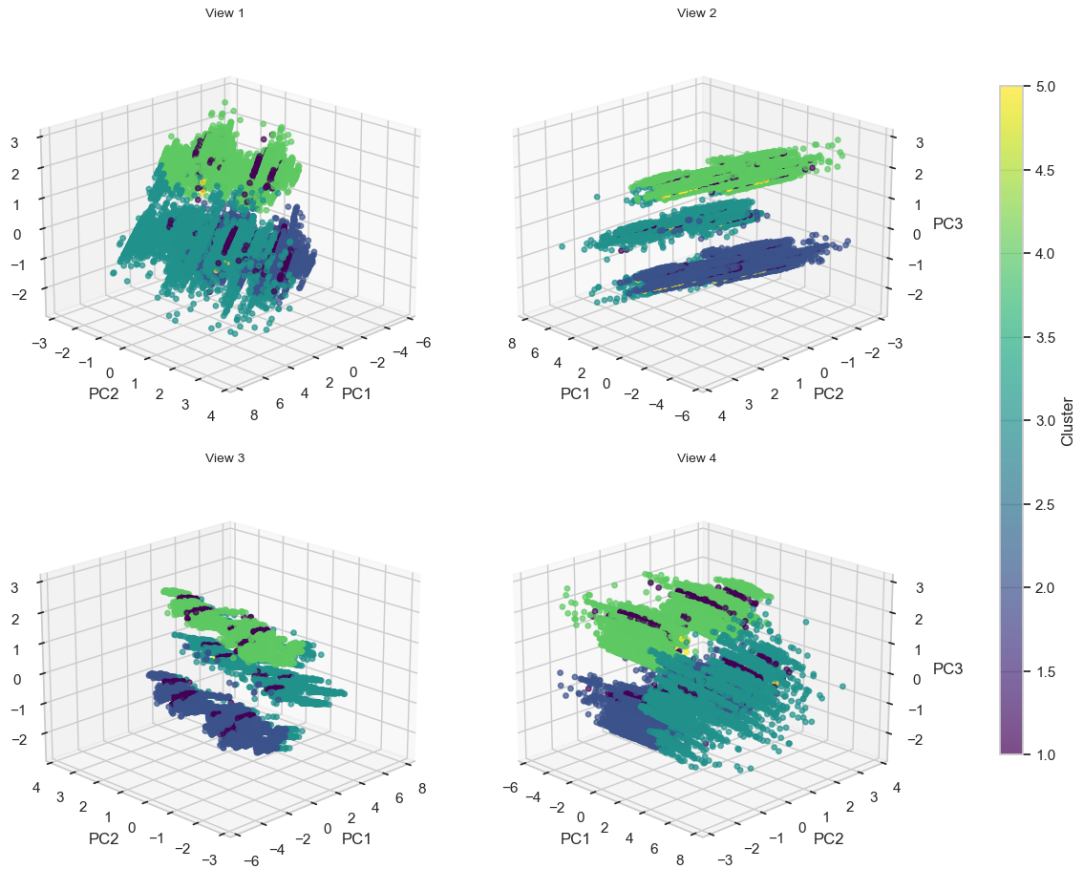
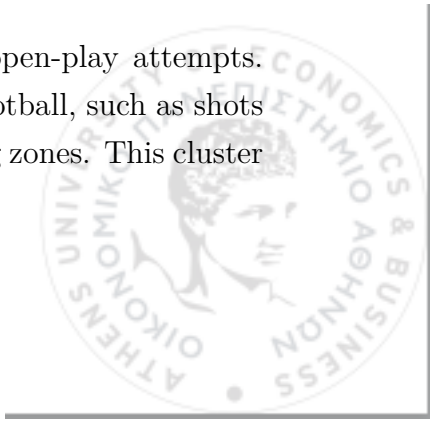


Figure 5.9: K-Means clustering projected on 3D from multiple perspectives

To interpret the clustering results in football terms, the centroids were projected back from the PCA space to the original feature space. This procedure allows us to recover the average profile of shots that belong to each cluster, expressed directly in terms of observable match characteristics such as distance, angle, body part, play pattern, and player position. In other words, the centroid of each cluster, as presented in Table B.13, represents the prototypical shot profile for that group, summarising its defining characteristics. This part of the analysis highlights distinct and interpretable patterns, further elaborated in the following discussion.

- **Cluster 1:** Composed of standard mid-range footed shots, often taken under moderate pressure and from central or slightly wide positions. This cluster reflects "average" shooting scenarios that form the backbone of general play but do not belong to specialised categories.
- **Cluster 2:** The largest cluster, dominated by right-footed open-play attempts. These include the typical finishing situations encountered in football, such as shots following through balls, cut-backs, or ball recoveries in attacking zones. This cluster represents the mainstream category of shooting behaviour.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

- **Cluster 3:** Defined by headed attempts from crosses and corners, usually taken under defensive pressure and from tight angles inside the box. This cluster corresponds to aerial duels and set-piece scenarios, where conversion is highly dependent on positioning and timing.

- **Cluster 4:** Consists primarily of left-footed strikes, often associated with players on their stronger side, for instance left-footed forwards or inverted wingers cutting in. These attempts are more technical in nature and reflect a subgroup of specialised finishing profiles.

- **Cluster 5:** A very small group of rare and atypical attempts, including one-touch finishes, unusual play patterns, or opportunistic efforts taken under extreme match conditions. This cluster highlights exceptional shot types that occur infrequently but are distinct from mainstream finishing.

Figure 5.10 compares the average xG and conversion rate across the five clusters. A key methodological insight is that the clustering model was trained exclusively on descriptive shot features, without incorporating xG values or goal outcome information. Remarkably, when these external performance indicators are reintroduced, the resulting clusters align with intuitive footballing patterns.

Cluster 5, despite being extremely rare in frequency, emerges as the group with the highest average xG and conversion rate, capturing elite-quality finishing scenarios. In contrast, Clusters 1 and 2, which encompass the bulk of attempts, are associated with routine or speculative shots, characterised by lower xG and modest success rates. This mirrors the reality of match play, where the majority of shots stem from less advantageous positions or occur under defensive pressure, while only a minority represent clear-cut chances with a high probability of scoring. This result reinforces that clustering based on shot descriptors can recover tactically meaningful profiles of finishing situations, providing an interpretable lens on chance quality.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

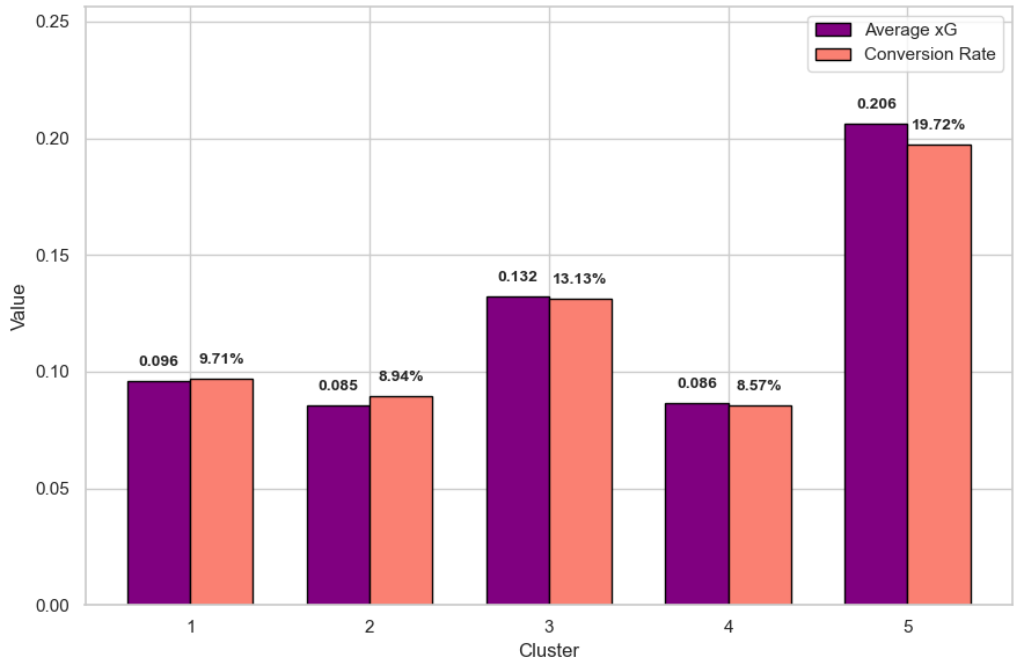


Figure 5.10: Comparison of average xG and conversion rates across clusters

Examining the distribution of shots across clusters, Cluster 5 emerges as a truly exceptional and rare category. It contains only 71 out of 50,677 attempts, a clear minority, yet its centroid profile sets it apart as the group of elite-quality chances. The defining features are substantially closer shooting distances, favourable central angles, markedly lower defensive pressure, a higher incidence of one–on–one situations and first-time finishes. Headers are scarce, reflecting a dominance of controlled footed shots rather than aerial duels. Play-pattern indicators show consistent negative values for structured restarts, combined with an extreme positive loading on "Other Play Pattern". This suggests that Cluster 5 chances often arise from highly dynamic or atypical phases of play: rebounds, goalkeeper parries, loose-ball recoveries, rapid counterpress regains, or unstructured scrambles in the box. In line with these descriptors, Cluster 5 records the highest average xG and the peak conversion rate, illustrating that when these rare contexts occur, they carry extraordinary scoring potential.

By contrast, Cluster 3 represents the most actionable high-quality category at scale. It comprises the 19% of the dataset and is the only common cluster to combine double-digit conversion with elevated xG (≈13%). Its centroid reveals a clear footballing identity: strong positive loading on headers, proximity to goal, but also favourable shooting angles and higher defensive involvement. Creation is strongly tied to set-piece or crossing situations, while Regular Play is less prominent. In practical terms, Cluster 3 captures aerial finishing in congested penalty-area environments, where the decisive factors are delivery quality, timing of runs, and separation from markers. This type of chance, while more difficult to convert than Cluster 5, occurs at a scale that makes it strategically significant.

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Teams can systematically target this cluster through repeatable attacking mechanisms, such as structured set-piece routines, wide overloads and service, or attacking patterns designed to generate crossing opportunities. Training emphasis on micro-details, such as body orientation, jump timing, and contact technique, can further enhance efficiency in these pressured aerial duels.

Taken together, Cluster 5 defines the absolute ceiling of chance quality but is too rare to constitute a reliable offensive strategy, whereas Cluster 3 provides a scalable and reproducible pathway to above-average shot quality, with clear tactical levers for coaching and team design.

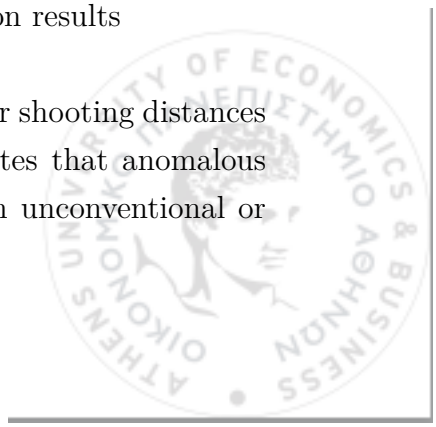
5.3.2 Anomaly Detection with Isolation Forest

Table 5.10 summarises the results of the Isolation Forest anomaly detection across three contamination thresholds, 1%, 2%, and 5%. At lower contamination levels, the model isolates only the most extreme and rare shot instances, while progressively higher thresholds expand the definition of anomalies to include a broader range of atypical attempts. As expected, the total number of detected outliers increases accordingly. However, the underlying behavioural patterns remain consistent across all settings. In every case, anomalous shots exhibit substantially shorter average distances, wider shooting angles, and considerably higher xG and goal conversion rates compared to the normal population. Moreover, the detailed distribution and feature breakdown of outliers across the different contamination levels are provided in Table B.14.

Metric	**1%**	**2%**	**5%**
# Outliers	507	1014	2534
Avg. Distance (Out)	9.54	11.12	12.83
Avg. Distance (Norm)	19.05	19.12	19.28
Avg. Angle (Out)	1.06	0.93	0.80
Avg. Angle (Norm)	0.47	0.47	0.46
Avg. xG (Out)	0.31	0.26	0.22
Avg. xG (Norm)	0.09	0.09	0.09
% Goals (Out)	39.3%	33.2%	25.8%
% Goals (Norm)	9.4%	9.2%	8.8%

Table 5.10: Summary of Isolation Forest anomaly detection results

The detected anomalies are systematically characterised by shorter shooting distances and wider angles compared to the normal population. This indicates that anomalous shots tend to originate from close proximity to goal, yet often from unconventional or



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

more acute angles, situations that are rare in frequency but potentially decisive in outcome. Correspondingly, the mean xG value among anomalies is over three times higher than that of normal shots, confirming that the model effectively isolates high-quality scoring opportunities. The goal conversion rate among anomalies reaches 39.3% at the 1% contamination level, compared to only 9.4% for the normal group, further validating the model's discriminative power.

Figure 5.11 illustrates the relative feature importances derived from permutation-based ΔAUC analysis. The results indicate that the most influential factors are situations initiated from set-piece restarts, particularly corners, often involving headers executed under defensive pressure. Spatial and execution-related features such as distance, angle, and first time shot also display substantial importance, indicating their combined role in shaping rare but high-value shooting contexts. These characteristics represent structured yet high-risk scenarios that differ substantially from open-play attempts and are therefore more likely to be isolated as anomalous by the model.

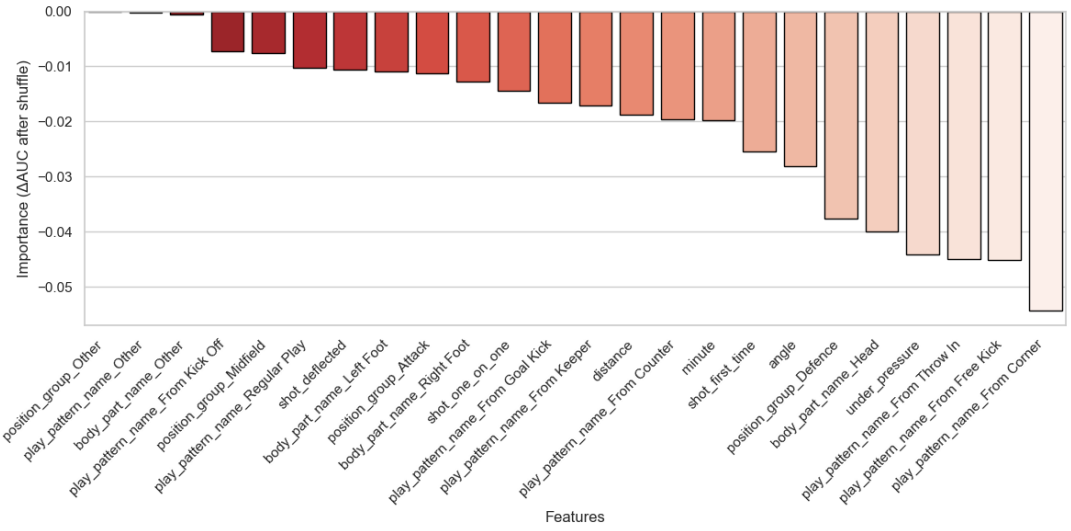


Figure 5.11: Feature importance (ΔAUC) for Isolation Forest

Taken together, the anomaly detection results reveal a distinct profile of finishing situations that deviate from the norm yet carry the highest offensive value. The Isolation Forest model consistently highlights close-range attempts that arise under exceptional or constrained match contexts, such as headers following corners or free kicks, one–on–one situations, and rebound opportunities emerging amidst defensive pressure inside the penalty area. These events represent structured or highly dynamic attacking moments rather than routine open-play shots. In essence, the anomalies correspond to rare but decisive actions where spatial advantage, quick decision-making, and situational chaos converge to produce disproportionately high xG and conversion outcomes. Within the broader shot landscape, they define the extreme upper tier of chance quality, the kind of opportunities that often determine the outcome of elite-level matches. In this

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

way, the model effectively highlights all underlying play patterns capable of generating not merely a high-quality shot on target, but an exceptionally clear and unrepeatable scoring opportunity.

To better understand how anomalies are distributed relative to the full dataset, two non-linear projections were employed: t-SNE and UMAP. The two-dimensional t-SNE representation, as illustrated in Figure 5.12, shows that anomalous shots (red) form compact and overlapping regions within the manifold rather than existing as completely detached edge cases. This indicates that these events share latent structural similarities with regular attempts, yet remain statistically rare in occurrence. In fact, anomalies are not confined to a single isolated zone but are dispersed across a wide portion of the shot space, often positioned on the periphery of multiple clusters. This spatial pattern captures the essence of what constitutes an anomaly in footballing terms: within the large variety of opportunities that can arise during play, most belong to recurring and well-defined contexts, while only a very small subset, those rare, high-impact situations, are singled out as truly exceptional by the model.

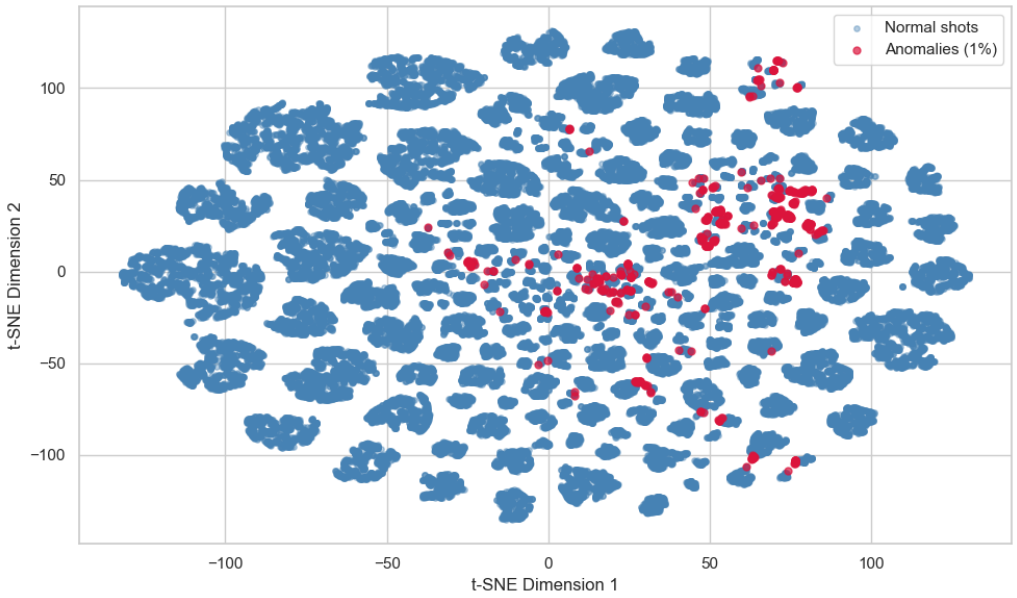


Figure 5.12: t-SNE visualisation of shot distribution (1% anomaly contamination)

Similarly, the UMAP projection, as illustrated in Figure A.14, reinforces and further clarifies this pattern. In contrast to t-SNE, which captures local density relationships, UMAP provides a clearer view of the global structure of the shot space. Here, anomalies appear as compact yet distinct subgroups embedded around the margins of larger shot clusters, outlining the boundary between typical and exceptional finishing situations. This spatial arrangement illustrates that anomalous attempts are not random noise, but coherent and contextually meaningful subsets of the data—events such as close-range headers, counter-attacking breaks, or one-on-one opportunities that occur infrequently

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

yet share a recognisable tactical signature. The clearer separation visible in the UMAP projection therefore emphasises the structural integration of anomalies within the broader landscape of shot creation, highlighting their role as rare but repeatable patterns of high offensive potential.

From a football analytics standpoint, these anomalous shots represent the upper tail of opportunity quality—situations that are statistically rare yet tactically decisive. They typically correspond to:

- Close-range finishes executed with minimal defensive pressure
- Aerial attempts originating from set-pieces such as corners or free-kicks
- One-on-one or rebound situations emerging from fast, transitional play

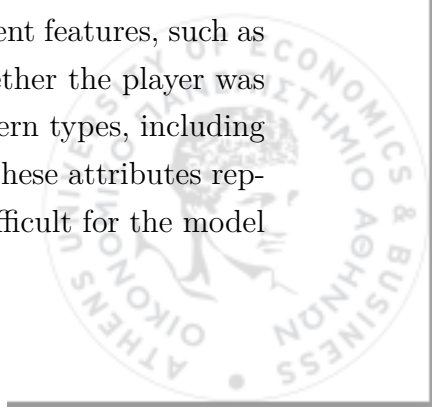
Collectively, these cases capture the kind of high-impact moments that often separate efficiency from dominance in attacking performance. The Isolation Forest model thus illustrates how unsupervised learning can extend traditional xG analysis by uncovering the underlying structures that give rise to exceptional chances. Rather than merely estimating shot probability, it detects the tactical and contextual irregularities that precede elite-quality finishing opportunities. In practical terms, this approach allows analysts and coaching staff to identify repeatable attacking patterns that are capable of generating disproportionately valuable outcomes, bridging data-driven evaluation with on-field tactical insight.

5.4 Causal Reasoning Approach

5.4.1 Counterfactual Autoencoder

The Counterfactual Autoencoder achieved a validation reconstruction error of approximately 1.02 (MSE), indicating a stable and non–overfitted generative mapping capable of reconstructing the underlying structure of shot events with reasonable precision. While the reconstruction error is higher than that of purely linear projections, it reflects the inherent trade–off between reconstruction fidelity and latent disentanglement imposed by the β–VAE regularization.

Figure A.15 reports the per–feature reconstruction error, providing insight into which shot descriptors are more challenging for the model to regenerate accurately. The highest reconstruction errors were observed for categorical or context-dependent features, such as the body part category labelled as “Other” and the indicator of whether the player was under pressure. Elevated errors were also found for certain play pattern types, including actions originating from kick-offs and regular open play situations. These attributes represent sparse or irregular match contexts that are inherently more difficult for the model



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

to encode within a smooth, continuous latent representation. In contrast, more structured numerical variable, such as shot angle and match minute, were reconstructed with considerably higher precision, indicating that the model captured spatial geometry and temporal dynamics of shot execution more effectively than complex contextual factors. From a football perspective, this distinction highlights that generic spatial geometry of attempts is easier to represent than contextual or psychological states, such as pressure or atypical play patterns.

The learned latent space exhibits a smooth, continuous manifold in which shot embeddings are clustered according to their underlying scoring potential. As shown in Figure 5.13, the projection of latent codes (first three principal axes) coloured by the external xG values reveals a clear gradient: low–xG attempts occupy the dense, compact core of the distribution, whereas higher–xG shots appear as sparser, peripheral points along specific directions. This spatial organization suggests that the CAE has internalized a representation of shot difficulty and conversion potential purely from unsupervised reconstruction objectives. In practical football terms, the model implicitly differentiates routine, low-threat attempts from rare, high-quality finishing opportunities without being told what "quality" means. It should be noted, however, that several data points appear visually overlapping due to the projection of the latent space from its original eight dimensions into a three-dimensional representation, which necessarily compresses and blends part of the encoded variability.

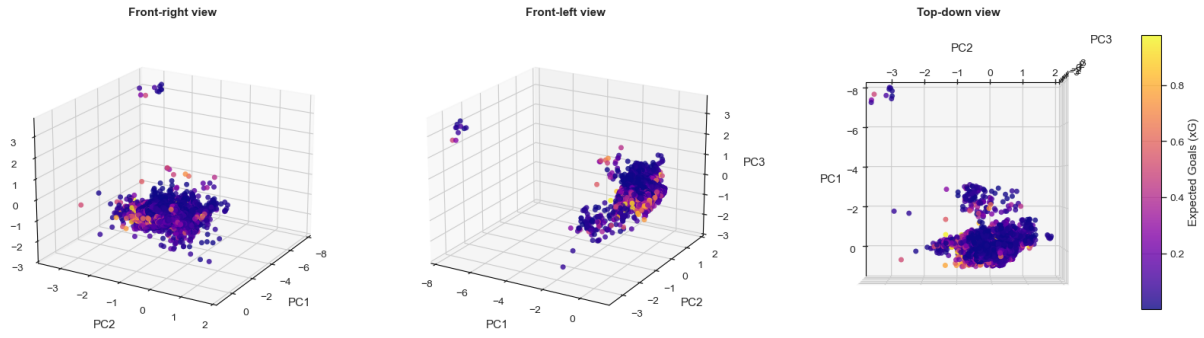


Figure 5.13: Latent space visualization of shot embeddings

The correlation matrix of latent variables (Figure A.16) indicates moderate but controlled independence across dimensions, with few strong pairwise correlations. This pattern reflects a partially disentangled representation, where certain dimensions capture orthogonal aspects of shooting behaviour, such as spatial geometry, body mechanics, and contextual play type, while others exhibit mild redundancy. The presence of structured but not perfectly independent factors aligns with the intended β–VAE regularization: promoting interpretability without oversimplifying the latent encoding. In football terms, these dimensions may correspond to distinct yet interacting traits of shot creation and execution, distance and angle, player positioning, or match context, each contributing to

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

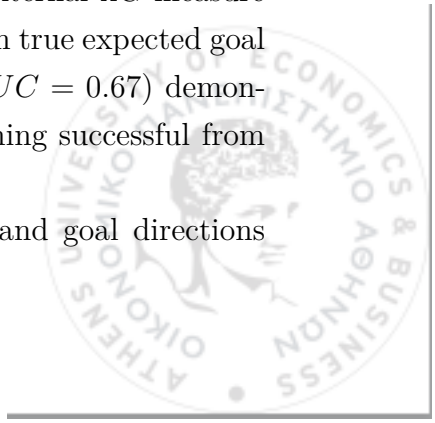
overall shot quality in an additive manner.

To assess whether the unsupervised latent space encodes football-relevant semantics, the external xG values and binary goal outcomes were reintroduced only after model training. Linear and logistic regressions fitted on the latent representations yielded $R^2_{\text{val}} = 0.19$ for expected goals and $AUC_{\text{val}} = 0.68$ for goal prediction. Although modest in absolute magnitude, these coefficients are notable given that the model was trained without any exposure to outcome information. An R^2 of 0.19 indicates that nearly one–fifth of the variance in external xG can be explained solely by the structure of the learned latent embeddings, demonstrating that the CAE successfully internalized underlying patterns of shot quality in an entirely unsupervised manner. In other words, the aim is not to reproduce or outperform an existing xG model, but to uncover the latent factors and directional dependencies through which variations in shot features causally affect expected goal value. Similarly, an AUC of 0.68 in discriminating goals from non–goals reflects a substantial degree of separability between effective and ineffective shots within the latent manifold—well above random chance and consistent with moderate predictive signal emerging from the learned representations.

Figure A.17 displays the directional loadings of each latent variable with respect to the derived xG and goal directions. Dimensions z_1 and z_5 show strong negative association with both targets, while z_6 and z_7 load positively, implying that traversals along these axes correspond to systematic increases or decreases in scoring likelihood. These directional patterns highlight that specific latent factors have become semantically aligned with offensive efficiency and conversion probability.

The distribution of projections onto the xG direction, shown in Figure A.18, follows an approximately Gaussian shape centered near zero. This symmetry indicates that most shots cluster around a neutral latent baseline, while only a limited fraction of attempts extend meaningfully toward the higher xG tail. In other words, the latent space encapsulates a continuous gradient of chance quality, with the majority of attempts corresponding to routine, low-probability situations and a sparse subset representing clear scoring opportunities. In practical terms, the histogram reveals that the autoencoder has learned an internal continuum of shot dangerousness, a latent xG scale that mirrors the underlying hierarchy of chance quality. It emerges from the inner product between each shot's latent representation and the learned xG direction within the latent space, effectively quantifying how strongly each embedding aligns with the semantic axis of increasing expected-goal probability. The observed correspondence with the external xG measure ($R^2 = 0.19$) confirms that this latent gradient aligns meaningfully with true expected goal variability, while the alignment with the goal outcome direction ($AUC = 0.67$) demonstrates that the same latent structure also contributes to distinguishing successful from unsuccessful attempts.

Finally, Figure 5.14 illustrates the orientation of both the xG and goal directions



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

within the latent manifold. The two vectors exhibit similar alignment, confirming that both outcome targets are governed by comparable latent mechanisms: shot geometry, positional context, and execution quality. From a football perspective, this alignment suggests that the CAE has learned an interpretable internal representation of what makes a shot dangerous: movement along these latent axes effectively traces the continuum from low quality, routine attempts to high quality, goal-bound situations.

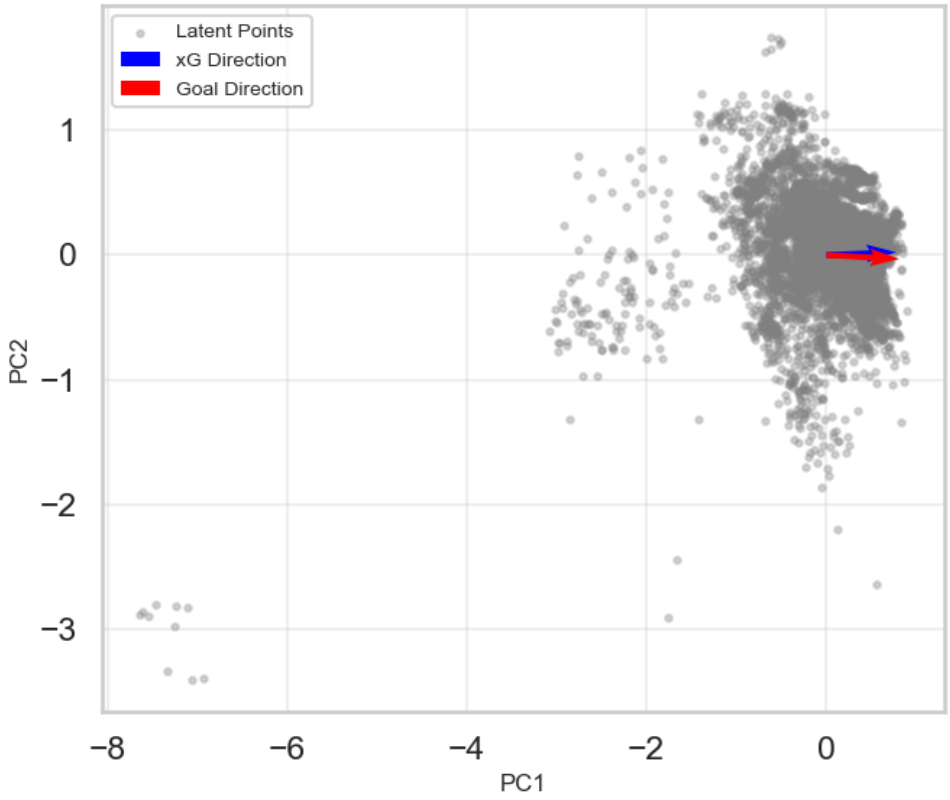
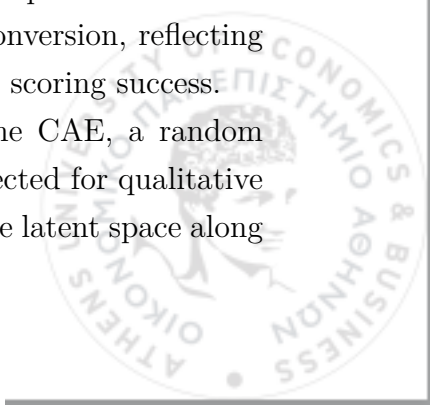


Figure 5.14: Visualization of the xG and goal directional vectors within the latent space

Although both arrows in appear to point towards a similar orientation within the latent manifold, the computed angular separation between the two vectors is approximately 23.30° in the full eight-dimensional latent space. This observation indicates that while the model has learned analogous underlying representations for shot danger and goal realization, these constructs are not perfectly aligned. The moderate angular offset suggests that the latent mechanisms governing expected goal estimation and actual goal outcomes share a common geometric and contextual basis, yet remain distinct in terms of stochastic execution and finishing efficiency. In essence, the CAE captures a coherent but non-degenerate relationship between chance creation and goal conversion, reflecting a realistic separation between the notions of opportunity quality and scoring success.

In order to further explore the causal structure captured by the CAE, a random representative shot instance from the test set (Shot #48720) was selected for qualitative analysis. The counterfactual generation process involves traversing the latent space along



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

the learned xG direction, decoding each perturbed point back to the feature space while restoring semantic validity of categorical variables. For this example, the latent representation of the shot was progressively displaced by three incremental steps, $\alpha = 0.5$, $\alpha = 1.0$, and $\alpha = 1.5$, along the positive xG axis, each representing a gradual movement toward configurations of higher scoring potential. Table 5.11 summarizes the original shot configuration and Table 5.12 its counterfactual reconstruction obtained after the largest displacement ($\alpha = 1.5$), corresponding to the most favourable latent modification.

Table 5.11: Original shot #48720

Feature	**Original Value**
Distance	8.505
Angle	0.937
Minute	41.000
Under Pressure	1.000
Body Part: Head	1.000
Play Pattern: From Free Kick	1.000
Expected Goal (xG)	**0.103**

Table 5.12: Counterfactual reconstruction shot #48720 ($\alpha = 1.5$)

Feature	**Counterfactual Value**
Distance	18.088
Angle	0.440
Minute	43.950
Under Pressure	0.232
Shot First Time	1.000
Body Part: Right Foot	1.000
Play Pattern: Regular Play	1.000
Expected Goal (xG)	**0.429**

The original shot corresponds to a header from a free kick, executed under pressure and from close range. The counterfactual reconstruction suggests that, in order to increase the expected-goals likelihood, the shot should transition into a configuration more typical of open-play finishing: a right-footed, first-time attempt taken in regular play and under substantially lower pressure. Although the reconstructed shot occurs from a slightly greater distance and narrower angle, these changes are counterbalanced by the improved control and reduced defensive interference, resulting in a higher latent-level xG estimation. In football terms, the model identifies that moving from an aerial, contested

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

scenario to a controlled ground strike in open play constitutes a minimal yet impactful modification that increases shot quality. Quantitatively, the reconstructed expected-goals value rises by approximately 0.33, indicating a fourfold increase in predicted scoring likelihood under the counterfactual configuration. This improvement is further visualized in Figure 5.15, which illustrates the corresponding trajectory of the shot in the latent space. The original embedding lies within the dense manifold of average-quality attempts, while counterfactual perturbations move the point smoothly along the positive xG direction, remaining within the manifold of realistic configurations. This confirms that the CAE generates semantically valid counterfactuals, hypothetical but plausible shots, rather than arbitrary noise. The latent displacement therefore represents a meaningful causal pathway from a low-probability to a higher-probability scoring scenario, grounded in interpretable football semantics.

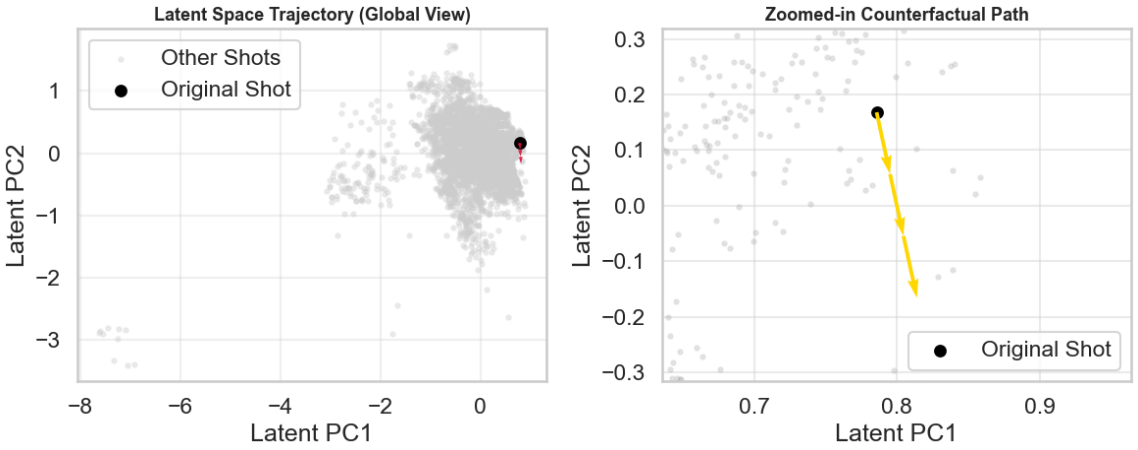
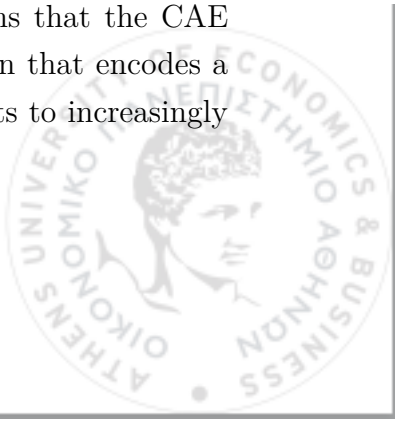


Figure 5.15: Counterfactual trajectory of Shot #48720 in latent space

As illustrated in Figure 5.16, counterfactual perturbations consistently lead to higher reconstructed xG estimates as the step magnitude ($\alpha = 0.5, 1.0, 1.5$) increases. Each point represents a shot instance from the test set, where the horizontal axis denotes the original expected-goals value and the vertical axis its counterfactual counterpart after traversing the latent space in the positive xG direction. All points lie clearly above the identity line ($y = x$), indicating that every step in the latent space corresponds to a plausible improvement in shooting potential. Moreover, the vertical separation between color groups reflects a proportional relationship between the displacement magnitude α and the degree of xG increase: larger movements in the latent space yield systematically higher scoring probabilities. This consistent, monotonic trend confirms that the CAE has successfully learned a structured and interpretable latent dimension that encodes a continuous gradient of chance quality, from routine, low-threat attempts to increasingly dangerous and high-probability finishing scenarios.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

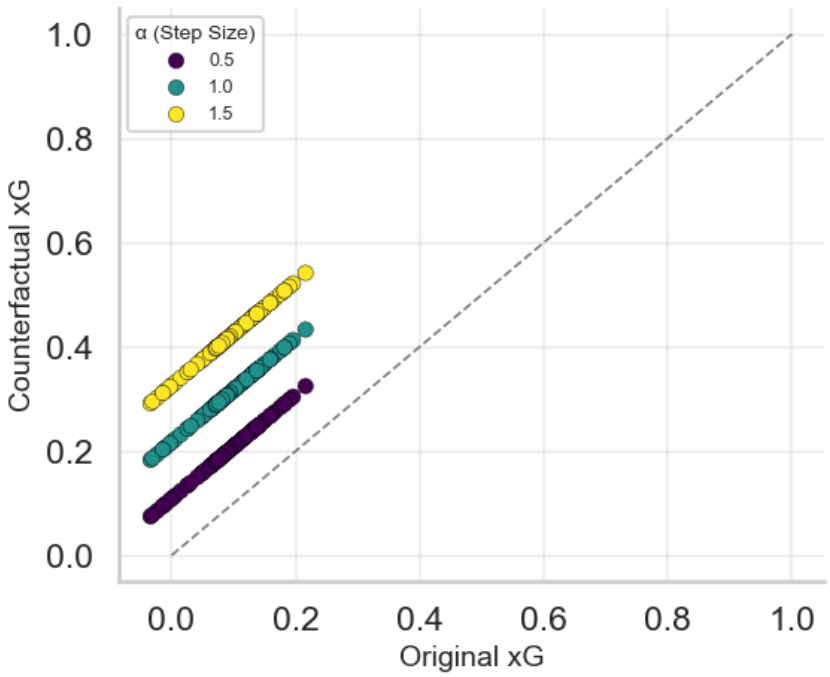


Figure 5.16: Latent-Step Impact on xG values

So as to ensure that the counterfactual findings are robust and not limited to a single instance, a random sample of 100 shots from the test set was selected. For each shot, counterfactual versions were generated along the learned xG direction, and the mean standardized feature change contributing to xG improvement was computed across all cases. As shown in Figure 5.17, the most influential adjustments correspond to shots taken from closer range, a reduction in defensive pressure, wider shooting angles, and transitions toward regular open-play situations. These modifications collectively describe scenarios in which the shooter faces less immediate opposition, has a clearer line of sight to the goal, and operates in more fluid attacking contexts—conditions naturally associated with higher-quality chances. Interestingly, temporal variables such as the minute of play exhibit a negative shift, suggesting that counterfactual improvements are marginally associated with earlier match phases. This effect is likely contextual rather than causal, possibly reflecting a statistical tendency for higher-quality chances to occur during open, high-tempo early-game periods, or minor noise captured in the latent representation. Conversely, features whose mean standardized change remains close to zero exert minimal direct influence on xG within the latent representation, although their slight variability indicates marginal, context-dependent effects that the model still captures in subtle ways.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

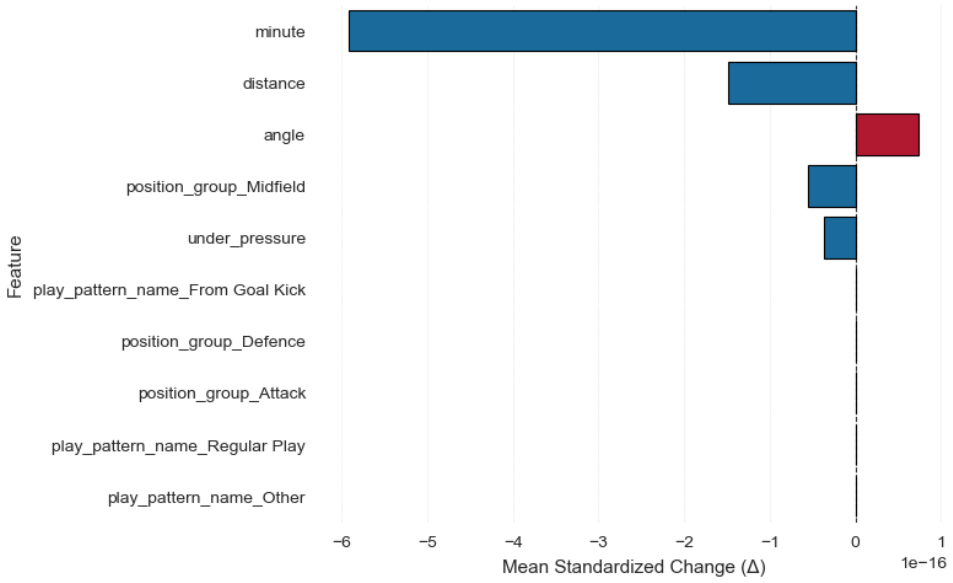


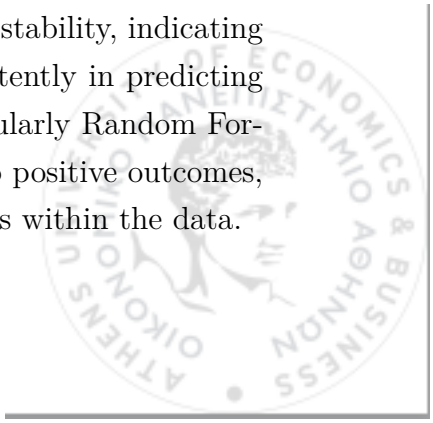
Figure 5.17: Mean Feature Change (sample of 100 Counterfactuals)

Taken together, these results suggest that the Counterfactual Autoencoder does more than replicate statistical correlations: it captures an interpretable causal geometry of shot creation and conversion. The model internalizes how variations in spatial positioning, execution dynamics, and situational pressure interact to shape the underlying probability of scoring. By generating coherent counterfactual trajectories, it reveals that improvements in offensive threat emerge not from isolated factors but from subtle, coordinated adjustments, such as achieving balance between proximity, shooting angle, and player control. In football terms, the CAE demonstrates that chance quality is a continuous, learnable field within which small, context-aware modifications can systematically elevate scoring potential. This highlights the model's ability to bridge descriptive analytics with causal insight, offering a generative framework for understanding how high-quality chances are formed rather than merely predicted.

5.5 Comparative Insights into Shooting Dynamics

5.5.1 Performance Assessment of Supervised Models

The comparative analysis of the supervised learning models reveals distinct behavioral patterns and trade-offs among different evaluation dimensions. Models such as Logistic Regression and Neural Networks demonstrate strong calibration and stability, indicating that they generate reliable probability estimates and perform consistently in predicting the overall outcome. In contrast, ensemble-based approaches, particularly Random Forest, show enhanced discriminative ability and improved sensitivity to positive outcomes, reflecting their strength in capturing complex non-linear relationships within the data.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

The Gradient Boosting and XGBoost models achieve a more balanced predictive performance, effectively mediating between precision and recall. Their behavior suggests an improved capacity to detect positive instances while maintaining acceptable calibration, a result that stems from the optimization of model hyperparameters and decision thresholds. These models combine predictive robustness with interpretability through feature importance and partial dependence analysis, providing meaningful insights into the underlying factors that influence goal likelihood.

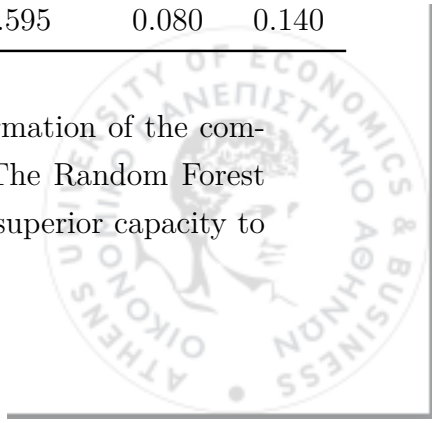
Overall, the results highlight the inherent trade-off between calibration accuracy and class sensitivity across the different algorithms. While simpler linear models remain effective in producing well-calibrated predictions, ensemble methods offer superior flexibility and adaptability in modeling complex patterns. Consequently, Gradient Boosting and XGBoost emerge as particularly suitable for expected goals estimation, where both predictive performance and interpretability are essential for practical and analytical applications.

Specifically, the XGBoost model demonstrates the most robust and consistent overall performance among the evaluated algorithms, as presented in Table 5.13. Its metrics remain well-balanced across multiple dimensions, reflecting reliable calibration, stable classification accuracy, and strong discriminative capacity. The model maintains low prediction error and well-calibrated probability estimates while simultaneously achieving competitive scores in both discriminative and composite evaluation measures. This equilibrium suggests that XGBoost effectively captures the underlying structure of the data without overfitting, producing predictions that are both accurate and generalizable. Moreover, its ensemble architecture and inherent regularization contribute to greater stability across validation folds, reinforcing its position as the most resilient and dependable model within this comparative framework.

Table 5.13: Comparison of Supervised Models performance

Model	**Log Loss**	**Brier**	**AUC**	**Accuracy**	**Precision**	**Recall**	**F1**
Logistic Regression	**0.260**	**0.074**	0.802	**0.907**	**0.601**	0.118	0.197
Decision Tree	0.272	0.076	0.782	0.906	0.576	0.098	0.168
Random Forest	0.375	0.116	**0.861**	0.830	0.276	**0.439**	0.339
Gradient Boosting	0.284	0.080	0.772	0.869	0.344	0.385	**0.363**
XGBoost	0.284	0.080	0.778	0.863	0.330	0.404	**0.363**
Neural Network	0.264	0.075	0.792	0.906	0.595	0.080	0.140

The ROC curves presented in Figure 5.18 provide a visual confirmation of the comparative discriminative ability of the supervised learning models. The Random Forest model clearly dominates the upper region of the plot, indicating its superior capacity to



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

distinguish between successful and unsuccessful goal events. Logistic Regression and the Neural Network exhibit similar and stable behavior, reflecting their consistent yet less flexible decision boundaries. In contrast, the Gradient Boosting and XGBoost curves maintain a balanced position between the linear and fully-ensemble approaches, combining sensitivity to positive cases with reasonable calibration. The overall proximity of the curves illustrates that all models achieve a satisfactory level of separability, yet the steeper ascent of the Random Forest underscores its advantage in true positive identification, aligning with its higher AUC value.

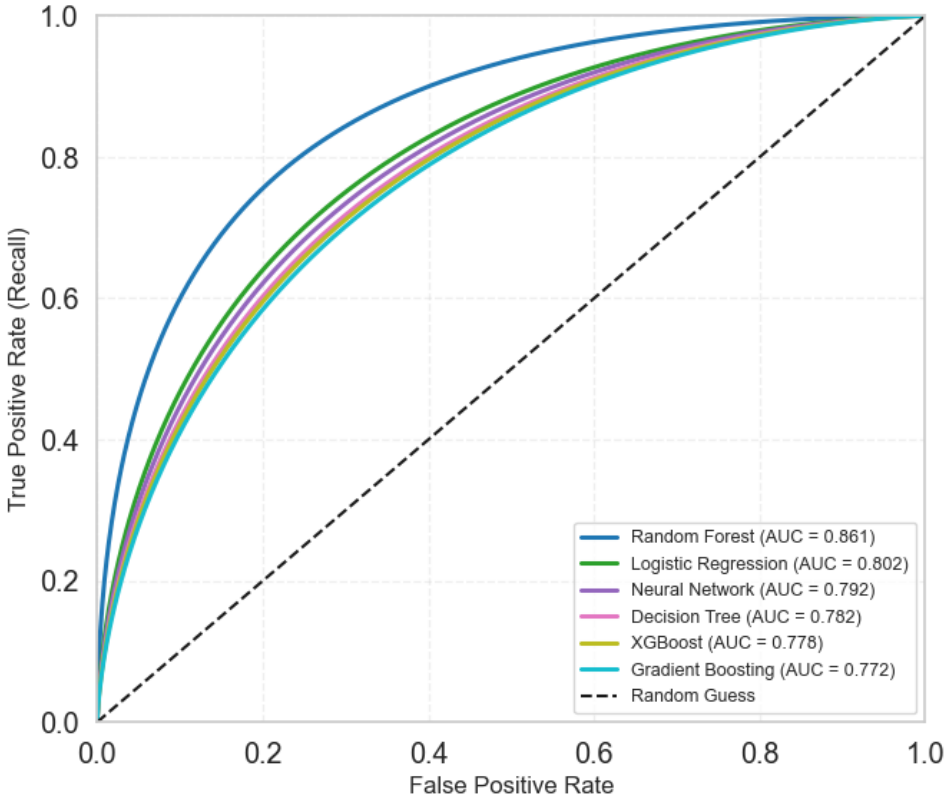


Figure 5.18: ROC curves comparison among supervised learning models

In the context of calibration analysis, the diagonal line in a reliability plot represents the ideal case of perfect calibration, where the predicted probabilities match the true event frequencies. When a model's curve lies exactly on this line, it indicates that the estimated probabilities are fully reliable. For instance, predictions of 0.3 correspond to approximately 30% of actual goals. Deviations below the diagonal reflect overconfident models that overestimate goal likelihoods, while curves above the line denote underconfident models that tend to underestimate scoring probabilities. Hence, the proximity of each model's curve to the diagonal provides a direct measure of the reliability of its probabilistic predictions, an especially critical property in expected goals modeling where interpretability and probability consistency are fundamental. The calibration plot in Figure 5.19 illustrates the degree of alignment between the predicted goal probabilities and

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

the actual observed frequencies across the supervised learning models. The curves of Logistic Regression, Gradient Boosting, and Neural Network remain closely aligned with the diagonal line, indicating well-calibrated probability estimates where predicted and true goal likelihoods correspond almost perfectly. Ensemble models such as XGBoost and Random Forest exhibit a slight underestimation for higher probability ranges, suggesting a conservative tendency in predicting goal occurrences. The Decision Tree displays a marginal deviation around mid-range values, reflecting limited stability and sensitivity compared to the more complex ensemble learners. Overall, the results confirm that all models achieve satisfactory reliability, with Logistic Regression and Gradient Boosting providing the most accurate calibration for expected goals estimation.

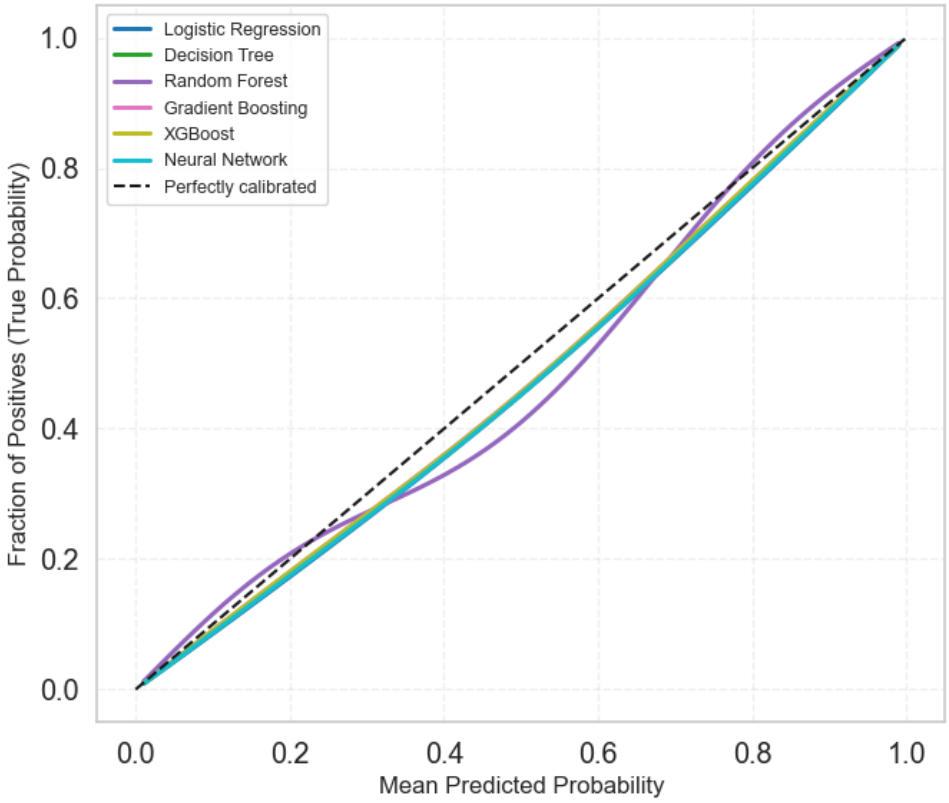


Figure 5.19: Calibration curves of the supervised learning models

Furthermore, the Precision–Recall curves illustrated in Figure 5.20 provide a detailed view of each model's ability to identify true goal events while minimizing false predictions. Ensemble-based methods, specifically Gradient Boosting and XGBoost, demonstrate the most favorable balance between precision and recall, maintaining high precision values across a wide range of recall levels. This indicates that these models can effectively detect goal events while preserving a relatively low rate of false positives. Random Forest follows closely, showing strong recall performance, though at the cost of lower precision in certain ranges. In contrast, Logistic Regression and the Neural Network exhibit more conservative predictive behavior, maintaining higher precision at low recall values but

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

quickly declining as recall increases. This pattern suggests that they are more cautious in assigning positive predictions, favoring reliability over coverage. The Decision Tree, on the other hand, presents the weakest overall performance, reflecting its limited generalization ability compared to the ensemble methods.

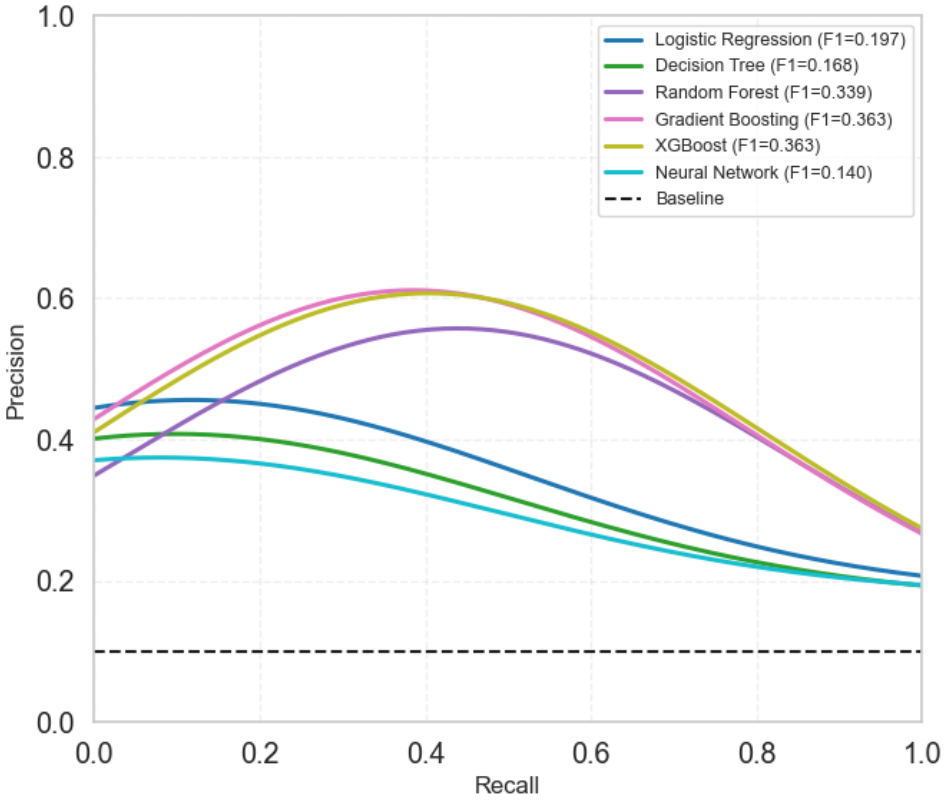


Figure 5.20: Precision–Recall curves for the supervised learning models

Overall, this part of analysis reinforces the conclusions drawn from the ROC and calibration evaluations: ensemble-based algorithms, particularly Gradient Boosting and XGBoost, achieve the most effective trade-off between identifying true goal events and maintaining prediction reliability, making them the most suitable approaches for Expected Goals estimation tasks.

Figure 5.21 illustrates the joint relationship between classification accuracy and F1 score across the supervised learning models. The dashed horizontal line at F1 = 0.30 serves as a reference threshold, distinguishing models with limited ability to balance precision and recall from those achieving a more effective trade-off. The results reveal two distinct performance groups. Logistic Regression (LR), Decision Tree (DT), and Neural Network (NN) cluster toward the right side of the plot, indicating high accuracy but low F1 values, suggesting conservative predictions that correctly classify most non-goal events but fail to capture many true positives. In contrast, ensemble methods such as Random Forest (RF), Gradient Boosting (GB), and XGBoost (XGB) occupy the upper-left and central regions, showing improved F1 scores and more balanced predictive

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

behavior. Among them, Gradient Boosting and XGBoost stand out as the most stable and robust models, both surpassing the reference line and demonstrating an optimal trade-off between predictive precision and sensitivity. Their position near the top of the diagram reflects a well-calibrated capability to identify goal-scoring opportunities without compromising overall classification reliability.

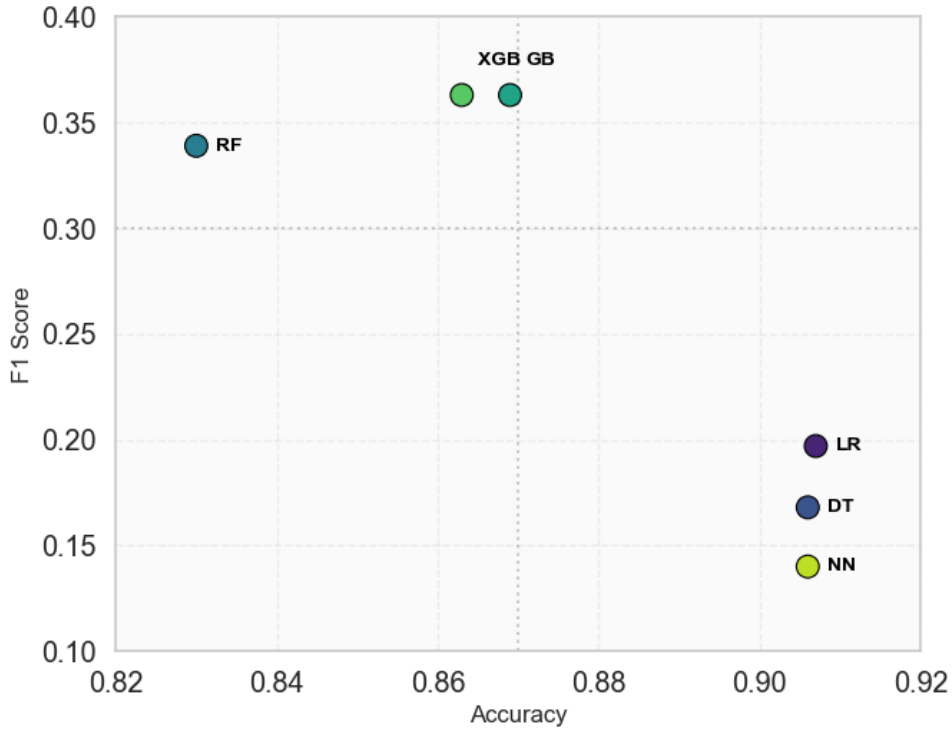
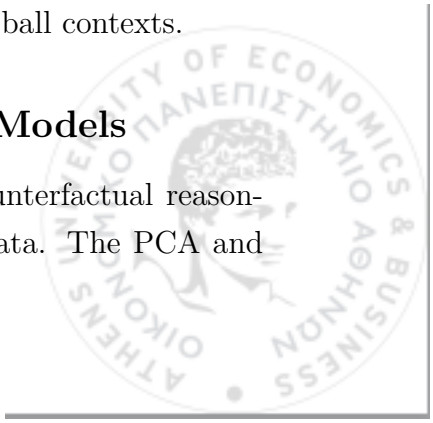


Figure 5.21: Accuracy–F1 relationship across supervised learning models

In summary, the comparative evaluation of supervised learning algorithms demonstrates that no single metric can fully capture model performance. While linear models such as Logistic Regression and Neural Networks offer strong calibration and interpretability, they struggle to detect positive goal events effectively. Decision Trees provide moderate performance but lack generalization. Ensemble-based methods consistently outperform individual learners, achieving superior discriminative power and a more balanced trade-off between precision and recall. Among them, XGBoost and Gradient Boosting emerge as the most reliable and analytically valuable models, combining robustness, calibration accuracy, and interpretability. Their capacity to provide well-calibrated probabilities while maintaining strong predictive performance underscores their suitability for Expected Goals modeling, particularly in analytical and tactical football contexts.

5.5.2 Performance Assessment of Unsupervised Models

The combined application of clustering, anomaly detection, and counterfactual reasoning reveals a coherent multilevel structure in the underlying shot data. The PCA and



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

K-Means framework identifies macro-level groupings of shots corresponding to distinct tactical and contextual archetypes, each characterized by unique spatial, temporal, and technical attributes. Within this structure, Cluster 3 reflects structured, repeatable situations, headed attempts from crosses and corners, typically executed under defensive pressure and from tight angles inside the box, where positioning and timing are decisive for success. In contrast, Cluster 5 represents a rare, high-quality category of one-touch finishes, rebounds, and opportunistic shots emerging from unstructured play, highlighting the upper extreme of finishing quality despite its low frequency.

At a finer level of analysis, the Isolation Forest model isolates micro-scale irregularities embedded within the broader landscape of shot behavior. The results indicate that the most influential predictors of anomalous, high-impact attempts are predominantly contextual and event-driven, rather than purely geometric. Specifically, variables related to play-pattern categories, notably restarts such as corners and free kicks, rebound, second-ball phases, and goalkeeper interventions emerge as key discriminators. Additional contributing factors include the presence of defensive pressure, the positional identity of the shooter, with defenders unexpectedly featuring among high-value anomalies, and finishing modes characterized by first-time or point-blank executions. Collectively, these findings suggest that anomalous but effective shots often arise from dynamic and unstructured match contexts rather than controlled build-up situations. In this respect, the anomaly detection framework complements the clustering analysis as presented in Table 5.14, by capturing the episodic, context-dependent deviations that define the upper extremes of shot quality beyond the systematic archetypes represented by Clusters 3 and 5.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Aspect	PCA + K-Means Clustering	Anomaly Detection with Isolation Forest
Objective	Identification of shot clusters	Detection of anomalous shots
Input	15 principal components (excluding is_goal, xG)	Standardized features (excluding is_goal, xG)
Internal Evaluation	Elbow method, Silhouette coefficient	Contamination variation, Feature importance
External Evaluation	Centroid interpretation in original space, xG and goal rate comparison	Comparison of outliers vs. normals (t-SNE/UMAP)
Findings	Distinct, interpretable shot profiles consistent with xG and conversion patterns	Outliers closer to goal, higher xG and goal rate, normals represent majority pattern
Overall Evaluation	Effectively groups shots into meaningful categories with distinct performance outcomes	Effectively identifies rare and unusual shots, validated both quantitatively and visually
Complementarity	Provides macro-level structure (clusters)	Provides micro-level exceptions (outliers)

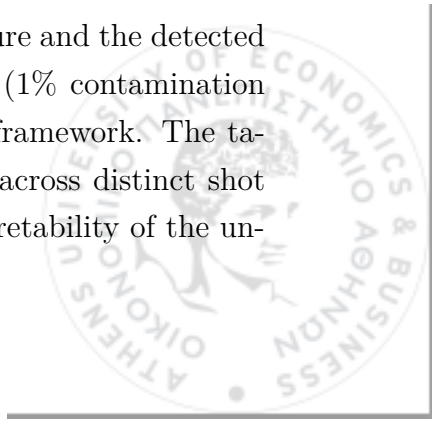
Table 5.14: Comparative summary of dimensionality reduction approaches

To cross-validate the findings of the clustering and anomaly detection approaches, a comparative analysis was conducted to examine how the 1% contamination level in the Isolation Forest, corresponding to 507 anomalous shots, distributes across the five clusters identified in the PCA–K-Means framework, as summarized in Table 5.15.

Cluster	Dominant Shot Profile
1	Baseline mid-range footed shots, balanced profile
2	Right-footed open-play finishing, typical attempts
3	Headers from crosses and set-pieces, under pressure
4	Left-footed strikes, controlled execution
5	Rare dynamic clear-cut chances, opportunistic contexts

Table 5.15: Interpretation of clusters based on centroid profiles

To further examine the relationship between the clustering structure and the detected anomalies, Table 5.16 presents the distribution of anomalous shots (1% contamination threshold) across the five clusters derived from the PCA–K-Means framework. The table highlights how the frequency and proportion of anomalies vary across distinct shot archetypes, providing an additional layer of validation for the interpretability of the un-



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

supervised models.

Cluster	Total Shots	% of Dataset	Absolute Anomalies	% Anomalies (of Cluster)
1	2,059	4.06%	43	2.09%
2	24,159	47.67%	34	0.14%
3	9,659	19.06%	356	3.69%
4	14,729	29.06%	74	0.50%
5	71	0.14%	0	0.00%

Table 5.16: Distribution of anomalous shots PCA–K-Means clusters

The results, illustrated in Figure 5.22, reveal clear asymmetries in the concentration of anomalies across different shot typologies. The majority of anomalous instances are concentrated in Cluster 3, followed by Cluster 1, while Clusters 2, 4, and 5 exhibit minimal anomaly rates below 1%.

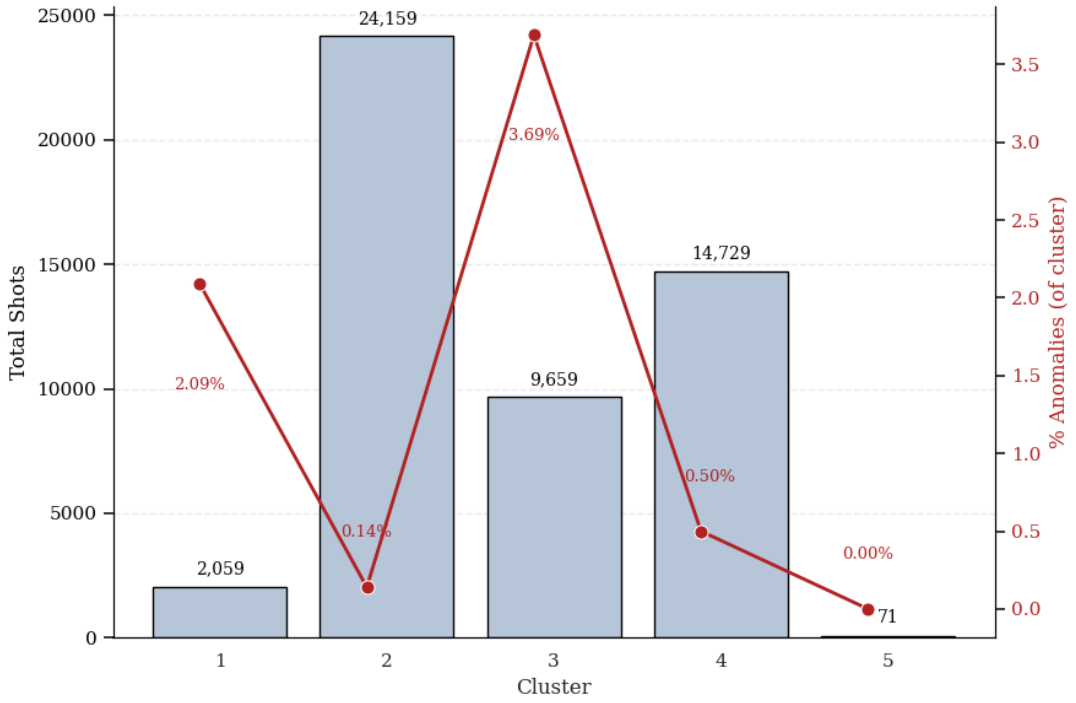


Figure 5.22: Cluster-wise distribution of anomalous (high-quality) shot instances

From a football analytics perspective, the distribution of anomalies across clusters highlights distinct behavioural asymmetries in shooting dynamics. Cluster 3, dominated by headed attempts from crosses and set-pieces under defensive pressure, exhibits the highest anomaly rate, reflecting the inherent volatility of aerial duels and second-phase headers within congested penalty areas. This observation aligns with the feature importance analysis, which identified contextual variables that introduce tactical instability and execution uncertainty.

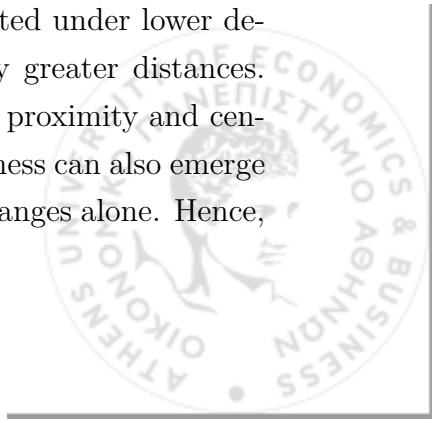
Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Cluster 1, consisting mainly of long-range and low-probability shots, also presents moderate anomaly presence, capturing isolated yet high-impact deviations from otherwise low-efficiency patterns. In contrast, Clusters 2 and 4 corresponding to right-footed and left-footed open-play finishes, display minimal anomaly rates, indicating stable shot mechanics and repeatable tactical structures. Interestingly, Cluster 5 representing rare but clear-cut one-touch and rebound finishes, contains no detected anomalies. This does not imply homogeneity within the cluster, but rather that such shots, despite their low frequency, align closely with the statistical profile of efficient finishing events in the dataset. In football terms, these actions embody "rare but optimal" situations: close-range, fast-executed, high-probability attempts that conform to the model's notion of normal, effective shot behaviour. Hence, while Cluster 5 captures the most uncommon match contexts, its internal dynamics reflect the archetype of efficient chance conversion rather than outlier behaviour.

To sum up, these findings show that anomalous and high-impact shots do not emerge from the most statistically rare clusters, but rather from those combining tactical congestion and executional volatility. In football terms, these situations correspond to crowded penalty-box contexts, such as aerial duels, deflected deliveries, or second-phase set-pieces, where defensive pressure, timing, and trajectory variation jointly amplify unpredictability. This explains why Cluster 3, defined by headed attempts from crosses and corners, contains the largest share of anomalies: small deviations in ball flight or player positioning can drastically alter the outcome. Conversely, clusters reflecting structured open-play shooting (Clusters 2 and 4) exhibit mechanical and situational stability, producing consistent, repeatable finishing patterns. Overall, the distribution of anomalies reveals that the most unpredictable and high-value opportunities emerge not from rarity itself, but from the dynamic, interaction-heavy phases of play where control, space, and execution collide.

5.5.3 Causal and Counterfactual Analysis

The Counterfactual Autoencoder extends the unsupervised findings by introducing a causal and prescriptive perspective on shot quality. Rather than merely identifying latent structures or deviations, it estimates the minimal and interpretable feature adjustments required to convert a low-probability attempt into a high-value opportunity. This approach reveals that improvements in shot quality are not exclusively geometric: in several reconstructed cases, shots became more effective when executed under lower defensive pressure or with faster, more direct play, even from slightly greater distances. Such outcomes contrast with the unsupervised results, where spatial proximity and centrality were dominant, but emphasize a crucial insight. Shot effectiveness can also emerge from contextual and behavioral adaptations rather than positional changes alone. Hence,



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

the model captures how tempo, fluidity, and execution context dynamically redefine the efficiency of a shooting opportunity beyond static geometry.

For the held-out test set, corresponding to the total sample of 7,602 shots out of 50,677 (15% of the dataset), a counterfactual version was generated for each original attempt using the trained autoencoder. Each counterfactual shot represents a minimally altered version of the original event, adjusted in the latent space by a step of $\alpha = 1.5$ along the learned xG direction. This procedure allows the model to estimate how much each type of shot could improve its scoring probability if subtle contextual or geometric modifications were applied. Figure 5.23 displays the average change in expected goals (ΔxG) across the five latent shot clusters. Each bar represents the mean uplift (Counterfactual xG – Original xG) for all test shots assigned to that cluster, while the annotations indicate the number of samples used in the calculation.

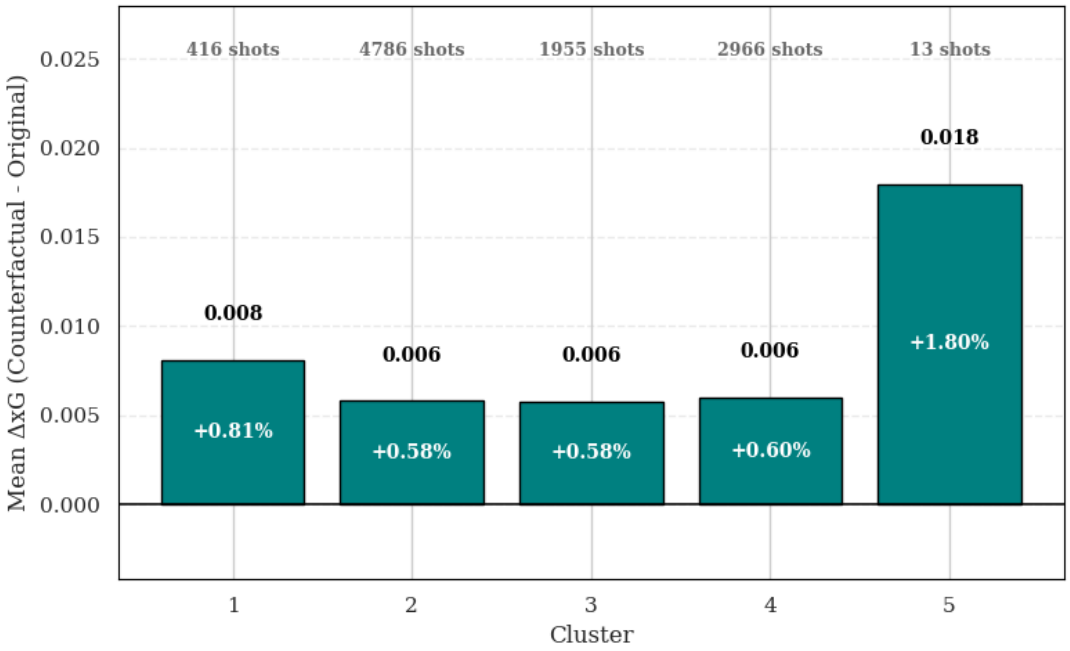


Figure 5.23: Average counterfactual xG improvement per latent cluster

Overall, all clusters display a positive counterfactual uplift, confirming that the model identifies realistic directions of improvement across shot types. However, the magnitude of change varies notably between clusters. Among all groups, Cluster 5 stands out by exhibiting the largest relative uplift, despite containing the fewest samples. This indicates that rare or atypical shot configurations possess the greatest capacity for improvement, suggesting that even small contextual or technical refinements could substantially enhance their scoring potential. Cluster 1 follows with a moderate yet consistent gain, corresponding to standard mid-range attempts typically executed under moderate pressure. These shots represent balanced, well-structured situations that are already technically efficient but still permit limited refinements in execution or positioning. By contrast, Clusters 2,

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

3 and 4 display only minimal uplift, ranging between +0.58% and +0.60%, implying that these common and geometrically stable patterns are already close to their inherent performance ceiling, with limited scope for further improvement.

Specifically, Figure 5.24 visualizes the dominant feature changes associated with counterfactual improvements in expected goals across the five latent clusters. Each cell represents the average directional change in a key shot attribute, with green values indicating positive contributions, features whose increase raises xG, and red values denoting negative ones, features that reduce xG when increased. It is worth noting that the feature corresponding to the minute of execution, consistently exhibited negative coefficients across all clusters and was therefore excluded from the visualization. This pattern suggests that the autoencoder implicitly learned that higher-quality opportunities tend to occur earlier in the match, when teams are physically fresher, cognitively sharper, and more capable of executing structured attacking patterns. In practical terms, this temporal signal reflects how early-game dynamics and rhythm control often create conditions for more effective shot creation, before fatigue and defensive compactness increase as the game progresses, and teams subsequently attempt to generate scoring chances through alternative, more opportunistic or less structured attacking approaches.

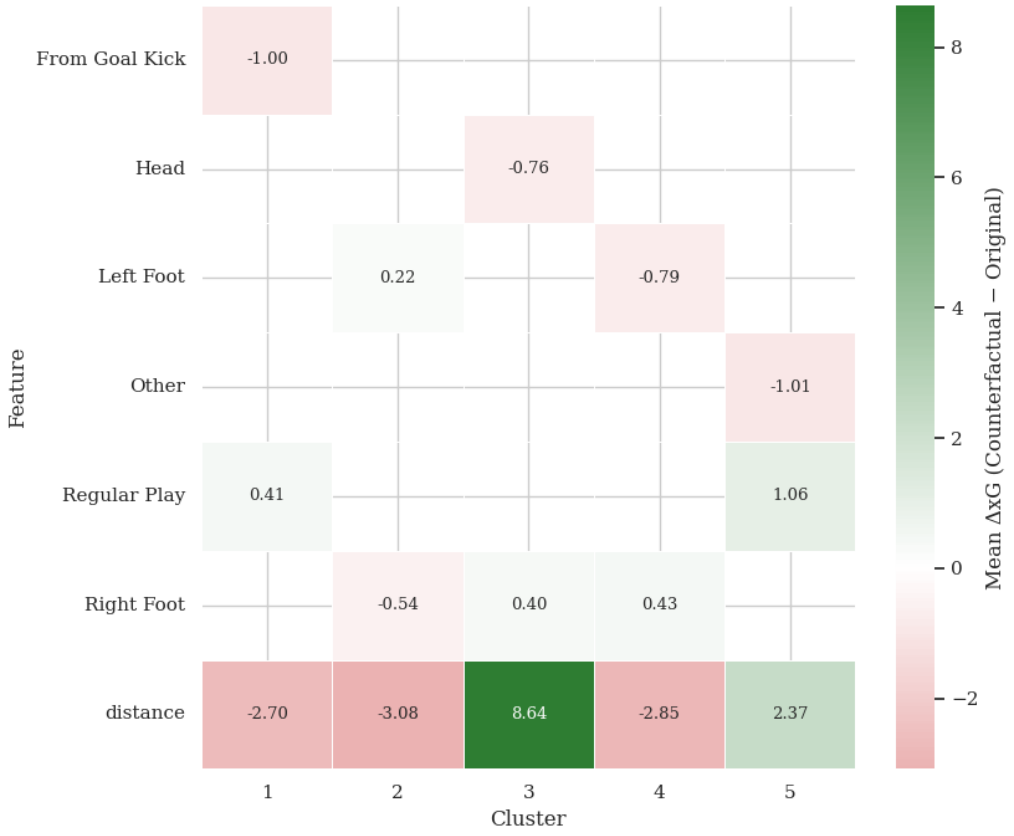


Figure 5.24: Mean feature Δchange between counterfactual and original shots

The most prominent and consistent driver across all clusters is shot distance. In

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Clusters 1, 2, and 4, the negative shifts in distance indicate that the model tends to move shots closer to goal when generating more effective counterfactuals, fully aligned with standard geometric intuition in football analytics. By contrast, Cluster 3 exhibits a positive distance change, indicating that some attempts become more effective even when simulated from slightly farther away. This cluster primarily corresponds to aerial or set-piece situations, often involving headed shots from mid-range areas. The counterfactual behavior suggests that in such contexts, improvements in timing, delivery, or body orientation can offset geometric disadvantages, allowing higher expected goal values despite increased distance. In Cluster 5, the counterfactual improvements are linked to an increase in Regular Play patterns and a decrease in rare or other scenarios, indicating that smoother, open-play contexts are more conducive to effective finishing. Similarly, small yet consistent changes in body part usage (Right Foot or Left Foot) across clusters suggest that the model implicitly adjusts for more natural-foot or balanced finishing conditions when enhancing shot quality. Together, these results demonstrate that while geometric proximity remains the dominant driver of shot quality, contextual and behavioral adjustments represent complementary levers for performance optimization.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Chapter 6

Conclusions and Recommendations

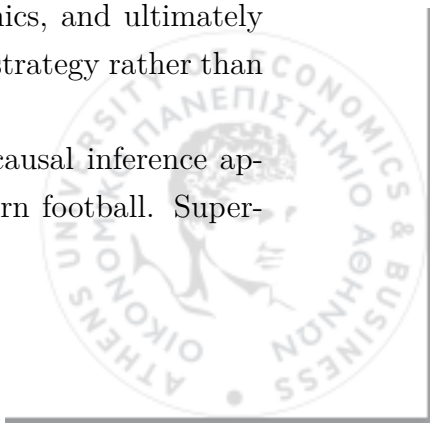
6.1 Conclusions

In the broader domain of sports and football analytics, modern machine learning methods provide a multilayered framework for understanding, predicting, and interpreting athletic performance. Supervised learning techniques form the foundation of predictive analytics, learning from historical data to estimate the likelihood or expected outcome of specific actions, such as the success probability of a shot, the expected goal value of a play, or the contribution of an individual player. These models transform the game into quantifiable probabilities and performance indicators, supporting evidence based decision-making, tactical evaluation, and strategic planning.

Conversely, unsupervised learning methods focus on pattern discovery, clustering, and representation learning without relying on predefined outcome labels. They reveal the latent structure and stylistic diversity of play, uncovering recurring tactical patterns, player roles, or contextual shot profiles that traditional metrics may overlook. By organizing and interpreting complex data spaces, unsupervised approaches enhance the descriptive and exploratory dimensions of analytics, offering novel perspectives on team identity, playing style, and decision-making behavior.

Beyond description and prediction, causal inference and counterfactual modeling introduce a third, more advanced analytical layer - one that aims to understand cause-effect relationships and simulate alternative scenarios. These methods shift the analytical focus from "what happened" to "what would have happened if conditions were different." Such causal frameworks enable simulation of tactical interventions, quantification of the impact of player positioning, defensive pressure, or temporal dynamics, and ultimately support the development of prescriptive analytics capable of guiding strategy rather than merely describing it.

Consequently, the integration of supervised, unsupervised, and causal inference approaches establishes a comprehensive analytical ecosystem for modern football. Super-



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

vised models quantify outcomes, unsupervised models reveal structure, and causal models explain mechanisms and simulate counterfactual realities. Together, they transform football analytics from a static description of events into a dynamic, interpretable, and decision-oriented science, that bridges data-driven modeling with tactical reasoning and real-world performance optimization.

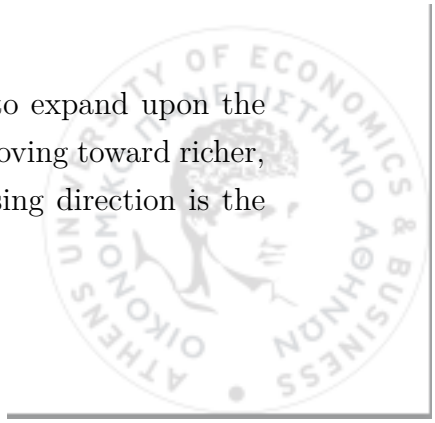
6.2 Practical Implications

The analytical process developed in this study can also be interpreted as a potential framework for practical performance evaluation. In applied contexts, analysts could follow a similar stepwise process to group and classify shot attempts based on spatial, contextual, and mechanical features, identify anomalous patterns that combine unusually not only high xG values, but also strong conversion rates, and characterise the specific attributes that distinguish these instances from the broader shot population. Such analysis can reveal not only the most effective shot profiles but also situational tendencies, such as preferred shooting zones, decision timing, or response under pressure, that influence a team's attacking efficiency.

In a subsequent stage, counterfactual simulation techniques could be employed to explore how adjustments in shot parameters might enhance the overall efficiency of similar attempts. By systematically varying factors, analysts can assess the potential performance gain associated with tactical or technical modifications. This analytical process enables a deeper understanding of how contextual and mechanical aspects of shooting behaviour influence team efficiency, linking quantitative model findings with applied tactical reasoning and the execution and adjustment of team strategies. While the approach presented here does not represent a definitive or exhaustive solution, it demonstrates how data-driven modelling, anomaly detection, and generative simulation can be combined into a practical methodology for tactical refinement, training design, and performance optimisation. Altogether, the proposed framework provides a structured view of attacking performance, depending on the analytical objective, it can be adapted to focus on specific aspects of offensive play, from overall shooting tendencies to in-depth examination of particular attacking patterns, offering a flexible basis for detailed, data-driven investigation of targeted elements within the offensive phase.

6.3 Future Directions

Future developments in sports and football analytics are expected to expand upon the integration of supervised, unsupervised, and causal methodologies, moving toward richer, context-aware, and explainable models of performance. One promising direction is the



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

incorporation of spatial-temporal and positional data, allowing models to capture dynamic interactions between players, ball movement, and game phases. Such multi-agent perspectives, supported by Graph Neural Networks (GNNs) and sequence-based architectures, such as Transformers or LSTMs, can provide deeper insights into coordinated team behavior and emergent tactical patterns.

At the same time, advances in representation learning and self-supervised pretraining could enable more generalizable models that learn directly from large volumes of match data without extensive labeling. These approaches may uncover latent structures of play that bridge individual actions with collective strategies, enhancing both interpretability and scalability. Furthermore, causal and counterfactual frameworks should continue to evolve as key instruments for prescriptive analytics, allowing analysts and coaches to model hypothetical tactical situations and evaluate the effects of possible interventions. Integrating causal reasoning with predictive and clustering models could enable systems that not only anticipate outcomes but also recommend optimal tactical adaptations in real time.

Finally, the increasing availability of multimodal data, combining event logs, tracking data, and even physiological or audio-visual inputs, opens the way for holistic, data-driven representations of performance that merge statistical accuracy with contextual understanding. The convergence of these approaches will drive football analytics beyond prediction, toward causal interpretation, tactical simulation, and decision optimization, aligning analytical models more closely with the complex realities of the sport.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Appendix A

Additional Figures

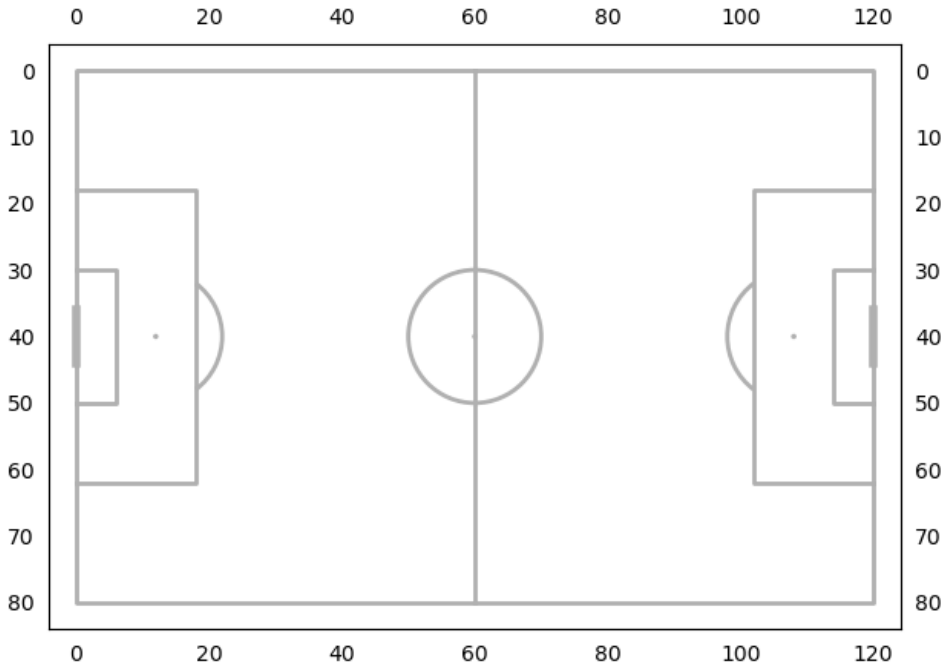


Figure A.1: Pitch dimensions and coordinate system

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

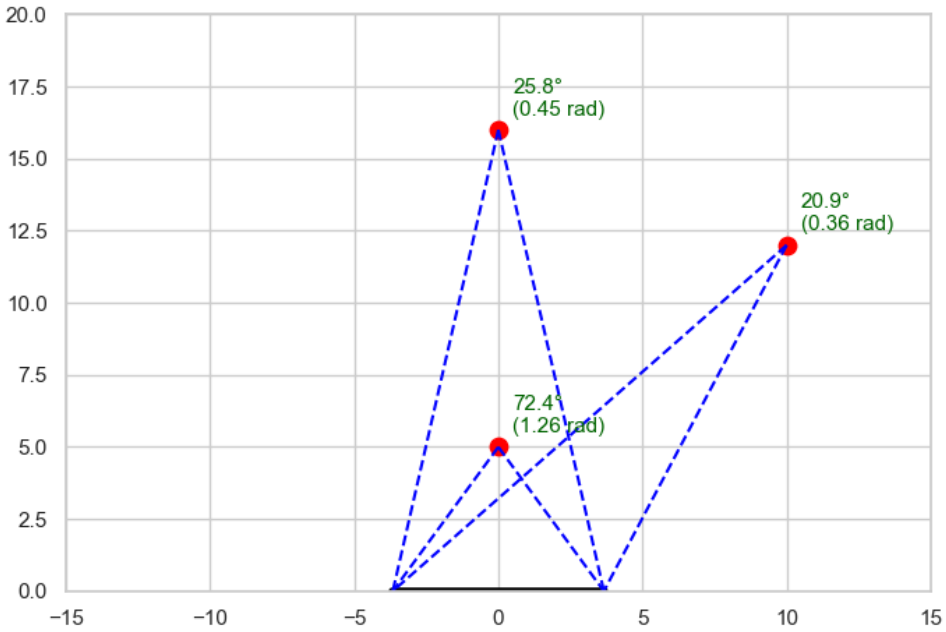


Figure A.2: Shooting angle for different shot positions on the pitch



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

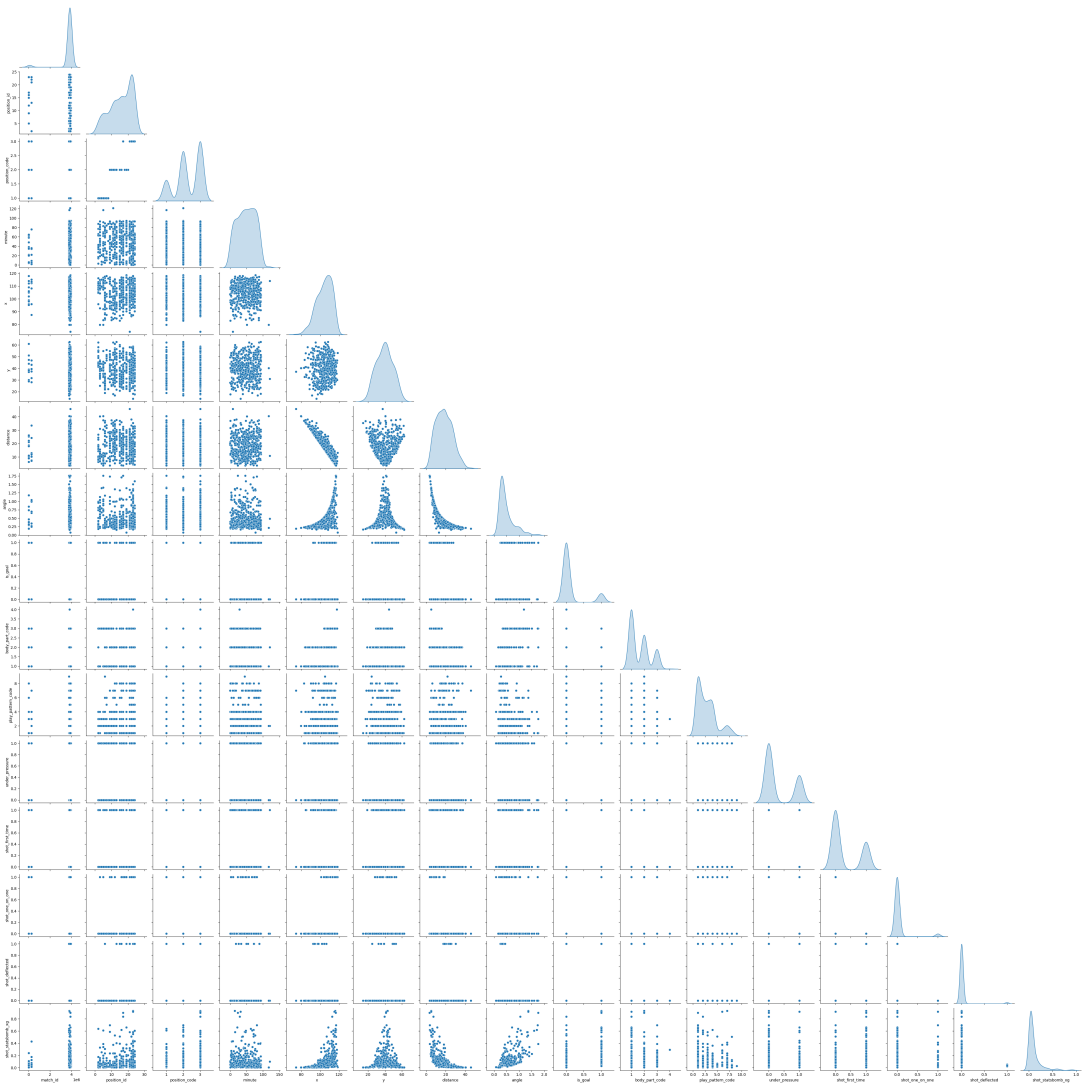


Figure A.3: Pairplot of numerical features (sample of 500 observations)

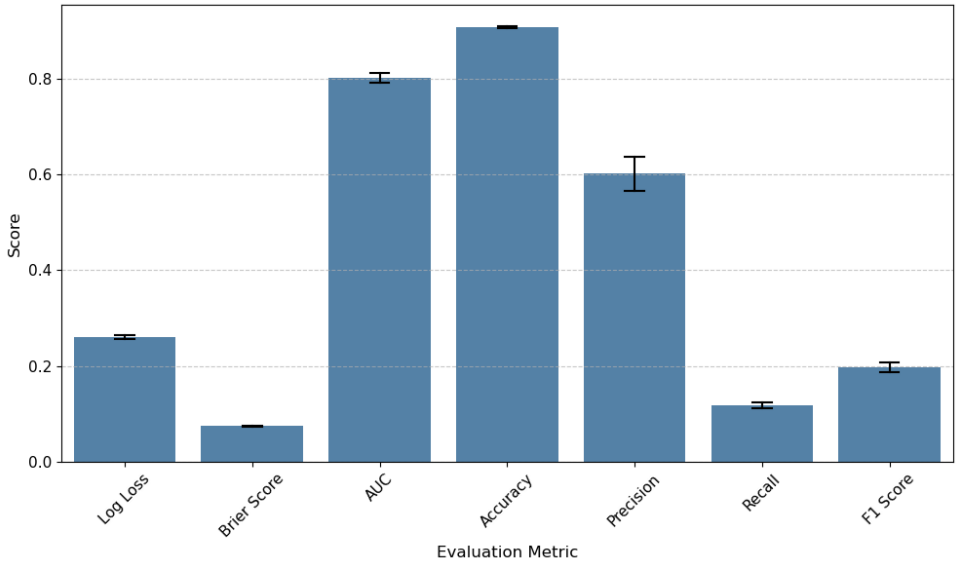


Figure A.4: Mean ± SD of evaluation metrics (5-fold CV) for Logistic Regression

ATHENS UNIVERSITY OF ECONOMICS & BUSINESS ΟΙΚΟΝΟΜΙΚΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΑΘΗΝΩΝ

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

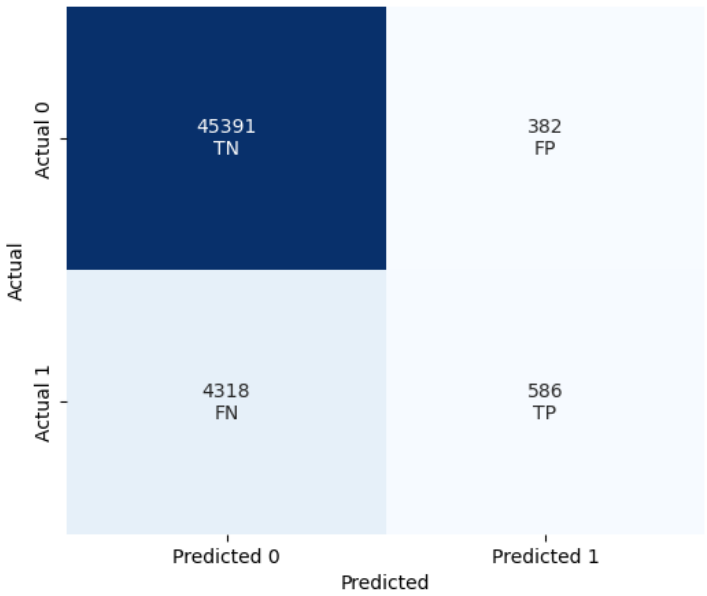


Figure A.5: Confusion matrix for Logistic Regression

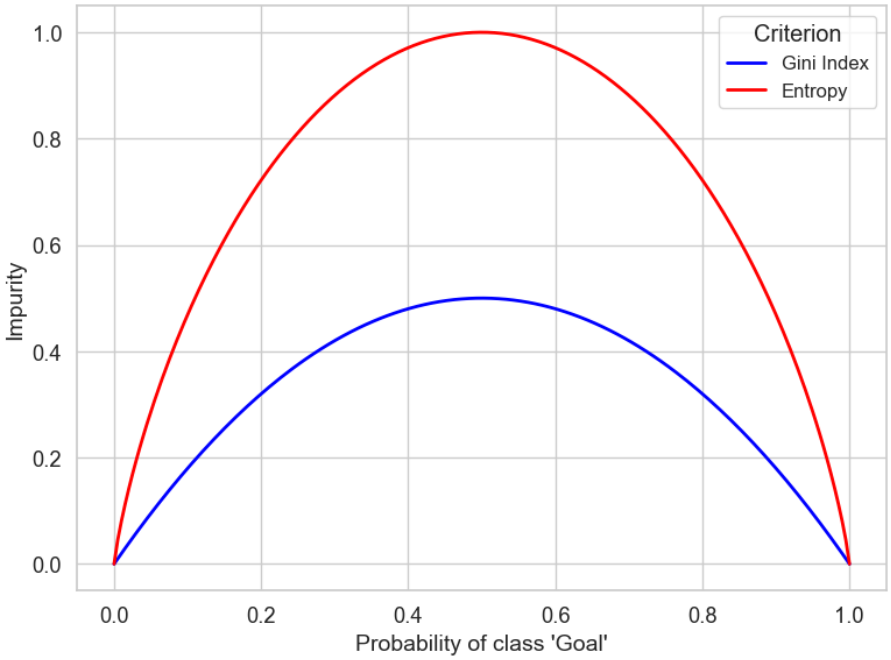


Figure A.6: Gini index and Entropy curves



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

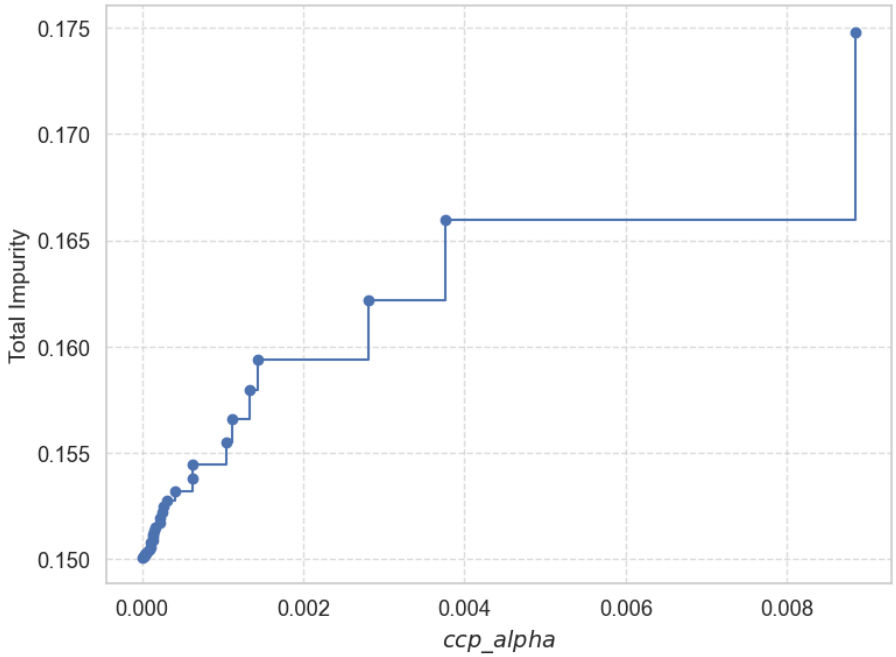


Figure A.7: Cost–complexity pruning path of the Decision Tree

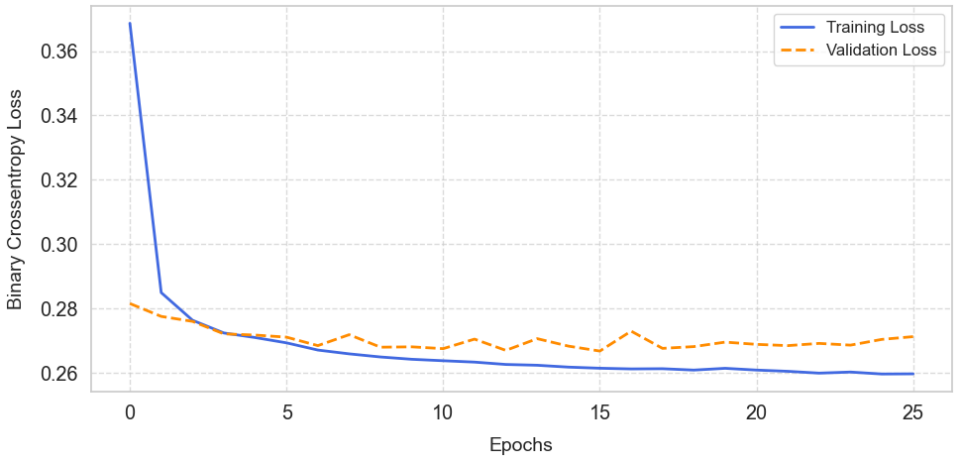


Figure A.8: Learning curves of the Neural Network model


ATHENS UNIVERSITY OF ECONOMICS & BUSINESS
ΟΙΚΟΝΟΜΙΚΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΑΘΗΝΩΝ

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

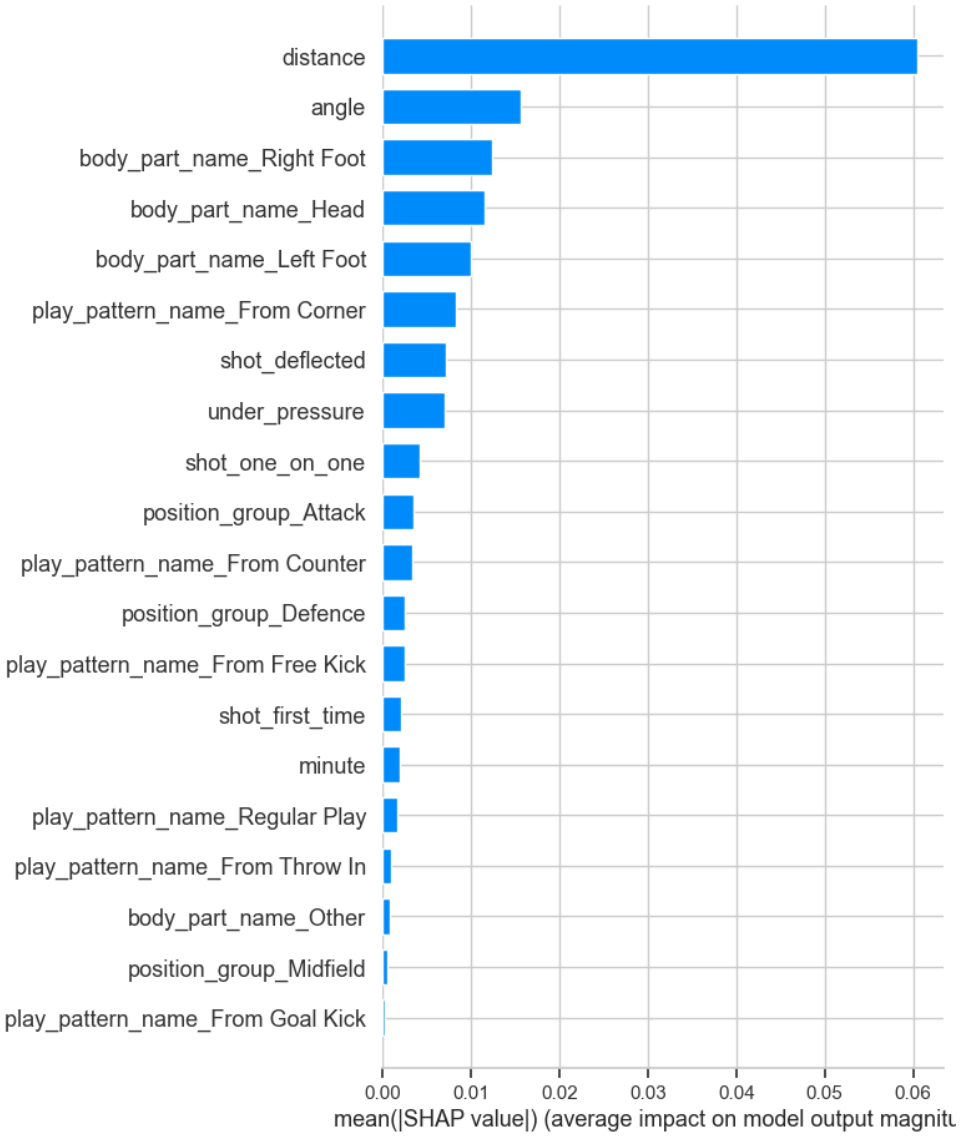


Figure A.9: Feature importance based on SHAP values for the Neural Network



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

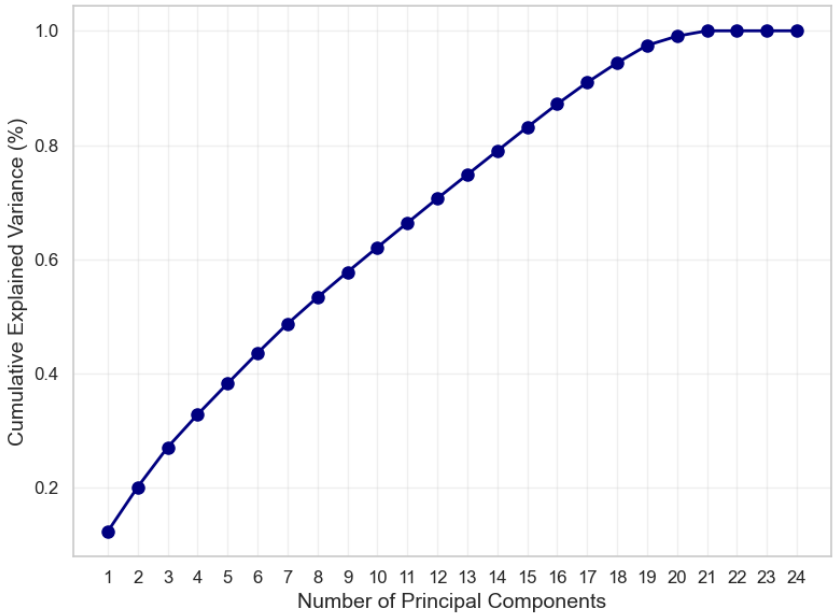


Figure A.10: Scree plot of Cumulative Explained Variance for PCA components

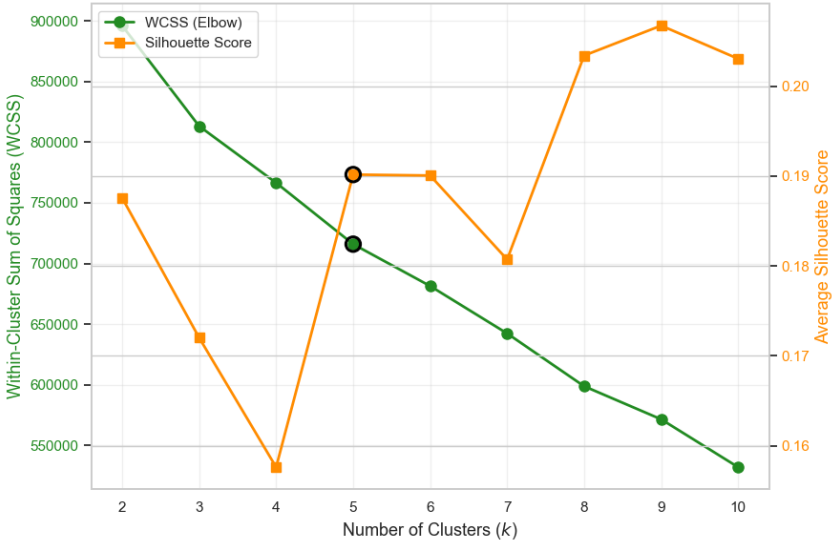


Figure A.11: Combined Elbow and Silhouette evaluation for optimal cluster selection



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

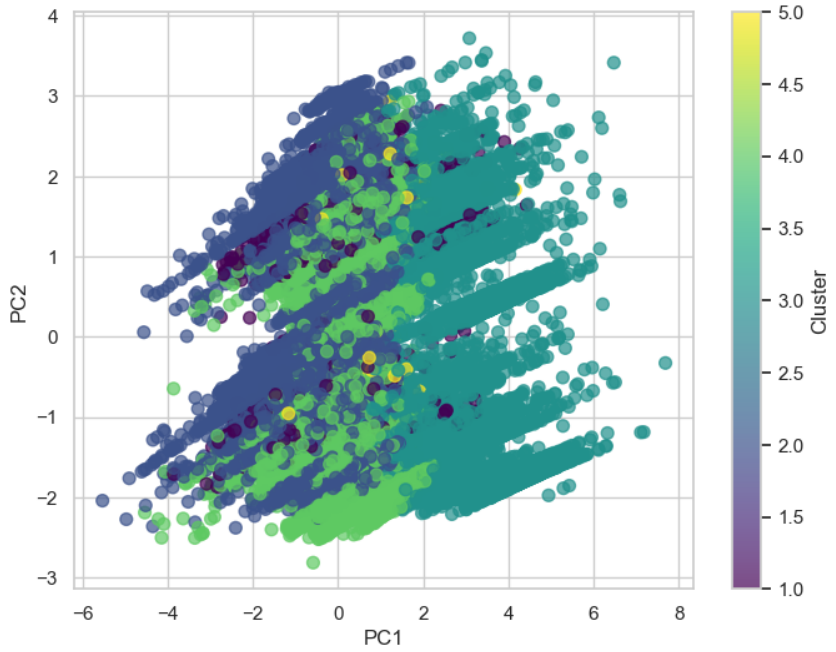


Figure A.12: K-Means clustering projected on 2D

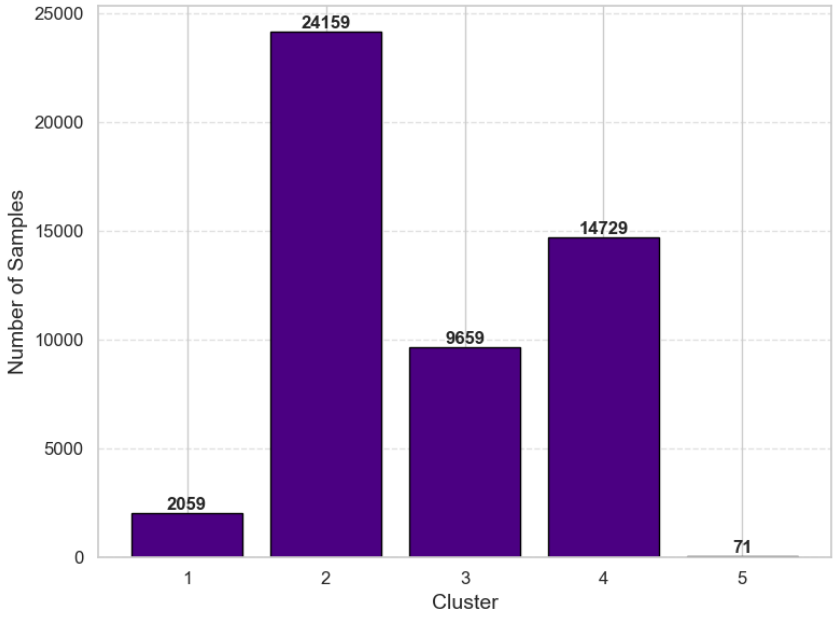


Figure A.13: Distribution of shot attempts across clusters



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

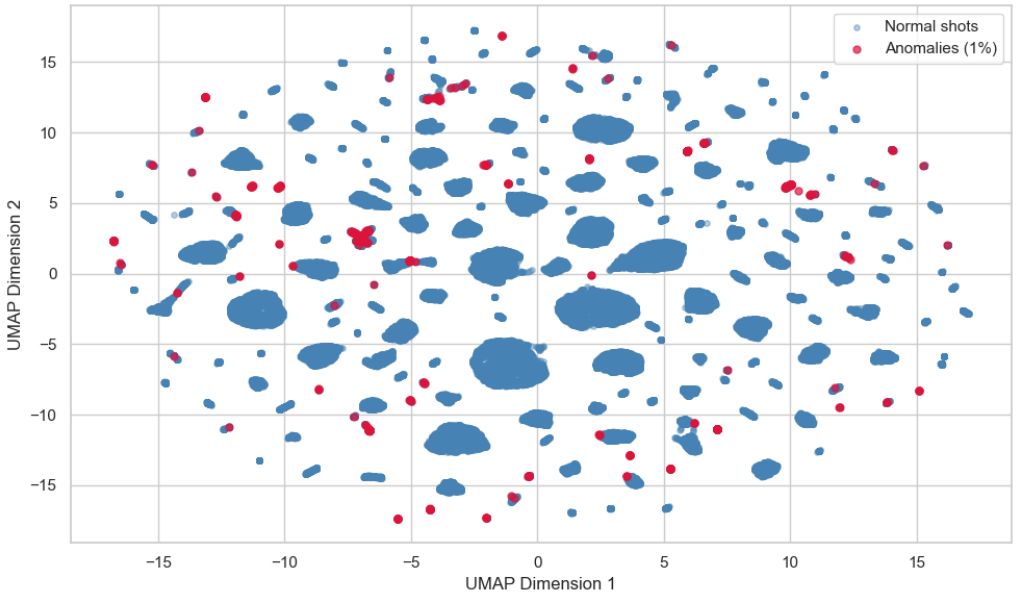


Figure A.14: UMAP projection of Isolation Forest anomalies (1% contamination)

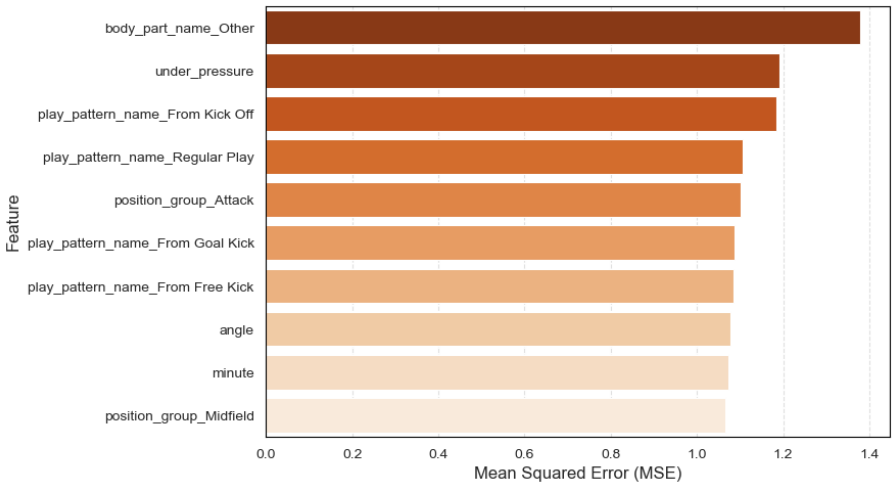


Figure A.15: Mean Squared Reconstruction Error across features for the CAE


ATHENS UNIVERSITY OF ECONOMICS & BUSINESS
ΟΙΚΟΝΟΜΙΚΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΑΘΗΝΩΝ

Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

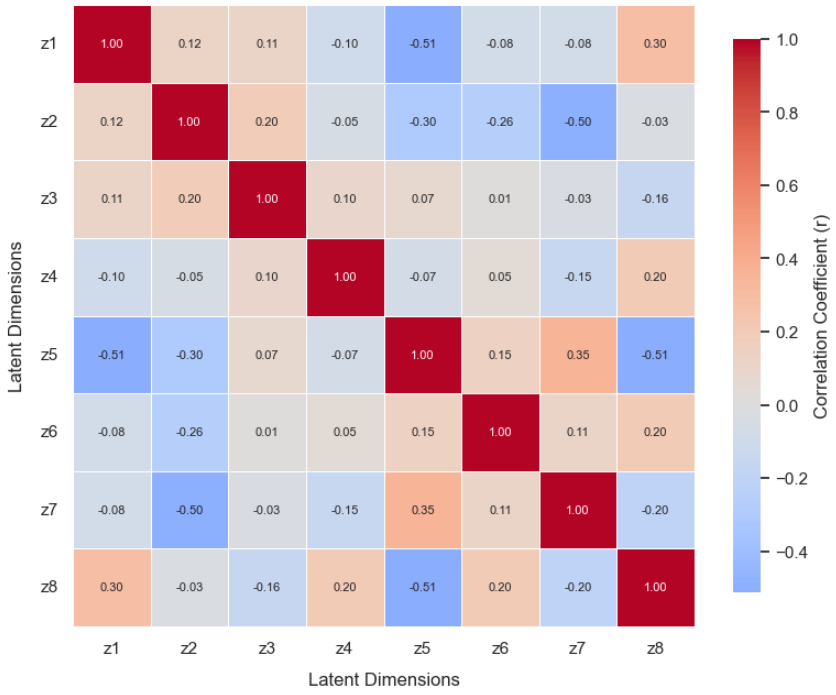


Figure A.16: Correlation matrix among latent dimensions

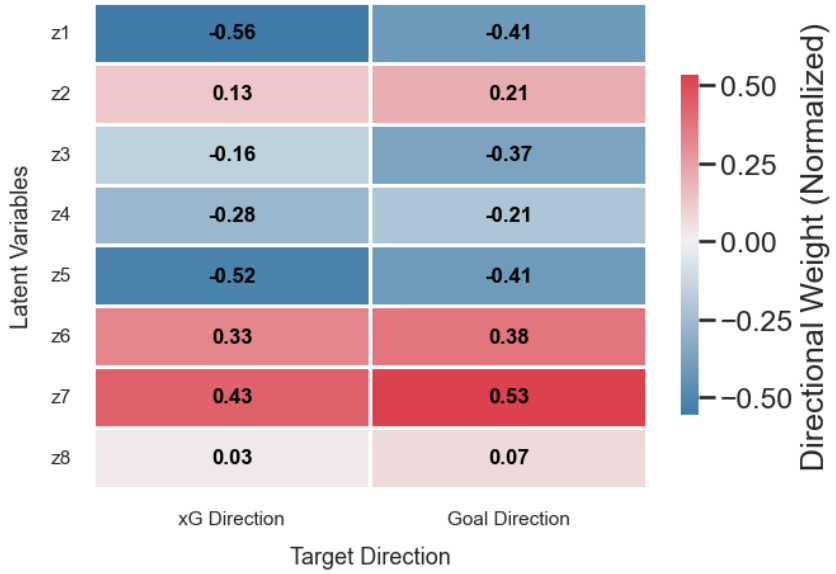


Figure A.17: xG and Goal latent direction weights



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

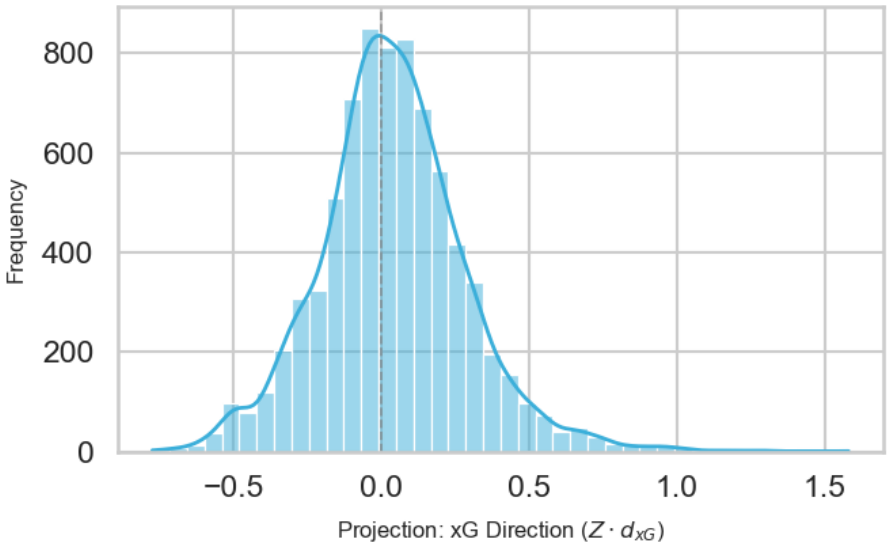


Figure A.18: Distribution of latent projections along the learned xG direction



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Appendix B

Additional Tables

Table B.1: Overview of the modeling approaches employed in this study.

Model Type	**Use of Outcome Variables**	**Analytical Objective**
Supervised Learning	Incorporates goal-related outcomes during training	Predicts goal conversion probability (xG) and evaluates model calibration and generalization
Unsupervised Learning	Excludes outcome variables from training	Identifies latent structures, clusters, and anomalies in shot characteristics
Causal Inference	Applies outcome information post-training	Simulates feature perturbations to infer causal effects on scoring probability



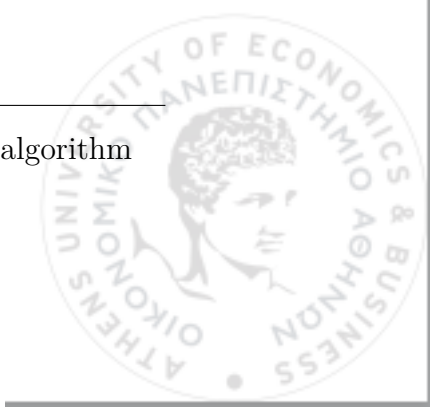
Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Parameter	**Description**	**Typical value**
n_estimators	Number of trees in the forest. A larger number increases stability but also computational cost.	100–500
max_depth	Maximum depth of each tree. Limits complexity and reduces the risk of overfitting.	10–20 or None
max_features	Number of features considered at each split. Commonly set to $\sqrt{p}$, where p is the total number of features.	sqrt
class_weight	Weights to address class imbalance. The option "balanced" scales weights inversely proportional to class frequencies.	balanced
criterion	Impurity measure used for splitting (Gini or Entropy).	gini

Table B.2: Key hyperparameters of the Random Forest algorithm

Parameter	Description	Typical values
n_estimators	Number of weak learners (trees) to be added in the ensemble. Larger values improve accuracy but increase computational cost.	100–500
learning_rate	Shrinks the contribution of each new tree. Lower values lead to better generalization but require more trees.	0.01–0.1
max_depth	Maximum depth of each tree. Controls model complexity and reduces risk of overfitting.	2–5
subsample	Fraction of samples used to fit each tree. Introduces randomness and reduces overfitting.	0.7–1.0
criterion	Splitting criterion used to measure node purity (e.g., deviance for classification).	deviance

Table B.3: Key hyperparameters of the Gradient Boosting algorithm



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Parameter	Description	Typical values
n_estimators	Number of trees in the ensemble. Larger values improve stability but increase computational cost.	100–500
learning_rate	Step size shrinkage controlling the contribution of each tree. Smaller values require more trees but enhance generalization.	0.01–0.1
max_depth	Maximum depth of each tree, controlling complexity and risk of overfitting.	2–6
subsample	Fraction of training samples used to grow each tree. Introduces randomness and reduces overfitting.	0.7–1.0
colsample_bytree	Fraction of features used to build each tree. Enhances diversity among trees.	0.7–1.0
reg_alpha	L1 regularization (lasso) on leaf weights. Encourages sparsity.	0–1
reg_lambda	L2 regularization (ridge) on leaf weights. Improves stability.	0–1

Table B.4: Key hyperparameters of the XGBoost algorithm



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Parameter	Description	Value
Learning rate (η)	Step size for gradient-based parameter updates. Smaller values improve convergence stability.	0.001
Batch size	Number of samples processed before each weight update. Balances memory efficiency and learning smoothness.	64
Hidden layers	Number of dense hidden layers in the network.	2
Neurons per layer	Number of units in each hidden layer, defining model capacity.	64, 32
Activation functions	Nonlinear transformations applied to neuron outputs.	ReLU (hidden), Sigmoid (output)
Dropout rate (p)	Fraction of neurons randomly deactivated during training to mitigate overfitting.	0.3
Early stopping patience	Number of consecutive epochs without validation improvement before stopping training.	10
Optimizer	Adam (Adaptive Moment Estimation) optimizer.	Adam
Loss function	Binary Cross-Entropy (BCE) for probabilistic binary classification.	BCE

Table B.5: Hyperparameter configuration of the Neural Network model



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Metric	**Description**
Log Loss ↓	Measures the accuracy of probabilistic predictions; lower values indicate better calibrated models.
Brier Score ↓	Assesses the mean squared error of predicted probabilities; lower values correspond to better calibration.
AUC ↑	Evaluates the discriminative ability of the model by quantifying the trade-off between true positive and false positive rates.
Accuracy ↑	Proportion of correctly classified observations out of all predictions.
Precision ↑	Proportion of predicted positive cases that are actually positive.
Recall ↑	Proportion of actual positive cases that are correctly identified.
F1 Score ↑	Harmonic mean of Precision and Recall, balancing both aspects in a single metric.

Table B.6: Evaluation metrics employed in the study

Parameter	Description	Value
n_components	Number of retained principal components. Determined from cumulative explained variance.	15
k	Number of clusters. Determined using Elbow method and Silhouette analysis.	Selected empirically
init	Initialization method for centroids.	k-means++
max_iter	Maximum iterations per run.	300
n_init	Number of random restarts to mitigate local minima.	10
random_state	Random seed for reproducibility.	42

Table B.7: Key hyperparameters of PCA and K-Means clustering



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Parameter	Description	Values considered
n_estimators	Number of trees in the ensemble. Controls the stability of isolation paths.	100 (default)
contamination	Proportion of samples assumed to be anomalies.	0.01, 0.02, 0.05
max_samples	Subsample size used to build each tree.	256 (default)
random_state	Random seed for reproducibility.	42

Table B.8: Key hyperparameters of the Isolation Forest algorithm

Table B.9: Key hyperparameters of β–VAE

Parameter	**Description**	**Value(s)**
latent_dim	Dimensionality of latent space (d_z). Controls embedding capacity.	8
β	Weight of KL divergence term. Balances disentanglement vs. reconstruction.	4.0
hidden_layers	Encoder–decoder layer widths.	[128, 64]
learning_rate	Adam optimizer learning rate.	10^{-3}
batch_size	Number of samples per training batch.	256
early_stopping	Validation patience for reconstruction loss.	8 epochs
activation	Hidden-layer nonlinearity.	ReLU
direction_vectors	Post-hoc linear/logistic regressors estimating xG and goal directions.	fitted post-training
random_state	Random seed for reproducibility.	42

Table B.10: VIF values for continuous predictors

Feature	VIF
angle	4.47
shot_statsbomb_xg	2.88
distance	2.78
shot_first_time	1.36
shot_one_on_one	1.18
under_pressure	1.12
minute	1.01
shot_deflected	1.00



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Table B.11: Full logistic regression results

	Coefficient	Std. Error
angle	1.628***	(0.081)
body_part_name_Head	-1.620	(307636.665)
body_part_name_Left Foot	-0.527	(307636.665)
body_part_name_Other	-1.511	(307636.665)
body_part_name_Right Foot	-0.449	(307636.665)
const	-4.108	(864238.334)
distance	-0.107***	(0.005)
minute	0.001	(0.001)
play_pattern_name_From Corner	-0.965	(274728.197)
play_pattern_name_From Counter	0.080	(274728.197)
play_pattern_name_From Free Kick	-0.568	(274728.197)
play_pattern_name_From Goal Kick	-0.432	(274728.197)
play_pattern_name_From Keeper	-0.254	(274728.197)
play_pattern_name_From Kick Off	-0.687	(274728.197)
play_pattern_name_From Throw In	-0.512	(274728.197)
play_pattern_name_Other	-0.362	(274728.197)
play_pattern_name_Regular Play	-0.408	(274728.197)
position_group_Attack	3.854***	(NA)
position_group_Defence	3.656***	(NA)
position_group_Midfield	3.780***	(NA)
position_group_Other	-15.397***	(NA)
shot_deflected	2.125***	(0.089)
shot_first_time	-0.035	(0.043)
shot_one_on_one	0.758***	(0.057)
under_pressure	-0.239***	(0.038)
Observations	50,677	
Pseudo R^2	0.184	

Note: *p<0.1; **p<0.05; ***p<0.01



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Principal Component	**Eigenvalue**	**Explained Variance (%)**
PC1	2.9600	12.33
PC2	1.8695	7.79
PC3	1.6700	6.96
PC4	1.3786	5.74
PC5	1.3011	5.42
PC6	1.2813	5.34
PC7	1.2266	5.11
PC8	1.1115	4.63
PC9	1.0638	4.43
PC10	1.0383	4.33
PC11	1.0273	4.28
PC12	1.0233	4.26
PC13	1.0025	4.18
PC14	0.9977	4.16
PC15	0.9948	4.14
PC16	0.9724	4.05
PC17	0.9081	3.78
PC18	0.8255	3.44
PC19	0.7352	3.06
PC20	0.3868	1.61
PC21	0.2262	0.94
PC22	0.0000	0.00
PC23	0.0000	0.00
PC24	0.0000	0.00

Table B.12: Eigenvalues and explained variance ratio for each principal component



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Feature	**Centroid 1**	**Centroid 2**	**Centroid 3**	**Centroid 4**	**Centroid 5**
Distance	0.244	0.320	-1.223	0.247	-0.728
Angle	-0.275	-0.307	1.286	-0.305	0.607
Minute	0.194	-0.015	0.005	-0.006	0.105
Under Pressure	-0.205	-0.196	0.823	-0.185	-1.037
Shot First Time	-0.128	0.187	-0.654	0.136	0.791
Shot One on One	0.104	-0.090	0.366	-0.110	0.513
Shot Deflected	-0.038	0.018	-0.054	0.010	0.178
Head	-0.278	-0.402	1.617	-0.363	-0.138
Left Foot	0.135	-0.694	-0.503	1.449	0.187
Other Body Part	0.007	-0.042	0.175	-0.042	-0.842
Right Foot	0.086	0.954	-0.787	-1.060	0.027
From Corner	-0.044	-0.188	0.818	-0.219	-0.652
From Counter	-0.434	0.079	-0.277	0.115	-0.477
From Free Kick	-0.394	-0.006	0.159	-0.038	-0.561
From Goal Kick	4.734	-0.202	-0.212	-0.190	-0.170
From Keeper	-0.253	0.032	-0.065	0.025	0.030
From Kick Off	-0.086	0.019	-0.051	0.014	0.232
From Throw In	-0.566	0.071	-0.176	0.081	-0.332
Other Play Pattern	-0.031	-0.037	-0.038	-0.038	26.603
Regular Play	-0.865	0.134	-0.386	0.158	-0.634
Attack Position	0.314	0.022	-0.108	-0.010	0.072
Midfield Position	0.071	0.114	-0.347	0.032	-0.210
Defensive Position	-0.507	-0.177	0.592	-0.028	0.179

Table B.13: Full cluster centroids across all shooting features (standardised values).



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Metric	**1%**	**2%**	**5%**
# Outliers	507	1014	2534
Avg. Distance (Outliers)	9.54	11.12	12.83
Avg. Distance (Normal)	19.05	19.12	19.28
Avg. Angle (Outliers)	1.06	0.93	0.80
Avg. Angle (Normal)	0.47	0.47	0.46
Avg. xG (Outliers)	0.31	0.26	0.22
Avg. xG (Normal)	0.09	0.09	0.09
Goals (% Outliers / Normal)	39.3 / 9.4	33.2 / 9.2	25.8 / 8.8
Under Pressure (%)	51.1 / 27.8	53.1 / 27.5	51.9 / 26.8
First Time Shot (%)	24.7 / 33.0	24.7 / 33.1	32.9 / 32.9
One-on-One (%)	64.3 / 4.5	60.0 / 4.0	45.1 / 3.0
Deflected (%)	15.4 / 1.3	13.9 / 1.2	10.1 / 1.0
Header (%)	46.5 / 17.4	42.8 / 17.1	35.4 / 16.7
Left Foot (%)	30.4 / 31.1	31.8 / 31.0	35.8 / 30.8
Right Foot (%)	15.4 / 51.3	20.8 / 51.6	25.7 / 52.3
From Corner (%)	19.9 / 17.1	19.3 / 17.1	21.3 / 16.9
From Counter (%)	8.7 / 4.6	12.7 / 4.5	12.4 / 4.3
From Free Kick (%)	27.0 / 17.3	26.3 / 17.2	21.9 / 17.1
From Goal Kick (%)	8.5 / 4.0	7.3 / 4.0	8.5 / 3.8
From Keeper (%)	5.9 / 1.8	6.0 / 1.7	7.5 / 1.5
From Throw In (%)	19.3 / 18.8	17.8 / 18.9	16.0 / 19.0
Regular Play (%)	8.1 / 35.0	8.7 / 35.3	10.4 / 36.1
Attack Position (%)	25.4 / 46.3	29.5 / 46.4	32.5 / 46.8
Defence Position (%)	46.2 / 16.5	42.5 / 16.3	40.6 / 15.5
Midfield Position (%)	28.4 / 37.2	28.0 / 37.3	26.9 / 37.7

Table B.14: Isolation Forest anomaly detection full-table results



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

Bibliography

[1] Damoulaki, A., Ntzoufras, I., & Pelechrinis, K. (2025). Lasso multinomial performance indicators for in-play basketball data. Computational Statistics, 40(4), 2157–2181

[2] Lampis, T., Ioannidis, N., Vasilenko, V., & Stavrianna, D. (2023). Predictions of European basketball match results with machine learning algorithms. Journal of Sports Analytics, 9(2), 171–190

[3] Bunker, R., & Susnjak, T. (2019). The application of machine learning techniques for predicting match results in team sport: A review. Journal of Artificial Intelligence Research, 73, 1285–1322

[4] Magaya, H., Liu, F., & Zhang, Y. (2021). A survey on applications of modern deep learning techniques in team sports analytics. IEEE Access, 9, 101234–101256

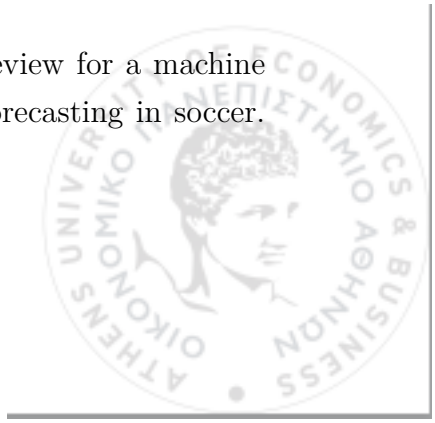
[5] Vec, V., Tomažič, S., Kos, A., & Umek, A. (2024). Trends in real-time artificial intelligence methods in sports: A systematic review. Journal of Big Data, 11(1), 148

[6] Peña, J. L., & Touchette, H. (2012). A network theory analysis of football strategies. Proceedings of the Euromech Physics of Sports Conference, 517–528

[7] Van Haaren, J. (2021). "Why would I trust your numbers?" On the explainability of expected values in soccer. In Proceedings of the Workshop on Artificial Intelligence for Sports Analytics (AISA 2021)

[8] Markopoulou, C., Papageorgiou, G., & Tjortjis, C. (2024). Diverse machine learning for forecasting goal-scoring likelihood in elite football leagues. Machine Learning and Knowledge Extraction, 6(3), 1762–1781

[9] Rossi, A., Pappalardo, L., & Cintia, P. (2022). A narrative review for a machine learning application in sports: An example based on injury forecasting in soccer. Sports, 10(1), 5



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

[10] Rodríguez, M. S., Hassan, A. R., & Coad, A. (2019). Uncovering value drivers of high performance soccer players. Journal of Sports Economics, 20(6), 819–849

[11] Mendes-Neves, T., Meireles, L., & Mendes-Moreira, J. (2024). Forecasting events in soccer matches through language. arXiv

[12] Pollard, R., Ensum, J., & Taylor, S. (2004). Estimating the probability of a shot resulting in a goal: The effects of distance, angle and space. International Journal of Soccer Science, 2(1), 50–55

[13] Mead, J., O'Hare, A., & McMenemy, P. (2023). Expected goals in football: Improving model performance and demonstrating value. PLOS ONE, 18(4), e0282295

[14] Cavus, M., & Biecek, P. (2022). Explainable expected goal models for performance analysis in football analytics. arXiv preprint arXiv:2206.07212

[15] Davis, J., & Robberechts, P. (2024). Biases in expected goals models confound finishing ability. arXiv preprint arXiv:2401.09940

[16] Lucey, P., Bialkowski, A., Monfort, M., Carr, P., & Matthews, I. (2015). Quality vs quantity: Improved shot prediction in soccer using strategic features from spatiotemporal data. In Proceedings of the 9th Annual MIT Sloan Sports Analytics Conference (pp. 1–9). MIT Sloan School of Management

[17] Pardo, M. (2020). Creating a model for expected goals in football using qualitative player information. Master's thesis, Universitat Politècnica de Catalunya

[18] Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32.

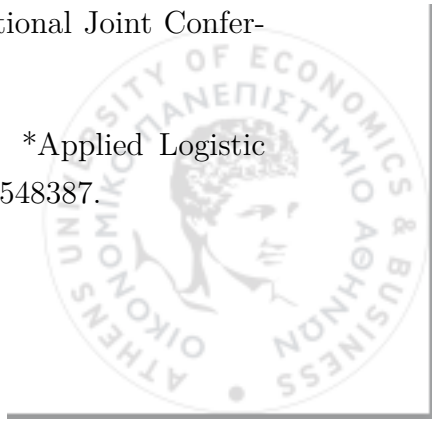
[19] Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning: Data mining, inference, and prediction (2nd ed.). Springer.

[20] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An introduction to statistical learning: with applications in R (2nd ed.). Springer.

[21] Pedregosa, F et. al, (2021). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

[22] Kohavi, R. (1995). A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Proceedings of the 14th International Joint Conference on Artificial Intelligence* (IJCAI), 2, 1137–1143.

[23] Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied Logistic Regression* (3rd ed.). Wiley. https://doi.org/10.1002/9781118548387.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

[24] O'Brien, R. M. (2007). A caution regarding rules of thumb for variance inflation factors. *Quality & Quantity*, 41(5), 673–690.

[25] Powers, D. M. W. (2011). Evaluation: From Precision, Recall and F-measure to ROC, Informedness, Markedness and Correlation. Journal of Machine Learning Technologies, 2(1), 37–63.

[26] Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. Monthly Weather Review, 78(1), 1–3.

[27] Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer.

[28] Friedman, J. H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. Annals of Statistics, 29(5), 1189–1232.

[29] Friedman, J. H. (2002). Stochastic Gradient Boosting. Computational Statistics & Data Analysis, 38(4), 367–378.

[30] Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794).

[31] Jolliffe, I. T. (2002). Principal Component Analysis. Springer Series in Statistics. Springer, New York.

[32] Kaufman, L., & Rousseeuw, P. J. (1990). Finding Groups in Data: An Introduction to Cluster Analysis. Wiley, New York.

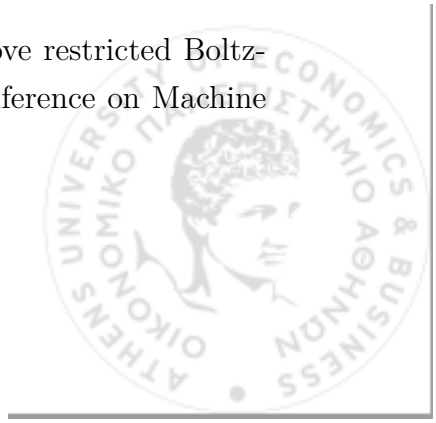
[33] Arthur, D., & Vassilvitskii, S. (2007). k-means++: The advantages of careful seeding. In Proceedings of the 18th Annual ACM-SIAM Symposium on Discrete Algorithms (pp. 1027–1035). SIAM.

[34] Abdi, H., & Williams, L. J. (2010). Principal component analysis. Wiley Interdisciplinary Reviews: Computational Statistics, 2(4), 433–459.

[35] Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. arXiv preprint arXiv:1412.6980.

[36] Goodfellow, I., Bengio, Y., & Courville, A. (2016).

[37] Nair, V., & Hinton, G. E. (2010). Rectified linear units improve restricted Boltzmann machines. In Proceedings of the 27th International Conference on Machine Learning (ICML), 807–814.



Attribution-NonCommercial-NoDerivatives 4.0 International https://doi.org/10.26219/heal.aueb.9726
http://creativecommons.org/licenses/by-nc-nd/4.0/

[38] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. Journal of Machine Learning Research, 15(56), 1929–1958.

[39] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In Advances in Neural Information Processing Systems (NeurIPS), 30.

[40] Louizos, C., Shalit, U., Mooij, J. M., Sontag, D., Zemel, R., & Welling, M. (2017). Causal effect inference with deep latent-variable models. In Advances in Neural Information Processing Systems (NeurIPS), 30.

[41] Kingma, D. P., & Welling, M. (2014). Auto-encoding variational Bayes. arXiv preprint arXiv:1312.6114.

[42] Kusner, M. J., Loftus, J., Russell, C., & Silva, R. (2017). Counterfactual fairness. In Advances in Neural Information Processing Systems (NeurIPS), 30.

[43] Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.

[44] LeCun, Y., Bottou, L., Orr, G. B., & Müller, K.-R. (2012). Efficient BackProp. In G. Montavon, G. B. Orr, & K.-R. Müller (Eds.), Neural Networks: Tricks of the Trade (pp. 9–48). Springer, Berlin, Heidelberg.

[45] Higgins, I., Matthey, L., Pal, A., Burgess, C., Glorot, X., Botvinick, M., Mohamed, S., & Lerchner, A. (2017). Learning basic visual concepts with a constrained variational framework. In International Conference on Learning Representations (ICLR).

[46] Burgess, C. P., Higgins, I., Pal, A., Matthey, L., Watters, N., Desjardins, G., & Lerchner, A. (2018). arXiv preprint arXiv:1804.03599.

[47] Locatello, F., Bauer, S., Lucic, M., Gelly, S., Schölkopf, B., & Bachem, O. (2019). Challenging common assumptions in the unsupervised learning of disentangled representations. In Proceedings of the 36th International Conference on Machine Learning (ICML), pp. 4114–4124.

[48] Alemi, A. A., Fischer, I., Dillon, J. V., & Murphy, K. (2017). Deep Variational Information Bottleneck. In International Conference on Learning Representations (ICLR).

